



Poster Abstracts

CATARACT

1. ROUTINE PROPHYLACTIC ANTIBIOTIC USE BY OPHTHALMOLOGISTS PERFORMING CATARACT SURGERY IN VICTORIA

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Purpose: To survey the current practice of prophylactic antibiotic use by ophthalmologists for preventing post-operative endophthalmitis following routine cataract surgery in Victoria, Australia.

Method: All RANZCO registered ophthalmologists in Victoria were invited by email to complete an online-based questionnaire.

Results: Invitations were sent to 195 consultant ophthalmologists in three rounds over 3 months. The response rate was 53.8% (105 responses). A total of 95 respondents indicated that they performed cataract surgery, with 82 (86.3%) performing at least 100 cataract operations per year, and 14 (14.7%) practicing in a rural location. Ten stated that they did not perform cataract surgery routinely. All surgeons used an antibiotic at the conclusion of surgery. Intracameral cefazolin is most commonly used, 74 (77.9%) currently, with 6 (6.3%) preferring to use intracameral vancomycin, and 15 (15.8%) indicating they did not routinely administer an intracameral antibiotic at all. Of the later group, 13 used subconjunctival cefazolin, one used subconjunctival gentamicin and one did not use a subconjunctival antibiotic either. Four (4.2%) used a combination of both intracameral cefazolin and subconjunctival gentamicin. Intracameral cefazolin use did not differ between rural and metropolitan locations (78.6% vs. 77.8% $p = 0.947$), but intracameral vancomycin was only used in metropolitan settings.

Conclusion: Most Victorian ophthalmologists performing cataract surgery routinely use intracameral cefazolin as prophylaxis in preventing endophthalmitis. Rural location does not affect the use of cefazolin in Victoria.

2. NASOLACRIMAL DUCT OBSTRUCTION AND ENDOPHTHALMITIS

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Purpose: To examine the role of unrecognised nasolacrimal duct obstruction in post-operative endophthalmitis. Although examination of the lacrimal system before undertaking intraocular surgery is recommended, there is little published data on the value of performing nasolacrimal duct (NLD) syringing.

Method: We carried out a prospective cohort study comparing NLD patency in a random sample of eyes undergoing modern phacoemulsification cataract surgery versus a series of eyes which developed endophthalmitis postoperatively. Control cases were recruited through convenience sampling. Endophthalmitis cases from the last 20 months were recalled for testing. Nasolacrimal syringing with normal saline was performed and in those clinically obstructed, a macro-dacryocystogram (MDCG) was performed.

Results: There were 33 control eyes from 28 patients (15 male) and nine eyes of nine patients (three male) with endophthalmitis who were age matched (mean 71.8 vs. 72.3 years). Rate of obstruction on NLD syringing was higher in the endophthalmitis group (4/33 vs. 5/9 $p = 0.013$). Of the four MDCGs in the control group, two showed mucosal thickening but patent ducts, one showed common canaliculal contraction but free drainage, and one demonstrated common canaliculal obstruction. Of the five MDCGs in the endophthalmitis group, two showed patent ducts, one showed NLD stenosis with free drainage, one showed common canaliculal obstruction, and one demonstrated NLD obstruction. Overall true obstruction rate between groups was 3.0% versus 22.2% ($p = 0.111$).

Conclusion: In view of the higher rate of NLD obstruction to syringing in the endophthalmitis group, screening for lacrimal symptoms and syringing if suspicious is warranted prior to cataract surgery.

3. INTRAOCULAR PRESSURE DURING FEMTOSECOND LASER-ASSISTED CATARACT SURGERY

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Purpose: To analyze the course of intraocular pressure (IOP) during femtosecond laser-assisted cataract surgery.

Methods: Interventional, prospective study. Twenty-five eyes underwent femtosecond laser-assisted cataract surgery using the Catalys Precision Laser System with Liquid Optics Interface (OptiMedica, Sunnyvale, CA). IOP was measured using a rebound tonometer (ICare PRO; Tiolat Oy, Helsinki, Finland) during different stages of surgery and analyzed by number of docking attempts, vacuum time, treatment time, and central corneal thickness (CCT).

Results: Mean baseline IOP was 17.5 ± 2.4 (SD) mmHg. During vacuum application, the mean IOP rise was 11.4 ± 3.3 (SD) mmHg. Peak IOPs were recorded immediately after laser capsulotomy and lens fragmentation (mean 36.0 ± 4.4 [SD] mmHg; mean increase from baseline 18.5 ± 4.7 [SD] mmHg) and remained above baseline at 2 minutes following the procedure (26.6 ± 4.0 [SD] mmHg; $P < 0.001$). Multiple regression analysis found no association between IOP rise and number of docking attempts, vacuum time, treatment time, or CCT.

Conclusion: Femtosecond laser-assisted cataract surgery using the Liquid Optics Interface is associated with a mean peak increase in IOP of 18.5 mmHg from baseline and appears safe and well tolerated.

4. CATALYS FEMTOSECOND LASER PRETREATMENT IS ASSOCIATED WITH LESS POST-OPERATIVE CORNEAL OEDEMA AND ENDOTHELIAL CELL LOSS COMPARED TO CONVENTIONAL CATARACT SURGERY

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Purpose: To investigate the effect of Catalys femtosecond laser pretreatment on corneal oedema and endothelial cell loss compared to conventional cataract surgery.

Methods: Prospective, interventional, non-randomised, single surgeon case-control study. Patients underwent either femtosecond laser (OptiMedica, Sunnyvale, CA) pretreatment followed by phacoemulsification cataract surgery (cases) or conventional phacoemulsification (controls). Central corneal thickness (CCT), central 3-mm

corneal volume, central endothelial cell density, and volume stress index (VSI) were measured pre- and post-operatively using Scheimpflug imaging (Pentacam; Oculus, Wetzlar, Germany) and noncontact specular microscopy (EM-3000; Tomey, Nagoya, Japan).

Results: Post-operative corneal oedema, as measured by an increase in central corneal volume, was significantly less with femtosecond laser pretreatment followed by cataract surgery compared to conventional cataract surgery alone at day 1 (5.1 mm^3 versus 6.8 mm^3 ; $P = 0.01$). Femtosecond laser pretreatment was associated with a significant reduction in endothelial cell loss compared to conventional surgery at week 3 (-113 cells/mm^2 versus -210 cells/mm^2 ; $P = 0.03$). There was no significant difference in post-operative intraocular pressure between cases and controls.

Conclusion: Femtosecond laser pretreatment followed by phacoemulsification cataract surgery is associated with a significant reduction in post-operative oedema and endothelial cell loss compared to conventional phacoemulsification.

5. CONTRALATERAL AMAUROSIS AND PARTIAL 3RD NERVE PALSY FOLLOWING PERIBULBAR ANAESTHESIA

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Purpose: We report the first ever known case of contralateral visual loss and partial nerve palsy following peribulbar anaesthesia.

Methods: Case report and review of literature.

Results: We report the case of a 72 year old woman who experienced transient complete visual loss and a partial 3rd nerve palsy following cataract surgery under local anaesthesia in her other eye. Symptoms and signs were assessed to result from the administration of a peribulbar block, which was presumably associated with ipsilateral trans optic nerve sheath spread. This appears to be the first report of contralateral amaurosis and oculomotor nerve palsy following peribulbar anaesthesia.

Conclusion: This is the first reported case of contralateral visual loss and partial third nerve palsy following peribulbar anaesthesia. We believe this is secondary to ipsilateral optic nerve sheath anaesthetic diffusion. This case brings attention to the complications associated with peribulbar anaesthesia, which should now include ipsilateral optic nerve sheath anaesthetic diffusion. This should be an important differential diagnosis in the event of contralateral cranial nerve palsies and visual disturbances following

ophthalmic surgery in which peribulbar as well as retrobulbar anaesthesia is administered.

6. ACUTE INTRAOPERATIVE ROCKHARD EYE SYNDROME – REPORT OF A NEWLY RECOGNISED SYNDROME AND ITS MANAGEMENT IN SIX CASES

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Purpose: To describe a newly recognised entity, the Acute Intraoperative Rockhard Eye Syndrome (AIRES), and its management in six cases. AIRES is characterised by anterior chamber (AC) shallowing and an acute, marked intraocular pressure (IOP) increase during phacoemulsification cataract surgery, precluding further surgery, but with no evidence of choroidal haemorrhage.

Method: Prospective, consecutive, non-comparative, interventional case series of phacoemulsification cataract surgery. Management was with emergent intraoperative pars plana needle aspiration (PPNA) of retrolenticular fluid with a 23-gauge needle attached to a 3 mL syringe. Follow-up was at 1 day, 1 week and 1 month. Resolution of AIRES, posterior segment integrity, and corrected distance visual acuity (CDVA) were documented.

Results: Six patients with AIRES were identified in 298 cases. Mean age was 81 years; all were female with nuclear cataracts of grade 2–4 (LOCS II). In each case, PPNA successfully and immediately normalised AC depth and IOP, and completion of surgery proceeded uneventfully. Follow-up was 100%. Postoperative funduscopy revealed transient mild vitreous haemorrhage in one patient; new posterior vitreous detachment in one patient; and localised vitreous syneresis in three patients. CDVA in 5 of 6 patients was 6/4 at 1 month.

Conclusion: AIRES presents with AC shallowing and marked IOP elevation during cataract surgery, occurring in 2% of cases in this series. PPNA effectively resolved each case immediately and was not associated with any significant complications.

7. REAL TIME VISUALISATION OF ANAESTHETIC FLUID LOCALISATION FOLLOWING INCISIONLESS SUB-TENON BLOCK

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Purpose: The sub-Tenon block is a commonly used procedure performed to provide ocular anaesthesia. We have previously developed a new minimally invasive technique of sub-Tenon's anaesthesia which allows access to the sub-Tenon space without prior conjunctival incision (Allman, Theron, & Byles, 2008). This reduces conjunctival trauma and improves both block quality and onset. This study was undertaken to investigate the spread of local anaesthetic following incisionless sub-Tenon's using B-scan ultrasonography and to compare it with standard techniques.

Method: Patients who were having routine cataract extraction by phacoemulsification were selected, and informed consent obtained. Dynamic B-scan ultrasonography was performed on the eye during administration of sub-Tenon's local anaesthesia administered either by the incisionless technique, or standard methods.

Results: The anaesthetic fluid was easily visualised on ultrasonography, as a dark outline tracking behind the globe in the retrobulbar space, before developing the characteristic 'T' sign around the optic nerve (see video). Preliminary results indicate there is no significant difference between techniques.

Conclusion: B-scan ultrasonography is a useful method for visualising the distribution of anaesthetic fluid following administration of a sub-Tenon block. Local anaesthetic spread appears to be similar irrespective of technique.

8. OPTICAL RESIDUAL ASTIGMATISM POST CATARACT SURGERY: WHAT IS IT AND DOES IT MATTER?

WITHDRAWN

9. PROSPECTIVE STUDY OF INTRAOPERATIVE FLOPPY IRIS SYNDROME (IFIS) INCIDENCE IN BARCELONA USING INTRACAMERAL PHENYLEPHRINE

WITHDRAWN

10. TRANS-SCLERAL FIXATION OF A TORIC SUPPLEMENTARY INTRAOCULAR LENS TO PREVENT ROTATION IN A PSEUDOPHAKIC PATIENT

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Purpose: To describe a method of scleral fixation of a toric pseudophakic supplementary intraocular lens (PSIOL). The existing PSIOL was a Rayner 653T with -3.00D of sphere and +5.50D of cylinder. Lid forces and ovalisation of the sulcus resulted in repeated anti-clockwise rotation off the desired axis.

Methods: A 61-year-old with penetrating keratoplasty and a pseudophakic supplementary toric intraocular lens (PSIOL) in the right eye presented to our practice multiple times for PSIOL re-alignment. The PSIOL was therefore sutured transsclerally to maintain axis orientation at 20 degrees. Haptics were externalised through corneal wounds, and sutures attached using a modified cow hitch technique.

Results: The lens remained stable at the desired axis post operatively. The postoperative uncorrected visual acuity was 6/7.5 with a residual refraction of +0.50 / -1.00 × 76.

Conclusions: Transscleral fixation of a supplementary IOL has been described and has been shown to result in potential long-term stabilisation with good visual rehabilitation.

11. THE DYNAMIC RISE FOR ASPIRATION AT OCCLUSION INFLUENCES THE PHACO TIME AND ASPIRATE VOLUME USING DTIP NEEDLES AND TORSIONAL PHACOEMULSIFICATION

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Purpose: Aslan presented a poster at the 2012 ASCRS meeting suggesting that setting the dynamic rise (DR) at -1 (a 25% reduction in the aspiration rate on full occlusion) reduced the phaco time 15% and aspiration volume 20% using a Kelman needle and continuous torsional phaco in 62 patients. This study assesses the effect of dynamic rise adjustment using a dTip needle and burst-mode torsional phaco.

Methods: Patients with grade NS2+ or more cataract had cataract extraction using machine settings of 75 cm irrigation, 25 cc/min aspiration, 250 mmHg vacuum, 60% burst mode (50 msec on, 80 msec off minimum). The DR was set at +3 for 93 consecutive cases, then -1 for 37 consecutive cases. The outcome

measures were: effective ultrasound time, aspirate volume, and *flowPhaco* (grams cataract cut per cc aspirate over the effective phaco time – this controls for the actual lens mass in each patient). These were assessed using t-tests.

Results: Reducing the DR to -1, increased the effective ultrasound time 24% (10.05 to 12.46 secs, $p = 0.03$), decreased aspiration volume 12% (33.90 to 29.68 cc, $p = 0.048$), but produced only a small reduction in *flowPhaco* (9%, 1.01 to 0.92, $p = 0.436$).

Conclusions: The DR function has a significant effect on phacoemulsification power and aspirate volume. The effect is different for different power modulations and the needle used. We found an increase in the DR enhanced phaco-emulsification, but used slightly more aspirate, resulting in only a modest improvement in *flowPhaco*.

12. SURGICALLY INDUCED ASTIGMATISM IN MANUAL SMALL INCISION CATARACT SURGERY IN FIJI

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Purpose: To estimate the surgically induced astigmatism using vector analysis in manual small incision cataract surgery.

Methods: This prospective study consisted of a consecutive series of 44 eyes of 56 patients who had uncomplicated manual small incision cataract surgery performed by a single surgeon between Jan 2011 to Jun 2012. The incision size was documented in all cases & the surgical technique was standardized. Outcome measures used to calculate the vector mean of induced astigmatism included preoperative & post-operative keratometry using the Nidek hand held auto keratometer at 6–10 weeks postoperatively. The data was entered into a spreadsheet & statistical analysis was performed using Microsoft excel.

Results: This article reports postoperative clinical outcomes at 6–10 weeks.

The age spectrum was between 42 to 90 years. 68% of patients were female & 55% belonged to the native ethnic group. 84% patients had visual acuity of 6/12 or better.

SIA calculated using the vector method had a mean of 1.10 diopters with a range of 0.09–4.62 diopters (SD = 0.7), Flattening effect at the surgical axis (90°) was -0.25 (SD = 0.84). A summated vector mean of the SIA was 0.45D at 117°.

Conclusions: These data suggest that this method of incision construction is astigmatically active to a small degree but, in common with many incisional

techniques, the axis of effect of the incision varies suggesting that a significant proportion of the observed effect is due to the common variations in all keratometric measurements.

13. CATARACT ORDER OF THE LIST (COOL) – IMPLEMENTING COOL IN CATARACT SURGERY TO IMPROVE SURGEON COMFORT AND FACILITATE OPTIMAL OPERATING CONDITIONS

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Purpose: A method for ordering the patient order on the cataract surgical list employs a document detailing medical, technical and psychosocial factors which may optimise operating conditions and surgeon comfort. This aims to allow the surgical team to commence the list with simpler cases before progressing to potentially challenging cases.

Method: The COOL document (see below) identifies twenty factors that are synthesised into three categories which can be used preoperatively to plan the list:

1. Early: Patients with insulin-dependent diabetes, chronic kidney disease, previous uneventful surgery or who are otherwise low risk/ low maintenance patients.
2. Middle: Shallow anterior chambers, ametropia, moderate grade nuclear sclerosis (LOCS 2 Grade 2–3), toric implants or with pseudoexfoliative syndrome.
3. Late: Higher grade nuclear sclerosis (LOCS 2 Grade 3–5), dark brown cataracts, toric implants, enophthalmos, difficult visualisation, previous eventful surgery, or otherwise high risk/high maintenance patients. Similarly placed are anxious patients, or those close to the surgeon as friends or colleagues.

Results: The COOL document has been used with subjective success by a single surgeon over four years. COOL can:

- Identify accurately an optimal patient surgical order of the list.
- Allow the surgical team to reach optimal mindset and dexterity prior to managing potentially challenging patients.
- Be time effective, is simple to implement and involves minimal cost.
- Be an educational tool for neophyte surgeons.
- Provide a platform for further exploration of the role of surgeon comfort.

Conclusion: COOL has provided an effective technique to optimise the order of cataract surgical lists.

14. EVALUATION OF MACULAR THICKNESS AFTER UNCOMPLICATED PHACOEMULSIFICATION USING OCT

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Purpose: To evaluate foveal thickness and macular volume before and after an uncomplicated phacoemulsification surgery by OCT to determine incidence of CME and its correlation with visual acuity.

Methods: A prospective study of 48 eyes of senile cataract [27 females, 21 males] undergoing phacoemulsification with foldable acrylic IOL was carried out between June 2010 to Jan 2011 [median age 58 years]. The patients of uveitis, glaucoma, diabetes mellitus, posterior capsular rent and retinal diseases were excluded from the study. Patients were subjected to detailed ocular examination including foveal thickness and macular volume determination by line and fast macular scan by Stratus OCT preoperatively and 1 and 3 month after surgery.

Results: The average foveal thickness preoperatively was $161.11 \pm 28.23 \mu\text{m}$ and at 1 month and 3 month was $181.74 \pm 51.64 \mu\text{m}$ and $173.23 \pm 35.04 \mu\text{m}$ respectively. The average macular volume preoperatively was $6.40 \pm .71$ and postoperatively at 1 and 3 month was $6.65 \pm .63$ and $6.64 \pm .62$. There was significant increase in foveal thickness and volume at 1 month [pvalue .004,.001] and 3 month [p value .004,.006]. Cystoid changes were seen in 4 patients [incidence 8.33%]. There was significant correlation of post operative thickness and volume of macula with logMAR visual acuity [SPEARMAN CORRELATION 1.00].

Conclusion: In uncomplicated phacoemulsification foveal thickness and macular volume increases after surgery and it has significant correlation with visual acuity. OCT is very reliable tool in determining sub-clinical CME, particularly in patients not achieving logMAR unit 0.

15. CORNEAL SIDEPORT DEPOSITION OF INTRAEPITHELIAL AND SUBEPITHELIAL OPHTHALMIC VISCOELASTIC DEVICE FOLLOWING PHACOEMULSIFICATION SURGERY

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Purpose: To assess the incidence and significance of subepithelial corneal deposits of Ophthalmic Viscoelastic Device (OVD) at the sideport incision site following phacoemulsification cataract surgery.

Method: Prospective, consecutive, non-randomized single surgeon study of 248 eyes undergoing phacoemulsification cataract surgery. Sideport OVD deposits were recorded at 1 week and 1 month postoperatively, as was duration of surgery. Confocal microscopy was done on one case, and in another, a sample of epithelium containing OVD vacuoles underwent Haematoxylin and Eosin staining using light microscopy. Electron microscopy and mass spectrometry were also attempted. A P value less than 0.05 was considered significant.

Results: 99.6% of patients were followed up for 1 month. There was OVD present in intraepithelial and subepithelial tissue at the sideport incision site in 38.1% of eyes at one week. At 1 month 67% of affected cases showed no residual OVD. 81.5% of the cohort achieved CDVA of 6/4 at one month, and each cornea was otherwise normal. There was a positive correlation between duration of surgery and presence of OVD, which was statistically significant ($p = 0.014$).

Conclusion: 1 month postoperatively, 33% of patients demonstrated residual subepithelial OVD at the sideport. This was significant correlated with duration of surgery. Residual corneal OVD at the sideport incision site appears to have no detrimental effect on the patient's final corrected visual outcome or corneal function. This is a novel observation and represents the first confocal imaging and light microscopy of Corneal OVD deposits.

CORNEA

16. PATTERN OF ETIOLOGIC AGENTS AND SENSITIVITIES FOR MICROBIAL KERATITIS AT CWM HOSPITAL – FIJI. A 4 YEAR RETROSPECTIVE STUDY (2007–2010)

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Purpose: To determine the etiologic agents responsible for microbial keratitis at CWM Hospital and their likely antibiotic sensitivities.

Methods: 4 year retrospective observational study of corneal scrape results taken from the Microbiology Laboratory Register at the Colonial War Memorial (CWM) Hospital (January 2007–December 2010). The minimum criterion for a positive culture was the growth of ≥ 5 colonies on a solid medium.

Results: 24 pathogens were isolated from 19/35 corneal scrapes. 16/35 corneal scrapes showed no growth due to incorrect temperature and incubation time. The average patient age was 39 years (range 4–61 years) and male to female ratio was 3.75:1. Bacteria accounted for 87.5% (21/24). Gram negative bacilli recovered in 58.3% (14/24) scrapes while gram positive cocci in 29.2% (7/24) and fungi (yeast) 12.5% (3/24) scrapes. No isolates of *Acanthamoeba* were found. Of the total bacterial isolates resistance to ciprofloxacin was zero percent ($n = 17$), chloramphenicol 14.2% (3/21), gentamycin 5.5% (1/18) and penicillin 83.3% (15/18).

Conclusion: The study supports use of ciprofloxacin as monotherapy for microbial keratitis in Fiji. This information helps in initial selection of therapy for microbial keratitis while waiting for culture results. However there is a need for accurate database to record corneal scrape details and a protocol for antibiotic sensitivity and resistance. This study will provide guidance for a larger prospective study.

17. ULTRA-THIN DSAEK

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Purpose: To evaluate the outcome of Ultra-Thin (UT) DSAEK performed in eyes with good visual potential.

Methods: Of 163 eyes undergoing UT-DSAEK, 76 eyes were deemed to have no ocular co-morbidity other than corneal decompensation secondary to Fuchs endothelial dystrophy or pseudophakic bullous keratopathy +/- cataract or previous cataract surgery. Visual acuity, refraction, and endothelial cell counts were assessed before and at 1, 3, 6, and 12 months after transplantation of an UT-DSAEK graft obtained by a double pass of the microkeratome. Phakic patients either underwent combined cataract surgery and implantation of lens, or remained phakic.

Results: In 45/76 Eyes (59%), best spectacle corrected visual acuity was $\geq 6/6$, and 75/76 (99%) achieved 6/9 or better. 24/76 (32%) achieved max BSCVA within 3 months of the surgery date. Mean endothelial cell loss at 12 months was 33.2%.

Conclusions: Outcomes of UT DSAEK continue to compare favourably with those of conventional DSAEK, and do not differ substantially from those of DMEK.

18. TWO STEP SURGERY FOR THE MANAGEMENT OF FAILED PK WITH HIGH DEGREE ASTIGMATISM

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Purpose: To evaluate a procedure alternative to repeat penetrating keratoplasty (PK) for failed PK with high-degree astigmatism (HA).

Methods: Best-corrected visual acuity (BCVA), refraction and corneal topography were assessed before and 1, 3, 6, and 12 months after 2-step surgery (1. Descemet stripping automated endothelial keratoplasty (DSAEK); 2. relaxing incisions in the PK wound up to the DSAEK) performed within 2 months in 6 eyes with failed PK and HA.

Results: In all eyes, 1 month after step 2, BCVA improved from $\leq 6/60$ to $\geq 6/9$. Mean topographic astigmatism decreased from 7.91 ± 1.94 diopters (D) to 1.33 ± 1.32 D ($p < 0.0001$).

Conclusion: In eyes with failed PK and HA, excellent BCVA is obtained much earlier with our 2-step procedure than with repeat PK.

19. LONG-TERM OUTCOMES OF DEEP ANTERIOR LAMELLAR KERATOPLASTY VERSUS PENETRATING KERATOPLASTY IN AUSTRALIAN KERATOCONUS PATIENTS

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Purpose: To compare long-term visual outcomes following deep anterior lamellar keratoplasty (DALK) and penetrating keratoplasty (PK) for keratoconus.

Methods: Single-surgeon retrospective chart review.

Results: A total of 31 DALK and 42 PK were included; mean follow-up was 3.9 yrs (SD 2.9) and 4.4 yrs (2.5) respectively. The proportion of patients achieving post-operative suture-out logMAR best correct visual acuity (BCVA) of 0.2 or better was not significantly different between DALK and PK cases ($p = 0.16$). However, PK cases were significantly more likely to achieve BCVA 0.0 ($p < 0.05$). Cases with post-op astigmatism $\geq 5D$ were similar between DALK and PK cases ($p = 0.24$). Complications (including rejection, cataract and glaucoma) were significantly higher in PK cases ($p < 0.05$).

Conclusion: Visual outcomes for DALK vs PK were not significantly different but complications were significantly higher in the PK group.

20. A SYNTHETIC CARRIER FOR CULTURED CORNEAL ENDOTHELIUM IMPLANTATION

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Purpose: A tissue engineered corneal endothelium could overcome potential problems associated with donor tissues. We have developed a thin (50 μ m) synthetic hydrogel film suitable for corneal endothelial cell (CEC) culture and implantation. An ovine model was used to develop and evaluate the film in vitro and in vivo.

Methods: Synthetic hydrogel films (SHF) with superior properties were developed using a novel approach. Ovine CEC (OCEC) density in vivo was determined by specular microscopy and compared to cell density achieved on tissue culture plastic and the hydrogel film as determined using Computer Assisted

Steriological Toolbox (Leica). Films were implanted into an ovine anterior chamber for 30d before histological evaluation.

Results: OCEC density in vivo was determined to be 3150 cells/mm² in 14 month old merino sheep (n = 3). Cell density at 28d was 3150 (SEM 459) on SHF, 2530 (SEM 526) on collagen I coated SHF and 3770 (SEM 201) tissue culture plastic (n = 4). The films had sufficient tensile strength for implantation and readily un-scrolled within the anterior chamber. Histology determined that PHF did not produce and inflammatory response in vivo (n = 4).

Conclusion: Hydrogel films suitable for implantation that support CEC cultivation have been produced. A combination of a SFH with a cultured CEC monolayer may be able to replace the donor tissue lenticule used in Descemet's Stripping Automated Endothelial Keratoplasty.

21. NON-TRAUMATIC PAECILOMYCES LILACINUS ENDOPHTHALMITIS-KERATITIS IN AN IMMUNOCOMPETENT CAUCASIAN MALE SUCCESSFULLY TREATED WITH VORICONAZOLE

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Purpose: Paecilomyces lilacinus keratitis/endophthalmitis is a rare but serious fungal infection, largely encountered in the setting of post-operative complication or ocular trauma. This case documents a deep keratitis with extension into the anterior chamber (AC), treated with intracameral, topical and oral voriconazole, resulting in an excellent visual outcome.

Method: Case report.

Results: A 49 year old man with a two month history of severe "knifing" OS pain, foreign body sensation, photophobia and watery discharge was referred for management of endophthalmitis. On presentation his best corrected visual acuity (BCVA) was OS 6/10. There was a moderate anterior chamber (AC) reaction with a fluffy white adherent mass from twelve to three o'clock which appeared to span the AC angle to involve corneal endothelium and stroma. He was treated initially with topical antibiotics, oral voriconazole and regular intracameral voriconazole. P. Lilacinus was cultured on repeat AC aspirates and treated with third daily intracameral for four weeks, topical for four weeks and oral voriconazole for 12 weeks. BCVA was hand movements two days after presentation, but

recovered to 6/6 four weeks after hospital discharge. Systemic workup, including echocardiography, did not yield a source. The patient remains disease free at four months post onset of symptoms.

Conclusion: This case highlights the importance of early recognition and referral of a potentially sight-threatening intra-ocular infection of which only a handful of cases exist in the literature and demonstrates that timely initiation of topical, oral and twice-weekly intracameral voriconazole can be curative.

22. MULTIPLE MYELOMA WITH CORNEAL PARAPROTEINEMIA

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Purpose: We report a case of corneal paraproteinemia in a patient that was diagnosed of multiple myeloma. This is a rare associated corneal finding, and it is necessary to exclude other diseases, syndromes and dystrophies with overlapping clinical features.

Methods: Case report and review of medical literature. A 65-year-old man was hospitalized by a septic shock from a pneumococcal pneumonia, he was in study cause of basement situation. He consulted in ophthalmology for orbital cellulitis. In the examination was observed edema and hyperemia of the margin of the lid, with inflammatory appearance. The biomicroscopy showed crystal deposits throughout the corneal thickness with corneal opacities in both eyes. The fundus was normal.

Results: The general studies of the patient determined the presence of 60% plasma cells with atypical morphology in Bone Marrow Aspiration; 3% plasma cells with atypical phenotype in Immunophenotypic study; IgG kappa monoclonal component was found increased in blood (58.6 g / L) and urine (0.94 g / day). The diagnose of multiple myeloma, in spite of IgG kappa was considered.

Conclusions: The corneal paraproteinemia are immunoproteins deposits that can appear in multiple myeloma, macroglobulinemia of Waldenström, lymphoproliferative disorders and leukemia. Corneal involvement may be the first manifestation of the disease. Severe corneal involvement may require penetrating keratoplasty. Corneal involvement in multiple myeloma is rare, retinal manifestations are more common. The corneal deposits must be differentiated from other causes of corneal crystals: cystinosis, drops and dystrophy of Schnyder.

23. ASSOCIATION BETWEEN FUCHS'S CORNEAL DYSTROPHY AND A SINGLE NUCLEOTIDE POLYMORPHISM ON CHROMOSOME 18 IN MEXICAN POPULATION

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Purpose: To determine the association of risk alleles of the single nucleotide polymorphism that has been most highly associated with the presentation of Fuchs's corneal dystrophy, rs613872 in Intron 3 of TCF4, in Mexican population.

Methods: A case control study was made. We recruited 20 subjects of Mexican population with the diagnosis of Fuchs's corneal dystrophy. Subjects were considered with this diagnosis if the clinical description of the corneal was consistent with grade 1 guttae or higher on a published scale. All subjects were of Mexican descent. All subjects provided written informed consent. We genotyped the single nucleotide polymorphism rs613872 in Intron 3 of TCF4 and confirmed the association with the disease.

Results: Three patients presented the polymorphism. None of the patients of the control group presented the polymorphism.

Conclusions: The polymorphism rs613872 in Intron 3 of TCF4 has been reported to be the most highly associated SNP in patients of European descent. This polymorphism was present in 15% of the patients with the diagnosis of FED, this is not statistically significant and suggests that in Mexican population other polymorphisms must be associated with the presentation of this disease, but the fact that none of the patients of the control group presented the polymorphism suggests that this could be confirmed in a bigger sample of population of the same descent.

24. IRIDOCORNEAL ENDOTHELIAL SYNDROME: LONG TERM FOLLOW-UP AND MANAGEMENT

WITHDRAWN

25. A RETROSPECTIVE ANALYSIS OF THE VISUAL OUTCOMES AND COMPLICATION OF CORNEAL ENDOTHELIAL TRANSPLANTATION

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Purpose: Retrospectively analyse visual outcomes and complications following corneal endothelial transplantation.

Setting: Anterior segment private practice.

Design: Case series.

Methods: 155 eyes of 132 patients following corneal endothelial keratoplasty with or without combined cataract surgery (triple procedure) between July 2007 to March 2012 were reviewed. Graft outcomes were compared with national data for endothelial keratoplasty and penetrating keratoplasty.

Results: Mean follow-up was 615 days (26 to 1533 days). 89 eyes had a corneal endothelial transplant and 66 had a triple procedure. Pseudophakic bullous keratopathy was the most common indication for transplantation (44.5%), and 112 patients had Fuchs' endothelial dystrophy. 148 eyes with greater than 60 days of follow-up had a mean BCVA improvement of 2.1 lines of Snellen visual acuity ($P < 0.0001$); improved or unchanged in 87.2%. 58.3% of transplantation only and 71.9% of triple procedure eyes had 6/12 or better visual acuity postoperatively; triple procedure eyes had 1.8 lines ($P = 0.0003$) better vision. 2.7% eyes lost 3 or more lines of Snellen visual acuity, most from glaucoma. Ocular hypertension occurred in 9.0%, half of these had a history of glaucoma. 14.2% had areas of detachment, 9.0% dislocated; 7.7% had an episode of rejection; 3.2% transplants failed and required regrafting. Results were significantly better than national figures for endothelial and penetrating grafts.

Conclusions: Endothelial keratoplasty is a relatively safe treatment for causes of corneal endothelial dysfunction. Earlier grafting in the triple procedure gave better visual outcomes, and there was significantly better results compared to national data and to penetrating keratoplasty.

26. TO SCRAPE OR NOT TO SCRAPE... THAT IS THE QUESTION

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Purpose: To report the utility and techniques of epithelial debridement in the diagnosis and management of corneal pathology.

Design: Retrospective Interventional Case Series.

Participants: All patients who underwent epithelial debridement or diamond burr polishing by a single surgeon between 1 Jan 2005 and 1 March 2012.

Methods: Data were collected regarding the symptoms and indications for epithelial debridement. Details regarding the techniques, adjunctive therapies and postoperative course were recorded.

Results: We performed corneal scrapes on 104 eyes of 100 patients suffering from pain, anterior surface abnormality or decreased vision. Thirty-three eyes had corneas scraped for diagnosis of microbial or neoplastic processes. Of the 71 eyes scraped or polished for therapeutic purposes, the majority (30, 42%) were for recurrent corneal erosions. A further 16 eyes (22.5%) were treated for band keratopathy, 10 (14.1%) for corneal scarring, 8 (11.3%) for foreign body or rust ring removal and the remaining 7 (9.9%) were for therapy of corneal dystrophies. Removal of corneal scarring and treatment of dystrophies often involved adjuvant treatment with Mitomycin C or alcohol.

Conclusions: Epithelial debridement and diamond burr polishing are important techniques for the diagnosis and management of anterior surface pathology. A variety of techniques and associated adjunctive therapies have been described. These techniques have been shown to be safe and effective for in-office treatment of superficial corneal disease.

28. IMPACT OF KERATOCONUS ON QUALITY OF LIFE

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Purpose: To assess the impact of keratoconus (KCN) on quality of life using Vision-related Quality of Life Index (VisQOL) questionnaire.

Methods: 131 participants with KCN were recruited from private and public clinics in Melbourne. Visual

impairment (VI) was categorised based on the best corrected visual acuity (BCVA) as none ($>6/12$), moderate ($<6/12-6/60$) and severe ($<6/60$). The utility score calculated using the six-item VisQOL questionnaire (ranging from 0 representing death to 1 representing full health) was the main outcome measure. The mean VisQOL utility scores of different groups were analysed using SPSS software.

Results: Participants mean age was 36.8 years (range: 14 to 75 years) and 56.5% (N = 74) were males. The mean VisQOL utility score was 0.61 ± 0.28 (range: 0.02 to 0.99). The mean utility scores were reduced for the 6 items in the questionnaire – injury (0.8 ± 0.25), coping (0.71 ± 0.28), friendships (0.68 ± 0.32), assistance (0.81 ± 0.29), roles (0.64 ± 0.30) and confidence (0.68 ± 0.30). The utilities reduced with increasing age ($r = -0.23$, $p < 0.05$) and were better for males (0.65 ± 0.27) than females (0.54 ± 0.28), $P < 0.05$. As keratoconus is an asymmetric condition, we considered the BCVA in the better eye and the worse eye separately. The mean VisQOL utility scores were reduced significantly with deteriorating BCVA in the worse eye: no VI (0.71 ± 0.26), moderate VI (0.56 ± 0.26) and severe VI (0.32 ± 0.22) respectively ($P < 0.001$).

Conclusion: This is the first study to show that VisQOL utility scores were low in participants with KCN and they were significantly reduced with increasing age and deteriorating vision in the worse eye.

29. EFFECT OF PTERYGIUM ON CONTRAST SENSITIVITY

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Purpose: To evaluate the effect of pterygium on contrast sensitivity.

Methods: 36 eyes with pterygium and 18 eyes without pterygium were included in the study. Size of pterygium was measured on the slit lamp vertically at limbus and categorized into 3 groups (≤ 3 , 3.1 to ≤ 5 , > 5 mm), horizontally on cornea (0.5 to ≤ 2 , > 2 mm) and area was calculated by multiplying the two and categorized into 3 groups (≤ 7 , 7.1 to ≤ 14 , > 14 mm²). Contrast sensitivity was measured using CSV-1000E charts (Vector Vision) at 3, 6, 12, 18 cpd under photopic and mesopic conditions.

Results: Mean of Vertical length of pterygium was 4.41 ± 1.12 , horizontal width was 2.09 ± 0.68 and area was 9.7 ± 5.01 . Significant difference in both photopic and mesopic contrast sensitivity at all frequencies was noted between normal eyes and Vertical length of pterygium > 3 mm, horizontal width ≥ 0.5 mm and

area > 7 mm². Contrast sensitivity had significant negative correlation with all 3 parameters.

Conclusion: Pterygium with vertical length > 3 mm, horizontal width ≥ 0.5 mm and area > 7 mm² caused a significant decrease in both photopic and mesopic contrast sensitivity at all spatial frequencies. Contrast sensitivity had significant negative correlation with all 3 parameters.

30. INTRACORNEAL RING SEGMENT IMPLANTATION, CORNEAL COLLAGEN CROSS-LINKING AND TORIC PHAKIC IOL IMPLANTATION: A THREE-STAGE PROCEDURE FOR KERATOCONUS

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Purpose: To present the results of staged combined treatment for keratoconus with intrastromal corneal ring segment (ICRS) implantation followed by corneal collagen cross-linking (CXL) with riboflavin/ultraviolet-A (UVA) then toric phakic implantable collamer lens implantation (ICL).

Methods: Prospective interventional case series of 14 eyes of 9 patients with progressive keratoconus. All patients underwent KeraRings ICRS implantation, followed by CXL, and then ICL implantation (with a minimum of six months between procedures). All patients were followed up for a minimum of 12 months after ICL implantation.

Results: Both mean uncorrected distance (decimal) visual acuity (UDVA) and mean corrected distance (decimal) visual acuity (CDVA) were significantly improved from 0.01 and 0.14 preoperatively, to 0.44 and 0.57 after the ICL implantation ($p < 0.0001$). Mean manifest refraction spherical equivalent (MRSE) reduced from -16.40 D (± 3.56 D) to -0.80 D (± 1.02 D) after the combined treatments ($p < 0.0001$), and the mean refractive astigmatism reduced from -4.73 D (± 1.32 D) to 0.96 D (± 0.35 D; $p < 0.0001$). Mean steep and mean flat keratometry values reduced from 60.57 D and 56.16 D to 54.48 D and 53.57 D ($p < 0.0001$), respectively. No intraoperative or postoperative complications occurred.

Conclusions: A combined three-stage approach to the treatment of keratoconus using Keraring ICRS implantation, followed by CXL and then ICL implantation appears to be effective in improving functional

vision, and reducing the progression of the disease. Further studies with longer term followup are required to further evaluate this treatment strategy, and compare with other approaches.

31. CLINICAL AND MICROBIOLOGICAL PROFILE OF POST-PENETRATING KERATOPLASTY INFECTIOUS KERATITIS IN FAILED AND CLEAR GRAFTS

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Purpose: To comparatively evaluate the clinical and microbiological profiles and treatment outcome of cases with post-penetrating keratoplasty (PK) infectious keratitis in failed and clear grafts.

Methods: All cases of infectious keratitis following penetrating keratoplasty admitted to the Royal Victorian Eye and Ear Hospital, Melbourne, between January 1998 and December 2008 were identified through a retrospective medical chart review. Cases without any surface sutures were selected and divided into two groups, microbial keratitis with pre-existing graft failure (FG) and microbial keratitis in clear graft on presentation (CG). Demographic, clinical and microbiological profiles of cases in both groups were analyzed and compared.

Results: Patients in FG group were older ($p = 0.004$) and had an early onset of graft infection ($p = 0.049$), compared to patients in the CG group. All patients in FG group were on long-term corticosteroid drops at the time of presentation (vs. 76% in CG group; $p = 0.005$). *Moraxella* sp. was more frequently isolated in FG group (30.4%) compared to CG group (8%). Surgical intervention in the form of corneal gluing and therapeutic corneal transplantation was performed more frequently in the CG group ($p = 0.03$).

Conclusion: Prolonged use of corticosteroid eye drops is a major risk factor for the occurrence of post-keratoplasty infectious keratitis. Infections in failed grafts occur earlier as compared to clear grafts and, indolent organisms like *Moraxella* are prevalent in this group of patients.

32. COMPARISON OF CORNEAL PARAMETERS AND CORRESPONDING BIOMECHANICAL MEASUREMENTS USING DIFFERENT ANTERIOR SEGMENT IMAGING SYSTEMS

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Purpose: To compare the Holladay reports comprising corneal elevation, curvature and thickness measurements obtained using two different imaging systems, the scheimpflug camera (Pentacam) and an optical coherence tomography device (Visante OMNI) in normal subjects and to study their correlation with corneal biomechanical measurement.

Methods: Corneal measurements of 50 eyes (50 normal subjects) were evaluated with the Pentacam, Visante OMNI and the Ocular response analyser (ORA) and the relationship between the values obtained with these devices was examined.

Results: No significant differences were noted between the corneal anterior elevations ($p = 0.41$), posterior elevations ($p = 0.73$), the anterior corneal minimum curvature (K1) ($p = 0.17$) and the anterior corneal maximum curvature (K2) ($p = 0.94$) between the two devices. Compared to the Pentacam, the Visante OMNI recorded thinner central corneal thickness (CCT) and the thinnest corneal pachymetry values by $14.08 \pm 17.77 \mu\text{m}$ ($p < 0.0001$) and $25.53 \pm 10.72 \mu\text{m}$ ($p < 0.0001$) respectively. However, these values were significantly correlated ($r = 0.88$, $p < 0.0001$ and $r = 0.95$, $p < 0.0001$ respectively). CCT measurements from both devices correlated positively with corneal hysteresis measurements from the ORA whereas no correlation was noted between anterior corneal maximum curvature (K2) and corneal hysteresis.

Conclusions: There is good agreement between the Holladay reports obtained from both the devices for corneal curvature and elevation data. The Visante OMNI measured significantly thinner corneal thickness measurements and the devices are not interchangeable. Corneal hysteresis appears to be influenced by CCT, but not by corneal curvature.

33. EVIDENCE FOR THE MANAGEMENT OF MEIBOMIAN GLAND DISEASE

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Purpose: To present the evidence supporting the available therapies for meibomian gland dysfunction,

a common cause of blepharitis producing troublesome ocular discomfort.

Methods: The 'International Workshop on Meibomian Gland Dysfunction (MGD): Report of the Clinical Trials Subcommittee' has been recently published. The results of this trial were appraised and a review of the literature conducted.

Results: Warm compresses with lid massage; lubricants; topical antibiotics; systemic tetracyclines and/or topical steroids were identified as empiric therapies for MGD. Clinical trials conducted to assess these management options are limited in number, have included small numbers of patients and are of short duration. Evidence is lacking to support the use of lubricants; topical antibiotics and steroids for MGD. However, lubricants may be of benefit when dry eye accompanies MGD. Newer therapies, including essential fatty acids, castor oil, topical cyclosporine, sex hormones and probing of meibomian gland orifices are awaiting further trials to establish their benefit.

Conclusions: Clinical trials have demonstrated that there is some evidence to recommend the use of warm compresses and systemic tetracyclines for MGD. Future studies are needed to investigate the optimal regime and dosing for the established and new therapies.

EPIDEMIOLOGY/PUBLIC HEALTH

34. ANTIOXIDANTS AND PREVALENT AGE RELATED MACULAR DEGENERATION: THE MELBOURNE COLLABORATIVE COHORT STUDY

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The use of antioxidant supplements has become the standard of practice in the USA for delaying the progression of age-related macular degeneration (AMD), but evidence for the efficacy of antioxidant intake in the primary prevention of AMD has been inconsistent.

Purpose: To evaluate associations between prevalent age-related macular degeneration and prior antioxidant intake.

Methods: The authors examined associations between AMD and antioxidant intake using 20,963 participants from the Melbourne Collaborative Cohort Study aged 40–69 years at baseline (1990–1994). Participants' nutrient intakes were estimated from foods reported in a food frequency questionnaire at baseline; digital macula photographs of both eyes were taken and evaluated for early and late AMD at follow up from 2003 to 2007.

Results: Antioxidant intake was not associated with the prevalence of AMD in this cohort (Odds Ratios (ORs) ranged from 0.91 to 1.16 $P > 0.05$), with the exception of vitamin E, where a moderate increased odds of 26% was demonstrated for a higher intake of vitamin E (OR 1.26 95%CI 1.08–1.47 $P = 0.03$). The use of multi-vitamin supplements was associated with higher odds of late AMD (OR 1.65; 95% 1.05–2.58, $P = 0.03$). Higher overall vegetable intake was associated with increased odds of late AMD (OR per quartile increase 1.13; 95% CI 1.03–1.22, P -value for trend = 0.007).

Conclusions: These results add weight to the message of caution in promoting supplement use in older people, where the benefits are not well demonstrated, and conversely, use may cause harm.

35. OpenClinica SOFTWARE: USE IN INVESTIGATOR INITIATED CLINICAL TRIALS

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Purpose: To describe the open access clinical trial management software "OpenClinica" and its implementation in investigator initiated clinical trials.

Methods: Laser Intervention for Early AMD (LEAD) study is an NHMRC-funded Randomised controlled trial (RCT) of a nanosecond laser intervention in AMD. We have constructed a database for this trial, based on 'OpenClinica' an open source, software package designed to capture, store and format data for analysis and monitoring, in a de-identified manner similar to a commercially sponsored clinical trial. OpenClinica facilitates building, modifying and perfecting individual Case Report Forms (CRF) for each type of data, according to specific trial requirements. It also allows the user to encompass questionnaires, and to import various data files and medical images to facilitate analysis and monitoring.

Results: For this trial, we have designed 26 trial specific CRFs, 18 of them use conditional displays, 4 of them used the specific rules regulating range and

quality of data on entry, 7 of them using the attachments of the images or measurement data files, and monitoring tools were implemented. Individual CRF pages will be demonstrated. No professional database writer was involved, as OpenClinica allows for multiple modifications of the design, according to trial needs and growing experience of users.

Conclusion: OpenClinica is an affordable and valuable resource for managing data for investigator initiated clinical trials to ensure quality, accuracy and security of data. It is now being used as the data management system for a number of trials within Centre for Eye Research Australia.

36. REVIEW OF CURRENT DIABETIC RETINOPATHY MANAGEMENT GUIDELINES IN THE CONTEXT OF LOW-RESOURCE SETTINGS

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Purpose: 70 per cent of diabetes is in low and middle income countries. This project is aimed at adapting evidence-based clinical practice guidelines for management of diabetic retinopathy (DR) in low resourced countries.

Methods: A review of the current practice guidelines for the management of diabetic retinopathy was conducted. The criteria for inclusion were based on questions set by the NHMRC Australia working group for guideline development. Guidelines included for review were required to meet the following component criteria. (i) Included the following key components related to DR: epidemiology, stages of DR, detection, and management; (ii) provided evidence-based recommendations; (iii) developed by an expert panel or an authority commissioned by a national authority. Guidelines were assessed using criteria adapted from the Conference of Guideline Standardisation (COGS).

Results: The database search revealed 123 references, of which 37 guidelines for different aspects of diabetic retinopathy management were available. Eleven guidelines were published in languages other than English. Of the eligible 26 guidelines, 13 (50%) included all key components; 22 (85%) discussed epidemiology; 20 (77%) discussed stages; 23 (88%) discussed detection of DR; 19 (73%) discussed management; 25 (96%) made evidence-based recommendations. Guidelines are consistent in the timing of first review; indication and timing for photocoagulation and vitrectomy.

Conclusion: Management of diabetic retinopathy in low-resourced countries requires consensus from evidence based recommendations at each level of care.

There is considerable variation between existing guidelines in recommended frequency of referral, methods for examination and personnel involved in screening and review. Evidence-based recommendations need to be adapted to resource constraints.

37. HOSPITALISATION RATES OF LEGALLY BLIND ADULTS

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Purpose: Compare the hospitalisation rates of blind adults with a matched group of sighted controls using linked hospital morbidity data.

Methods: A cohort of confirmed legally blind adults identified from the state blind register was age and gender matched to a cohort of normally sighted adults from the electoral roll. The rates of hospital admissions, length of stay and co-morbidities of the two cohorts were compared over an 11 year period (1999–2010).

Results: The combined blind and control cohorts comprised 4407 adults who had 50,457 separate hospital admissions totalling 216,610 hospital bed days. There were 2215 legally blind adults aged 18–104 years with the majority being female (59%). Ninety one per cent of the blind cohort and 88% of controls had at least one hospital admission during the study period. Hospital admission for renal dialysis accounted for 30% of all admissions by blind adults compared to 19% by the control cohort. Regression analysis estimated that blind adults had 30% (95% CI: 20%–40%) more hospital admissions than the control adults after adjusting for the non-linear effects of age. The predicted mean length of stay in hospital was 11 days/year for blind people compared to 7 days/year for controls. Socio-economic and remoteness scores were not associated with the outcome measures. Almost twice as many blind people ($n = 65$) had periods of psychiatric hospital care, compared to controls ($n = 34$).

Conclusions: The use of linked health data sets has enabled us to identify special health care needs of people who are blind.

38. PATIENT PREFERENCES – WHAT COUNTS, VISUAL ACUITY IN THE BETTER EYE, WORSE EYE OR BOTH?

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Purpose: To assess the impact of visual acuity (VA) in the better and worse eyes on patient reported preferences using generic health-related (EuroQoL EQ-5D) and vision-specific (Vision and Quality of Life Index (VisQoL)) multi-attribute utility instruments.

Methods: 1085 patients and 274 controls underwent a complete ophthalmic examination and rated preferences with the EQ-5D ($n = 868$) and the VisQoL ($n = 837$). Utilities were calculated for six levels of VA loss in the better and worse eyes, as well as for 21 different levels of vision impairment based on combinations of the better and worse eye VA categories.

Results: Using the VisQoL, utility scores decreased significantly with deteriorating vision in both the better and worse eyes when analysed separately. When stratified by the 21 levels of vision impairment, VisQoL utilities decreased as VA declined in the worse eye despite stable VA in the better eye. Differences in VisQoL scores were statistically significant for cases where the better eye had no vision impairment ($VA < 0.1$ LogMAR) and the worse seeing fellow eye had mild, moderate, severe or profound vision impairment. In contrast to the VisQoL, EQ-5D utilities were not associated with better eye VA, worse eye VA or any of the bilateral vision states.

Conclusion: Calculating utilities based only on better eye VA is likely to underestimate the impact of vision impairment, particularly when the better eye has no or little VA loss and the worse eye is moderately to severely visually impaired. These findings have considerable implications for economic evaluations within eye health.

39. SURVIVAL AFTER PHACOEMULSIFICATION SURGERY TO CORRECT VISUAL IMPAIRMENT: THE CATARACT SURGERY AND AGE-RELATED MACULAR DEGENERATION STUDY

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Purpose: To assess mortality rates associated with visual impairment (VI) correction via cataract surgery in an older surgical cohort.

Methods: We followed cataract surgical patients aged ≥ 64 years for five years after phacoemulsification surgery. Consecutive cataract surgical patients were recruited during 2004–07. Mortality data was obtained from the Australian National Death Index (2011). Preoperative and one-month postoperative VI status in the surgical eye was defined as none if visual acuity (VA) $\geq 6/12$, mild if VA $< 6/12$ – $6/18$, or moderate-severe (VA $< 6/18$). Patients with mild or no VI one-month postoperatively were compared to those with persistent moderate-severe VI at the same visit. Cox regression models were used to estimate mortality risk (HR).

Results: Of 905 patients with moderate-severe VI before surgery, 60.4% ($n = 547$), 15.5% ($n = 140$) and 24.1% ($n = 218$) had no, mild and moderate-severe VI, respectively, at one-month postoperative visit. Age-standardized mortality rates were non-significantly lower in patients with either mild (24.7%; 95% confidence interval (CI), 16.5–32.9%) or no VI (23.9%; 95% CI, 19.7–28.2%) compared to those with moderate-severe VI (30.4%; 95% CI, 23.2–37.7%). After adjusting for age, sex, body mass index, smoking, number of medications used and number of comorbid conditions (angina, myocardial infarction, stroke, TIA, hypertension, hypercholesterolaemia, asthma and kidney disease), patients with mild (0.63, 95% CI, 0.40–0.99) or no VI (HR 0.68, 95% CI, 0.49–0.94) had reduced mortality risks compared to those with persistent moderate-severe VI.

Conclusion: Correction of moderate-severe VI via cataract surgery was associated with a reduced mortality risk among older patients with cataract-related VI.

40. DIETARY PATTERNS DEFINED WITH PRINCIPAL COMPONENT ANALYSIS IN RELATION TO AGE-RELATED MACULAR DEGENERATION: MELBOURNE COLLABORATIVE COHORT STUDY

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Purpose: To identify dietary patterns by use of factor analysis and to evaluate associations of dietary patterns with age-related macular degeneration (AMD) in the Melbourne Collaborative Cohort Study (MCCS).

Methods: The MCCS recruited 41,514 people either born in Australia, or migrated from UK, New Zealand, Greece, Italy or Malta, aged 40 to 70 years at baseline (1990 to 1994). Of them, 21,132 participants were

assessed in 2003–2007 for prevalent AMD with non-mydriatic 45–7 digital colour fundus photography. Dietary data were collected at baseline with a self-administered 121-item food frequency questionnaire. Factor analysis (principal component analysis, PCA) was performed to identify dietary patterns. Logistic regression models, adjusted for covariates were used to assess associations between these patterns and both early and late AMD.

Results: In the study population, a total of 2508 (12.8%) had early AMD and 108 (0.6%) had late AMD. Six factors representing dietary patterns were identified, with predominant intakes of fruits (F1), vegetables (F2), mixed foods (boiled rice, mushrooms, zucchini, yoghurt, avocado, chicken and fish) (F3), meat products (F4), dairy food (F5) and salad (F6). These 6 factors taken together explained 33% of the total variance. After adjustment for confounders, F3 was significantly associated with lower prevalence of both early AMD (OR 0.92, 95% CI, 0.85–0.99) and late AMD (OR 0.63, 95% CI 0.41–0.97).

Conclusion: Factor analysis revealed six food patterns in the ethnically diverse cohort of older Australians. The factor with predominantly mixed dietary intakes was associated with lower AMD prevalence.

41. OCULAR INJURY IN ORBITAL FRACTURES – A RETROSPECTIVE CASE SERIES

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Purpose: To document the range of ocular injuries associated with orbital fractures, as well as their management at a tertiary referral hospital.

Method: The records of 183 patients admitted to Royal North Shore Hospital with orbital fractures between January 2009 and June 2011 were reviewed. Variables examined included demographics, mechanism, fracture details, presence of concurrent facial fracture or other serious injury, emergency department and ophthalmologic examination findings.

Results: Seventy percent of patients were male with a median age of 47 years. Inferior orbital wall fracture was the most frequent (68%). Eye injuries were documented in 50 patients (27%), the commonest being diplopia due to extraocular muscle or soft tissue damage, in 23 (13%). Eight had potentially serious injuries: traumatic optic neuropathy, 4 (2%), lens subluxation, 2 (1%), vitreous haemorrhage, 3 (2%) and hyphaema, 2 (1%). Other injuries were extraocular muscle entrapment, 4 (2%), ocular cranial nerve

paresis, 4 (2%), traumatic mydriasis, 8 (4%), comotio, 7 (4%), traumatic iritis, 3 (2%), conjunctival laceration, 2 (1%) and corneal abrasion, 2 (1%). Most patients (82%) sustained at least one additional corporal injury. Visual acuity was recorded by an emergency department doctor in 33 patients (18%) and slit lamp examination in 9 (5%). Of 5 patients with serious ocular pathology, 3 had reduced VA but only one was identified in ED (a further 3 were not examinable due to reduced GCS).

Conclusion: Patients with orbital fractures were infrequently assessed for eye injuries by emergency doctors, may assist with prognostication.

42. EXUDATE FINDER

WITHDRAWN

43. OPHTHALMOLOGY IN PAPUA NEW GUINEA. A SHORT TERM SURGICAL TRIP PROVIDES INSIGHTS INTO THE CURRENT LEVEL OF EYE CARE AND HOW WE CAN BEST HELP

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Purpose: Papua New Guinea has a low standard of public health and in many districts there is no Ophthalmology service. This paper looks at current conditions and how RANZCO can help to improve eye care in PNG.

Method: For two weeks in June/July of this year I worked in the Gulf Province of PNG Operating on cataracts and teaching Small Incision Cataract Surgery (SICS) to the senior registrar from Port Moresby.

Results: The Gulf province has a population of 106,000 and has a low standard of primary health care and no Ophthalmology services. Most of the population live by subsistence agriculture and are unable to afford the trip to Port Moresby for eye and health care. The outreach was operated by the YWAM group who arranged the surgical team, venue in a local hospital, and the necessary equipment. Accommodation was provided on their ship.

Two weeks of surgery proved to be enough time to teach most of the SICS procedure and was a valuable part of the registrars training.

Conclusion: The trip gave valuable insights as to the lack of reasonable eye care in PNG and the usefulness of short term help and teaching. The paper discusses needs in PNG and how RANZCO and its members can help our neighbouring country.

44. DOES CATARACT SURGERY REDUCE FALLS IN OLDER PEOPLE? A WHOLE POPULATION LINKED DATA STUDY

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Purpose: Our aim was to undertake a whole-population investigation of the association between first-eye cataract surgery and the incidence of falls requiring hospitalisation.

Methods: Data linkage was used to identify all first-eye cataract surgeries in those aged 60+ years in 2004 to 2008. In this group, hospital admissions for a fall in the year before and after surgery were identified. Poisson regression was used to compare the incidence of falls before and after cataract surgery after adjustment for age, gender, marital status, Indigenous status, socioeconomic status, and comorbid medical conditions.

Results: There were 15,295 first-eye cataract surgeries between 2004 and 2008 in Western Australians aged 60+ years. There were 600 people (3.92%) hospitalised for 625 falls in the year before (n = 273) or after (n = 352) cataract surgery. A third of those who fell sustained neck of femur fractures. The group that fell after cataract surgery were older, more likely to be married and have more comorbidities. The incidence of a fall after first-eye cataract surgery was 27% higher (95% confidence interval 1.04–1.56) after adjustment in the multivariate Poisson model for age, gender, presence of comorbid conditions, marital status, residential location, Indigenous status, and socioeconomic status. None of these other factors significantly affected the incidence of a fall after first-eye cataract surgery.

Conclusion: First-eye cataract surgery was associated with an increase in falls requiring hospitalisation in the year after, compared to the year before surgery. Further work is needed to identify the underlying causes and reasons.

45. CHANGE IN DRIVING DIFFICULTY AFTER FIRST-EYE CATARACT SURGERY

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Purpose: To investigate self-reported driving difficulty before and after first eye cataract surgery and determine the visual measures affecting change in driving difficulty after surgery.

Methods: A cohort of 99 older drivers awaiting first-eye cataract surgery was recruited. Data were collected the week before and 12 weeks after first eye cataract surgery. Visual measures including visual acuity, contrast sensitivity, stereopsis and useful field of view were assessed. Self-reported driving difficulty was measured with the Driving Habits Questionnaire. Cognitive status was assessed using the Mini Mental State Examination. Regression analysis was undertaken to determine the association between changes in visual measures and self-reported driving difficulty after first eye cataract surgery.

Results: Self-reported driving difficulty improved after first eye cataract surgery. However, 16% of participants did not improve and driving difficulty worsened in 11% following surgery. Improvement in driving difficulty score after first eye cataract surgery was associated with improved contrast sensitivity in the operated eye ($p < 0.001$). New glasses after surgery ($p < 0.001$) and the number of chronic health conditions ($p = 0.016$) also affected change in driving difficulty.

Conclusion: Contrast sensitivity rather than visual acuity was a significant factor affecting change in self-reported driving difficulty after first eye cataract surgery for bilateral patients. This has implications for licensing standards that rely heavily on visual acuity for visual fitness to drive.

46. EPIDEMIOLOGY OF EYE TRAUMA PRESENTING TO A REGIONAL HOSPITAL EMERGENCY DEPARTMENT AND OPHTHALMOLOGY SERVICE

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Purpose: To describe the epidemiology of eye trauma presenting to a regional referral health service.

Method: A two stage study was conducted. A 1-year retrospective case series (RCS) was developed using relevant SNOMED CT and ICD-10 codes applied to medical records. Data was extracted on date of presentation, patient demographics, injury details, diagnosis, and management. A prospective case series (PCS) recruited patients presenting with eye trauma to the emergency department of a regional hospital or its ophthalmology service over 80 days with additional data collected by telephone interview.

Results: 411 and 117 eye injuries were identified for the RCS and PCS respectively. Mean age was 35.54 ± 18.55 (RCS) and 34.12 ± 17.09 (PCS). Both groups demonstrated male predominance, 77.86% ($X^2 = 127.59$, $p < 0.001$) (RCS) and 89.74% ($X^2 = 73.92$, $p < 0.001$) (PCS). Grinding-related injuries were the most frequently encountered mechanism, 20.68% (RCS) and 27.35% (PCS). Corneal foreign bodies were the most common diagnosis, 36.01% (RCS) and 36.75% (PCS). A large proportion of injuries were work-related, 40.23% (RCS) and 45.83% (PCS), with trades workers being the most affected occupational group, 23.93% (PCS). Protective eyewear use in work-related injuries was low, 36.36% (PCS). A small number of injuries required surgical intervention, 2.19% (RCS) and 2.56% (PCS); 3 cases were referred for tertiary ophthalmological services.

Conclusion: Eye trauma remains a significant public health concern. Protective eyewear compliance is inadequate in work-related eye injuries. There are clear patient demographic and occupational factors which may be targeted to reduce the burden of disease.

47. TRACHOMA IN THE SOLOMON ISLANDS: EVIDENCE FROM TRACHOMA RAPID ASSESSMENTS

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Purpose: To determine the rates of trachoma in the Solomon Islands.

Methods: Trachoma Rapid Assessment (TRA) Workshops for community health nurses were conducted to enable them to perform Trachoma Rapid Assessments in their allocated health zones. High-risk communities were chosen from each health zones and all children 1–9 years were examined for active trachoma and adults more than 40 years were examined for trichiasis.

Results: A total of 1,336 adults >40 years and 5,408 children 1–9 years were screened at 73 sites. Trachoma Rapid Assessments were conducted in nine of the ten provinces in the Solomon Islands with rates of active

trachoma ranged from 15.9% to 64.7% with an average of 38.9%.

38.3% of children screened were found to have dirty faces.

The rate of trichiasis in the country was 0.3%.

3,304 households were assessed for environmental factors. These included the availability of water source more than half an hour walk away; households at risk of solid waste and animal waste and the absence of functional latrines being on average 26%, 67.4% and 73.3% respectively.

Conclusion: Trachoma is a public health problem in the Solomon Islands and indicates priority areas for implementation of trachoma control measures.

48. YOUNGER SIBLINGS, C-REACTIVE PROTEIN AND RISK OF AGE-RELATED MACULAR DEGENERATION

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Purpose: To determine the relationship between sibling exposure and 1) the risk of Age-related Macular Degeneration (AMD) and 2) high-sensitivity C-reactive protein (hsCRP) level.

Methods: Retrospective analysis of the pooled cross-sectional data from two studies: the Cardiovascular Health and Age-Related Maculopathy study (2001–2002) and the Age-Related Maculopathy Statin Study (2004–2006). Associations between sibling number and AMD were assessed using multinomial logistic regression. Association between sibling number and CRP was examined using generalized linear model for negative binomial distribution.

Results: In the multinomial regression analyses, higher number of younger siblings was associated with significantly lower odds of early AMD in those with a family history of AMD (OR = 0.2; 95%CI: 0.1, 0.8; P = 0.022), but was unrelated to AMD for those who had no family history (OR = 1.0; 95% CI: 0.9, 1.2; P = 0.874). Increased number of younger siblings correlated with lower CRP levels (B = -0.19, 95%CI -0.38, -0.01; P = 0.036).

Conclusion: Exposure to younger siblings reduced the risk of early AMD in patients with a family history of the disease. CRP levels were inversely related to the number of younger siblings. This supports the theory that immune modulation contributes to AMD pathogenesis and suggests younger sibling exposure might be protective where there is a family history of AMD.

49. MAGNITUDE AND RISK FACTORS OF POOR DIABETES CONTROL IN PATIENTS WITH AND WITHOUT DIABETIC RETINOPATHY: THE DIABETES MANAGEMENT PROJECT

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Purpose: To identify the independent risk factors of poor diabetes control in patients with diabetic retinopathy (DR) and diabetic macular oedema (DME).

Methods: Patients with diabetes (n = 609) were recruited from the Royal Victorian Eye and Ear Hospital as part of the Diabetes Management Project (DMP). Diabetes control parameters were defined as: (1) poor diabetes control (glycosylated haemoglobin [HbA_{1c}] ≥7%, systolic blood pressure [SBP]/ diastolic blood pressure [DBP] ≥130/80 mmHg), poorly controlled dyslipidaemia (total cholesterol [TC] to high density lipoprotein [HDL] ≥4.0); and (2) poor diabetes control but without poorly controlled dyslipidaemia.

Results: There were 364 patients with DR. Forty-four (13.5%) patients had poor diabetes control and 114 (33.5%) patients recorded poor diabetes control without poorly controlled dyslipidaemia. Independent risk factors for poor diabetes control with poorly controlled dyslipidaemia were younger age (OR = 0.95; 95% CI 0.92–0.98), other diabetes complications (OR = 2.57; 95% CI 1.32–4.99) and poorly controlled dyslipidaemia (OR = 1.21; 95% CI 1.01–1.44). There were 176 patients with DME. Twenty-four (15.2%) patients had poor diabetes control with poorly controlled dyslipidaemia and 58 (34.9%) patients had poor diabetes control without poorly controlled dyslipidaemia. Independent risk factors were obesity (OR = 1.08; 95% CI 1.01–1.16) and poorly controlled dyslipidaemia (OR = 1.4; 95% CI 1.0–1.95) for poor diabetes control; younger age (OR = 0.94; 95% CI 0.89–0.99), obesity (OR = 1.1; 95% CI 1.01–1.2) and other health issues (OR = 0.28; 95% CI 0.09–0.88) for poor diabetes control without poorly controlled dyslipidaemia.

Conclusion: Targeting dyslipidaemia, younger patients and obesity may help to improve optimal diabetes management parameters in patients with DR and DME.

50. EFFECT OF PESTICIDES ON COLOUR VISION AND ANTERIOR SEGMENT OCULAR STRUCTURES OF FARMERS

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Purpose: The objective of this study was to determine the prevalence of anterior ocular structure abnormalities and colour vision defects among agricultural workers using pesticides.

Methods: Agricultural workers in Cameron Highlands, Malaysia using pesticides were randomly recruited. The inclusion criteria were having no systemic or ocular disease, not on medication, non-smokers and non-alcoholic. Slit lamp biomicroscopy was carried out to assess anterior ocular structures. Colour vision was assessed binocularly using Ishihara plates, D-15 test and FM100Hue test and performed using daylight illumination.

Results: Seventy-four workers of mean age 32.24 ± 1.23 years were recruited. The only anterior ocular changes observed were bulbar conjunctival hyperemia with a prevalence of 25.7%. The mean working hours was 8.83 ± 3.58 and the mean working years were 7.69 ± 8.35 . Only 23% of farmers wore goggles to protect their eyes. There was no statistically significant difference found for means of working years ($t = 0.019$, $p = 0.985$), working hours ($t = 1.014$, $p = 0.314$) or use of goggles ($\chi^2(1, N = 74) = 2.238$, $p = 0.135$) among workers with and without hyperemia. The prevalence of colour vision defects was 5% using the Ishihara Plates, 29% using the D15 test and 49% using the FM100Hue test. Majority of the colour defects were of non-polar type. The mean total error score of the colour vision defectives was 170.67. There was a significant correlation between the duration of working hours ($r = 0.46$, $p = 0.000$) with the total error score. **Conclusion:** Exposure to pesticides can cause hyperemia and colour vision defects in farmers and the colour vision defects correlates highly with exposure duration.

51. CHALLENGE IN SCREENING FOR RETINOPATHY OF PREMATURITY IN THE INDIGENOUS POPULATION AT THE TOP END

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Purpose: To determine the incidence and compliance with follow up of the indigenous population in infants

screened for and diagnosed with retinopathy of prematurity (ROP) at the Royal Darwin Hospital.

Method: A retrospective statistical analysis and descriptive study that included all 200 infants (400 eyes) who underwent ROP screening at Royal Darwin Hospital in the 4-year period from January 2008 to 2012.

Results: Infants were seen at least once, for a total of 713 examinations and an average of 3.5 per infant, in the 4-year. Of the 200 infants screened for ROP, 57% were indigenous. 47(23.5%) infants were diagnosed with some stage of ROP. 27 babies with ROP (57%) were indigenous. The percentages of ROP among indigenous and non indigenous populations were clinically non significant.

The average birth weight of ROP infants was 983.6 g and the average gestation was 27 weeks, compared to 1340 g and 29 + 6 weeks for the non-ROP infants. The indigenous population was over represented in the negative outcomes. Of the total infants lost to follow up, 72% were indigenous.

Conclusion: While indigenous infants may be over represented in infants screened for ROP, they were no more likely to develop ROP than their non-indigenous counterparts. They were over-represented in infants lost to follow up, most likely due to social and cultural reasons.

52. OUTCOMES OF NONATTENDANTS IN THE DIABETIC OPHTHALMOLOGY CLINICS

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Purpose: No known study to date has assessed the impact of nonattendance to diabetic ophthalmology clinics on additional ophthalmology related services. The aim of this study was to investigate the additional impacts and costs associated with non attendance.

Method: Retrospective review of ophthalmology clinic appointments over a 36 month period in a hospital. 731 eligible patients were followed to assess their attendance to appointments and use of additional services during the study period.

Results: Demographics: 53.2% males, 46.8% females, 18.1% were new referrals, mean age 63.1 ± 14.4 years (range 16–96). Mean follow-up period was 31.8 ± 3.2 months, minimum follow up 24 months.

For diabetic ophthalmology appointments, 10.8% did not attend their first appointment, 39.4% had missed at least one, 22.3% repeatedly did not attend.

Appointment use for nonattendants versus (vs) patients who always attended was: 0.78 ± 2.3 vs

0.31 ± 0.93 ($p < 0.001$) for emergency, 7.5 ± 3.5 vs 6.1 ± 2.8 ($p < 0.001$) for routine diabetic ophthalmology and 1.1 ± 2.5 vs 0.54 ± 1.4 ($p < 0.001$) for laser. No significant difference was found for optical coherence tomography, fundus fluorescein angiography, intravitreal bevacizumab and other ophthalmology subspecialty clinics.

Positive correlations were found between the number of non attended appointments and the number of: emergency ($r = 0.144$, $p < 0.001$), laser ($r = 0.226$, $p < 0.001$) and additional diabetic ophthalmology appointments ($r = 0.273$, $p < 0.001$).

Conclusion: Improving patient attendance to diabetic ophthalmology clinics is crucial to eliminating costs from resultant additional clinics. An estimated annual total of \$60,650 was spent on nonattendance at this study centre with half contributed by additional clinics generated from non attendants.

GENETICS

53. THE AUSTRALIAN INHERITED RETINAL DISEASE REGISTER AND DNA BANK PART II – GENETIC ANALYSIS FINDINGS

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Purpose: To identify disease-causing mutations in families participating in the Australian Inherited Retinal Disease Register (AIRDR) and DNA Bank.

Methods: *Microarray:* DNA was presented to the following genotyping microarrays (Asper Ophthalmics) from participants with the following diagnoses:

ABCA4: Stargardt disease ($n = 77$), cone-rod dystrophy ($n = 22$)

ADRP: RP dominant ($n = 50$), RP of unknown inheritance ($n = 21$)

ARRP: arRP ($n = 21$), RP of unknown inheritance ($n = 50$)

Usher: Usher syndrome Type I ($n = 6$), Type II ($n = 22$), Type III ($n = 9$)

LCA: Leber congenital amaurosis ($n = 11$)

Sequencing: For selected participants, the following genes were bi-directionally sequenced: *ABCA4*, *BEST1*, *NR2E3*, *PDE6B*, *PROM1*, *PRPH2*, *RHO*, *RPGR*, *RS1* and *USH2A*.

Results: Using the above methods, disease-causing mutations were found in participants with the following presenting diagnoses:

adRP: *PRPH2* ($n = 5$), *PRPF3* ($n = 2$), *RHO* ($n = 6$), *RP1* ($n = 4$), *RP9* ($n = 1$)

arRP: *CRB1* ($n = 5$), *NR2E3* ($n = 2$), *PDE6B* ($n = 4$), *USH2A* ($n = 9$)

xlRP: *RPGR* ($n = 1$)

RP of unknown inheritance: *ABCA4* ($n = 3$), *PDE6A* ($n = 1$), *PDE6B* ($n = 3$), *NR2E3* ($n = 1$), *PROM1* ($n = 5$), *RP1* ($n = 2$), *USH2A* ($n = 19$)

Stargardt disease: *ABCA4* ($n = 51$)

cone-rod dystrophy: *ABCA4* ($n = 13$), *PRPH2* ($n = 1$)

Usher syndrome: *MYO7A* ($n = 5$), *USH2A* (Type I $n = 1$, Type II $n = 19$, Type III $n = 3$)

retinosis: *RS1* ($n = 10$)

Leber congenital amaurosis: *CEP290* ($n = 3$), *CRB1* ($n = 2$), *GUY2CD* ($n = 1$)

Best disease: *BEST1* ($n = 5$)

In the case of microarray analysis, disease-causing mutations were identified with the following frequencies: *ABCA4* (56%), *ARRP* (43%), Usher (63%), *LCA* (41%) and *ADRP* (15%).

Conclusion: DNA from 220 families was analysed. Disease-causing mutations were found in 187 individuals. These mutations were observed in 18 different genes.

54. PREVALENCE OF MACULAR DYSTROPHY ASSOCIATED WITH MATERNALLY INHERITED DIABETES AND DEAFNESS DETECTED ON ROUTINE RETINAL PHOTOSCREENING

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Introduction: Maternally inherited diabetes and deafness (MIDD) is a rare monogenic cause of diabetes. Macular dystrophy has been described in a majority of cases. This study investigates the utility of searching for macular dystrophy on retinal photoscreening images in order to guide MIDD genetic testing.

Methods: All diabetic photoscreening images acquired between 1st January 2010 and 30th June 2010 in a single New Zealand centre were reviewed retrospectively. Patients with macular dystrophy on these images were invited to attend a clinic appointment. Visual acuity, slit lamp fundoscopy, SD-OCT and FAF were performed. Buccal mouth wash samples were taken and the m.3243A > G mutation responsible for MIDD was screened for using automated bidirectional dideoxy sequencing.

Results: Diabetic photoscreening images from a total of 2615 patients were reviewed. Twenty patients were identified to have non-specific macular changes

potentially consistent with early MIDD maculopathy. Of these, 13 patients were available for examination. This confirmed a disturbance in the macular region in all 13 patients but none were phenotypically consistent with MIDD maculopathy. Molecular genetic testing excluded the presence of the m.3243A > G mutation in all 13 patients.

Conclusion: We failed to identify any new cases of MIDD macular dystrophy on routine diabetic retinal photoscreening suggesting that in our study population, either MIDD is very rare or the association of macular dystrophy with MIDD is low. Further studies are needed to confirm the prevalence of MIDD in the New Zealand population.

GLAUCOMA

55. LOW LUMINANCE ACHROMATIC (LLA) AND BLUE-ON-YELLOW (BONY) MULTIFOCAL VISUAL EVOKED POTENTIALS (mfVEP) IN EARLY GLAUCOMA

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Purpose: (a) To examine and compare the sensitivities of LLA and Bony mfVEP stimulation in identifying visual loss in early, including pre-perimetric, glaucoma. (b) To examine the role of each, in combination with other tests, in identifying such glaucoma.

Methods: 54 patients with at least one glaucomatous optic disc and normal or early defect(s) (MD < 6 dB) on achromatic Humphrey visual fields (HVF) were examined. VA < 6/9, cataract and/or other ocular abnormality were excluded.

LLA and Bony mfVEP were performed on all participants. Other tests included SITA-SWAP, FDT-Matrix, Spectralis OCT (RNFL) and dilated optic disc stereophotography. Optic disc photographs were assessed by 2 independent examiners in a masked fashion.

Results: 87 eyes of 54 patients were analyzed. 48(55.2%) eyes demonstrated significant abnormality LLA mfVEP, while 36 (41.4%) eyes demonstrated abnormalities on Bony mfVEP. The proportions were significantly different ($p < 0.0001$). 31 eyes showed abnormalities on both tests.

Among 48 eyes with LLA defects, 37 (77.1%) had abnormalities on at least one of the other perimetric tests. Of 36 eyes with Bony defects, 26 (72.2%) had such defects.

The presence of LLA and FDT-Matrix defects were significantly correlated ($p = 0.002$).

29 subjects were normal on HVF, Matrix and SITA-SWAP. Of these 14 eyes (48.3%) showed mfVEP defects as the only evidence of functional loss. 10 eyes were positive on both LLA and Bony.

Conclusions: LLA mfVEP demonstrated better sensitivity than Bony mfVEP in identifying early glaucomatous losses. Together, they identified defects in about 50% subjects who are normal on SWAP, FDT and achromatic VF.

56. SAFETY AND EFFICACY OF MITOMYCIN C IN PATIENTS UNDERGOING TRABECULECTOMY FOR VERY HIGH PRESSURE (>35 MM HG) GLAUCOMA

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Aim: To study safety and efficacy of mitomycin c in patients undergoing trabeculectomy for very high pressure (>35 mm hg) glaucoma.

Methods: This interventional prospective cohort study included patients undergoing primary trabeculectomy with IOP > 35 mm hg between June 2008 and October 2010.

Results: 31 eyes of 28 patients were evaluated of which 21 was male. The age were between 42 to 76 years, average being 65 years. Pre-operative IOP was between 38 to 51, MEAN IOP 41. 16 eyes of POAG group and 15 eyes of PACG. Arcuate scotoma in 15 eyes and double bjerrums in 16 eyes. pre-operatively none of the patients had macular split. Final IOP ≤ 14 mm of Hg in 90% patients and unrecordably low in 2 and more than 14 in one patient. In our study we achieved a reduction of 60 % of Initial pressures. Diffuse bleb was seen in 23 eyes, cystic in 4, shallow in 2 and overfiltering bleb was noted in 2 eyes, which was treated by bleb revision. IOP elevation more than 25 seen in one case which was managed medically. No patient developed macular wipe out syndrome.

Conclusion: Trabeculectomy with mmc is an extremely effective and safe in this group of patients who have more than 35 mm hg. mmc related complications were not seen, but these patients are being followed upon regularly.

57. LINKING SYSTEMIC OXIDATIVE STRESS WITH RETINAL GANGLION CELL DAMAGE USING METABOLOMICS: A NOVEL APPROACH

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Purpose: Metabolomics refers to the study of small molecule metabolite profiles and provides a “finger-print” of cellular processes within an organism. Oxidative stress has been implicated in the pathogenesis of glaucoma with increased reactive oxygen species (ROS) generation contributing to retinal ganglion cell (RGC) degeneration, glial dysfunction, and activation of immune response in glaucoma.

Methods: We employed a metabolomics approach to assess metabolite changes in serum between POAG patients following a strict inclusion criteria and healthy volunteers. Low molecular weight metabolites were analyzed by liquid chromatography-time of flight-mass spectrometry analysis (LC/MS Q-TOF). Further identification of metabolites was conducted using METLIN Personal Metabolite Database and analyzed with Mass Profile Professional (MPP) software to compare metabolite profiles amongst these two groups. Statistical analysis included principal component analysis (PCA), clustering tests and partial least squares analysis.

Results: Thirty eight different metabolites were detected by global (non-targeted) approach. Sixteen of them were significantly different between glaucoma and controls at an adjusted p-value of 0.001. Glycine-conjugated bile acids, palmitoylcarnitine, Inosine, and β -ureidoisobutyric acids showed the greatest increase in POAG patients. These metabolites indicate an increase in systemic metabolites of oxidative stress pathways in glaucoma.

Conclusions: Our findings provide further evidence of increased oxidative stress in POAG. This is the first report using metabolomics to profile POAG patient's serum. These results may give more understanding about the pathogenesis, diagnosis and the management of POAG.

58. GENETIC POLYMORPHISM OF ANTIOXIDANT GENES: IMPLICATIONS ON OPTIC NERVE HEAD ABNORMALITY IN PRIMARY OPEN ANGLE GLAUCOMA PATIENTS

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Purpose: Oxidative stress has been implicated in the pathogenesis of retinal ganglion cell loss in primary open angle glaucoma (POAG). The purpose of this study was to develop and conduct a targeted approach to examine genetic variants in candidate antioxidant genes with POAG patients and healthy controls using polymerase chain reaction (PCR).

Methods: We developed and validated allele specific-PCR to examine polymorphisms of antioxidant genes (catalase, glutathione peroxidase, and superoxide dismutase). The variants were detected based on the band size using gel electrophoresis and validated by direct sequencing.

Results: The technique was efficient to detect genetic variations of antioxidant genes. In our preliminary findings of twenty three POAG patients with one hundred and sixty five healthy volunteers screened, no polymorphisms of catalase (C-262T) and glutathione peroxidase (P200L) were detected. Manganese superoxide dismutase polymorphism (Val16Ala), however showed a higher frequency in POAG patients. There was increased probability for POAG to occur when the patient is carrying the Ala allele (RR = 1.214, OR = 1.4103), (CI 95% 0.7611 to 1.9365 and 0.5875 to 3.3855 respectively). The method is now ready for further recruitment of patients for a larger study.

Conclusion: Oxidative stress has been implicated in POAG. PCR is a rapid and effective method to detect genetic variants in glaucoma patients. We have developed an efficient method to detect genetic variations of antioxidant genes. Preliminary findings, though underpowered, has revealed that there is an increased Mn-SOD polymorphism in POAG suggesting a state of altered antioxidant system which can end by oxidative stress.

59. OUTCOMES OF TRABECULECTOMY VERSUS PHACO-TRABECULECTOMY IN SURGICAL MANAGEMENT OF GLAUCOMA

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Purpose: To compare the results of trabeculectomy with two-site phaco-trabeculectomy in patients requiring surgical management of glaucoma.

Method: Retrospective analysis of patients in a prospective database who underwent trabeculectomy or phaco-trabeculectomy for POAG, PXF-related or angle closure glaucoma between June 2008 and June 2011. Apart from the concomitant phacoemulsification cataract extraction via a separate corneal incision, all patients underwent an identical fornix-based trabeculectomy. Patients were excluded if they did not have a minimum of 1-year follow-up and, if glaucoma was secondary to uveitis or anterior segment neovascularisation.

Results: 41-trabeculectomy patients and 24-phaco-trabeculectomy patients had a mean pre-operative IOP of 29.5 mmHg and 23.2 mmHg, respectively. This was reduced to 13.3 mmHg and 13.0 mmHg at 6-months, and 12.8 mmHg and 13.0 mmHg at 12-months. Absolute success rates at 6-months were 76.3% and 68.1% respectively for IOP < 18 mmHg, 68.4% and 63.6% for IOP < 16 mmHg, and 36.8% and 40.9% achieving IOP < a predetermined target IOP. At 12-months, the trabeculectomy group had 70.5% with IOP < 18 mmHg; 67.6% with IOP < 16 mmHg; and 44.1% with IOP < target IOP, and phaco-trabeculectomy had 55%, 50% and 30% respectively. Qualified success rates for trabeculectomy at 6-months were 86.8% with IOP < 18 mmHg, 76.3% with IOP < 16 mmHg; and 42.1% with IOP < target IOP, and for phaco-trabeculectomy at 6-months were 90.9%, 81.8%, and 54.5% respectively. At 12-months qualified success for trabeculectomy was 94.1%, 88.2% and 55.8% respectively, and 85%, 80% and 45% for phaco-trabeculectomy.

Conclusion: Phaco-trabeculectomy is a useful technique in managing patients with concomitant cataract and glaucoma requiring surgical intervention, as similar IOP outcomes can be achieved when compared to stand-alone trabeculectomy.

60. GLAUCOMA EYE DROPS: A LIFELONG CHEMICAL INJURY?

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Purpose: Preservatives in ophthalmic preparations are known to cause ocular surface damage. Excipients can also contribute to oxidative stress in the dysfunctional ocular surface. Latanoprost has recently come off patent in the UK and generic products are becoming more widely available. Companies often accentuate the positive properties of their formulations (eg physiological pH), but such claims are often made without good comparative information about other products.

We evaluated commonly used topical glaucoma medications to ascertain pH levels and the intrinsic presence of free radicals.

Methods: 27 topical glaucoma preparations were analysed for Total Antioxidant Status (Randox). Analytical grade indicator paper was used to ascertain pH levels.

Results: Free radical concentrations in these samples ranged from 0 to 4.54 mmol/l, with median 0.66 mmol/l (mean 0.662 mmol/l, S.D. 0.839). Levels of pH ranged from 4.0 to 7.4, with median 6.5 (mean 6.252, S.D. 0.826). There was no evidence of a direct correlation between these 2 variables ($r = 0.232$, $p = 0.275$).

Conclusion: This study documents the range of pH and concentration of free radicals intrinsically present in glaucoma medications. Preservatives, free radicals and pH levels can all contribute to ocular surface damage following long term exposure from eye drops. These factors should be considered when prescribing glaucoma therapy with a compromised ocular surface. Differing excipients in generic products could be a future source of patient intolerance and subsequent poor compliance with treatment.

61. ONG EYE SPECULUM – FROM DESIGN TO MANUFACTURE

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Purpose: In glaucoma surgery such as trabeculectomy, access to the superior sclera and limbus is needed

A superior rectus bridle suture or preferably, a traction suture through the superior peripheral cornea is used to rotate the eye to look down, thus exposing the superior sclera. Although the superior corneal traction suture causes minimal trauma, it may contribute to mild inflammation and can leave a small scar in the superior cornea.

Method & Results: The Ong Eye Speculum has a larger inferior inner blade of about 15 mm compared to the standard speculum inner blades of about 5 mm. The larger inferior blade pushes on the inferior conjunctival fornix and hence rotates the eyeball down. During the speculum development, various designs were tried and modified. The Ong Speculum will now be manufactured with the adjustable mechanism of the Lieberman speculum and this improves its functionality as it enables the amount of infraduction to be varied.

Conclusion: This new eye speculum design facilitates exposure of the superior sclera and limbus by rotating the eyeball to look down without the need for a traction suture through the superior peripheral cornea. The Ong Eye Speculum will be manufactured by Amann Ophthalmic Instruments (Germany) and distributed by Designs for Vision (Australia). The author-designer has no financial interest in the product.

62. A PATIENT WITH NORMAL TENSION GLAUCOMA AND MOYAMOYA SYNDROME

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Purpose: To report a 64 year-old man of Chinese descent with normal tension glaucoma who was found to have Moyamoya Syndrome. This is the first case report of a patient with Moyamoya Syndrome and normal tension glaucoma in the literature. A case of congenital glaucoma and Moyamoya Disease in a 16 month old female, and a case of acute angle closure glaucoma following cerebral angiography in Moyamoya Disease, have been reported in the literature.

Method: Neuroradiological imaging is routinely performed on patients who have advanced visual field defects with intraocular pressures in the normal range when making a diagnosis of normal tension glaucoma.

Results: This patient had cerebral MRI and MRA, the latter which demonstrated Moyamoya Phenomenon. "Moyamoya" literally means "puff of smoke" and is the term used to describe the sudden brightening on angiography due to abnormal net-like vessels at the base of the brain in combination with usually bilateral

stenosis or occlusion of the basal cerebral arteries. Moyamoya Syndrome describes the condition when Moyamoya Phenomenon is demonstrated on neuro-radiological imaging, while Moyamoya Disease is a progressive, occlusive disease originally described in the Japanese.

Conclusion: This case illustrates the well-known association of normal tension glaucoma and cerebrovascular disease. In this case, Moyamoya Syndrome could be regarded as a compensatory mechanism to his cerebrovascular disease. Patients with normal tension glaucoma should be investigated for cerebrovascular disease with neuroradiological imaging, as it may influence their management. In this case there was a dilemma with the management of his cerebrovascular disease: whether to commence anticoagulation or not, as this could cause bleeding from the collateral circulation.

63. SURGICAL OUTCOMES OF COMBINED TRABECULECTOMY-CYCLODIALYSIS FOR GLAUCOMA

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Purpose: To determine the performance of combined cyclodialysis-trabeculectomy surgery over 25 years.

Methods: A retrospective cohort study of 55 eyes (39 patients) treated between 1987 and 2012 was performed.

The following data were collected: age, gender, glaucoma aetiology, visual data, intraocular pressure (IOP), topical medications, laser trabeculoplasty, 5-fluorouracil application, lens status, follow-up and complications. Patients were subdivided based on pre-operative IOP into groups: 10–20 mmHg, 21–30 mmHg, and >30 mmHg.

Failure was defined as reoperation for glaucoma, progression to legal blindness, IOP ≤ 5 mm Hg, >21 mm Hg or less than 20% reduction below baseline after 3 months. All others were a "complete success" if no additional topical medication was required and a "qualified success" if medication was required.

Results: Mean IOP fell from 25.2 to 11.9 mm Hg over a mean 11.3 years. 32 eyes (58.2%) achieved an unqualified success, 18 (32.7%) a qualified success and 5 (9.1%) a failure. No patient progressed to legal blindness. Cases with greater pre-operative IOP had a greater reduction in IOP ($p < 0.0001$). On univariate analysis no intra-operative application of antimetabolite (odds ratio (OR) 0.74, 95% confidence interval (CI) 0.49 – 0.94, $p = 0.022$) and a longer follow-up (OR 1.16, 95% CI 1.02 – 1.33, $p = 0.025$) predicted a

greater final IOP; these were not significant on multivariate analysis. 36 of 47 phakic eyes developed cataract; the majority in the first four years.

Conclusion: Combined trabeculectomy-cyclodialysis produces sustained lowering of IOP for long periods of time, despite a cataractogenic effect.

64. HIGH-DEFINITION OPTICAL COHERENCE TOMOGRAPHY IN THE ASSESSMENT OF IRIDOCORNEAL ANGLE CLOSURE

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Purpose: To compare the use of high-definition optical coherence tomography (HD-OCT) with standard anterior segment optical coherence tomography (AS-OCT) in the assessment of iridocorneal angle closure.

Method: Subjects with normal tension glaucoma (NTG), primary open-angle glaucoma (POAG), narrow angles, primary angle-closure (PAC), or primary angle-closure glaucoma (PACG) underwent gonioscopic assessments and anterior chamber imaging with AS-OCT and HD-OCT adapted with anterior segment lens mounted over the imaging aperture. Gonioscopy was performed, with angles graded open or close by a single examiner. The angles seen on OCT images were independently graded open or closed by 2 other examiners.

Results: 33 eyes from 17 subjects were studied. There were 5 eyes (15.2%) with NTG, 15 eyes (45.5%) with POAG, 2 eyes (6.1%) with narrow angles, 7 eyes (22.6%) with PAC, and 4 eyes (12.9%) with PACG. The majority of the subjects were Chinese (76.5%) and male (58.8%). The scleral spur was visualized in 45.5% and 84.8% of patients in the HD-OCT and AS-OCT images respectively ($P < 0.01$). The agreements between the 2 OCT image readers were $k = 0.36$ and $k = 0.55$ for the HD-OCT and AS-OCT respectively. The agreements between gonioscopy and HD-OCT and AS-OCT were $k = 0.24$ and $k = 0.32$ respectively.

Conclusion: The adapted HD-OCT allows an additional means to assess for iridocorneal angle closure. Our study shows that AS-OCT currently fares better than HD-OCT in terms of agreements with gonioscopy and inter-reader agreements. This may be related to the difficulty in identifying the scleral spur in HD-OCT images in our series.

65. 7,8 DIHYDROXYFLAVONE PROTECTS RETINAL GANGLION CELLS AGAINST EXCITOTOXIC AND OXIDATIVE STRESS

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Purpose: To evaluate neuroprotective effects of 7,8-dihydroxyflavone (7,8-DHF) on isolated rat retinal ganglion cells (RGCs) and RGC-5 cells to assess for the activation of TrkB and downstream survival signaling pathways.

Methods: To determine the optimal concentration for TrkB signalling activation, we initially used RGC-5 cell line treated with various concentrations of the 7,8-DHF (0–500 nM). Cells were also incubated with the drug (100 nM) for 0–16 hrs to study the time-course of TrkB signaling. The effects on the TrkB receptor and its downstream signaling on isolated RGCs and RGC-5 were then monitored by western blotting. Cells were exposed to Glutamate+H₂O₂ for stress induction. Changes in cell density and neurite outgrowth were analyzed by Thy-1 immunostaining. TrkB receptor kinase activity was assessed by immunoprecipitations followed by ELISA. Cell viability and apoptosis were evaluated using Quick Cell proliferation assay and caspase-3 activation respectively.

Results: 7,8-DHF treatment resulted in the phosphorylation of TrkB receptor and activation of TrkB kinase activity ($p < 0.001$) as well as downstream Akt and Erk signaling in both RGC-5 and isolated rat RGCs. 7,8-DHF also induced neuritogenesis by stimulating neurite outgrowth in the RGC-5 cells. Cell counting, proliferation assay and Caspase-3 activation demonstrated that 7,8-DHF treatment protected RGCs significantly ($p < 0.03$) against glutamate and H₂O₂ induced cell death in culture.

Conclusions: 7,8-DHF suppresses caspase-3 activation and improves RGC viability under excitotoxic and oxidative stress thereby suggesting a possible therapeutic strategy for protection of RGCs in glaucoma. The results obtained with RGC-5 cells were confirmed in isolated rat RGCs.

NEURO-OPHTHALMOLOGY

66. TEMPORAL ARTERITIS: RECOGNITION OF SIGNS AND SYMPTOMS IN A MAJOR TERTIARY EMERGENCY CENTRE

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Purpose: Temporal arteritis (TA), also referred to as giant cell arteritis, is a potentially irreversible cause of acute vision loss. Many patients will first present to an Emergency Department. Thus it is vital that Emergency Physicians be able to recognise the symptoms and signs of TA, as delay in diagnosis and appropriately prompt treatment may have devastating consequences.

Method: Patients from Prince of Wales Hospital, Sydney, between the years 2000 and 2011 inclusive who had had temporal artery biopsy-proven TA were reviewed through examination of their case notes. Data regarding patient demographics, medical and ophthalmic history, symptoms, signs and laboratory findings were recorded. As a priority, those who first presented via ED were definitively analysed in terms of a standardised proforma of symptoms and signs of TA.

Results: 62 patients were shown to have a superficial temporal artery that was positive for TA. Patients were classified in terms of 23 localised or systemic symptoms and 14 signs which had been identified. The symptoms and signs were both typical and atypical. Laboratory investigations carried out included full blood count, erythrocyte sedimentation rate and C-reactive protein levels.

Conclusion: TA is a disease that has which may present with typical features, but atypical features may also manifest. These must be recognised to avoid the major visual sequelae of TA. A high index of clinical suspicion is required to make the diagnosis.

67. COBALT-CHROMIUM TOXIC RETINOPATHY

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Purpose: To report the clinical case of a 65 year old male who developed retinal dysfunction following Cobalt-Chromium Toxicity.

Methods: A review of the clinical, haematological, radiological and electrophysiological investigations of a single patient was performed in order to form a case report.

Results: A 65 year old male presented to his ophthalmologist with a one year history of worsening vision on the background of a multisystem illness including motor axonopathy, pericardiomyopathy and bulbar palsy. Past medical history included hypertension, hypercholesterolaemia and a metallic hip prosthesis. Ocular examination revealed significantly reduced visual acuity bilaterally along with very poor colour vision. Cornea, fundi and optic discs all appeared normal. Bilateral moderate nuclear sclerosis was noted. Basic investigations including mitochondrial studies, auto-immune screen and MRI brain were unremarkable. Further investigations showed significantly elevated plasma cobalt and chromium levels. Electrophysiological studies revealed an abnormality in all phases of the ERG including a negative B-Wave form, suggestive of inner retinal pathology. Following subsequent revision of the hip, cobalt and chromium levels decreased and the patient's vision improved. Further electrophysiological testing indicates a persistent ERG abnormality despite a significant improvement in both the patient's visual acuity and colour vision.

Conclusions: These results suggest that Cobalt-Chromium toxicity can cause inner retinal dysfunction.

68. MUNCHAUSEN AND THE WEIRD AND WONDERFUL

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Purpose: A careful history with probing questions often allows a correct provisional diagnosis that is then confirmed by examination and investigational findings. However, at times the given history can be a figment of the patient's imagination whilst at times some apparently fantastic claims can be proved true!

Methods & Results: Examples of clinical cases of Munchausen and Munchausen-by-proxy syndrome

are described together with cases with seemingly unbelievable symptoms that proved to be true. Often patients are puzzled by their symptoms and attempt to help their clinician by searching the internet and bringing in data (usually unhelpful); an intriguing example is given.

Conclusion: The clinician must be alert to the nuances of history taking, in order to arrive at a timely provisional diagnosis. At times symptoms are a "cry for help". However, we will still be fooled some of the time.

69. AN UNUSUAL CASE OF BILATERAL CONSECUTIVE ORBITAL APEX SYNDROME

WITHDRAWN

70. RETINAL REARRANGEMENT CAUSING RETINOTOPIC PLASTICITY

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Purpose: To describe a remarkable case of retinal detachment, with implications for models of cortical plasticity.

Method: Case report of a 56 year old man with retinal detachment.

Results: After vitrectomy with scleral buckle, the right macula slipped nasally, creating a sharp vertical retinal fold between the fovea and the disc. This macular heterotopia created binocular diplopia, and this appears to be the first reported case in which strabismus surgery was used to restore binocularity after retinal slippage. Secondarily, but of wider significance, the retinal slippage fold led to a scotoma in the other eye, which matched the retinal fold in visual space. That is, there was a curved narrow vertical linear scotoma in the left eye, separate from the blind spot, which perfectly matched the right retinal fold in visual space. The specificity of the contralateral scotoma led us to theorise that this represented plasticity of the retinotopic map. This illuminates a type of cortical plasticity known as perceptual learning.

Conclusion: We propose that this phenomenon was driven by recognition of shapes from higher centres, and we discuss implications for models of perceptual learning.

71. CLINICAL SACCADIC ASSESSMENT IN NEUROLOGISTS COMPARED WITH OPHTHALMOLOGISTS: A PROSPECTIVE CONSECUTIVE STUDY

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Purpose: To study the ability of two separate cohorts of Ophthalmologists and Neurologists to assess clinical saccadic eye movement slowing in a single video recording of a patient with a mild left sixth nerve palsy.

Methods: In this study, we evaluated prospective consecutive cohorts of Ophthalmologists and Neurologists. The Ophthalmologists were assessed at the 43rd RANZCO Annual Scientific Congress in November 2011. The Neurologists were assessed at the 13th Asian Oceanian Congress of Neurology in May 2012. Each Ophthalmologist/Neurologist was asked to assess the saccadic velocity of the patient's slow saccade in a small amplitude compared with a large amplitude saccadic excursion. Each doctor was given the opportunity to review the video as many times as he/she wished.

Results: Of the 53 Ophthalmologists, 36 (65.5%) believed that the small amplitude saccade provided best recognition of the patient's slow saccade.

Of the 55 Neurologists, 43 (78.2%) recognized that the patient's reduced abducting velocity was best recognized with a small amplitude saccade.

Conclusion:

- 1) The majority of Ophthalmologists and Neurologists were able to recognize saccadic slowing.
- 2) The clinical evaluation of small amplitude saccades allows reduced saccadic velocity to be recognized more readily than large amplitude saccades. Although this is contrary to some traditional teaching, the utilization of small amplitude saccades permits velocity of slowed saccadic velocity to be most readily recognized.
- 3) In the course of this study, we have been able to document 10 theoretical and practical reasons that best recognition of slow saccades is achieved using small saccadic amplitudes.

OCULAR ONCOLOGY

72. SMALL CELL NEUROENDOCRINE CARCINOMA OF 'SMALL BLUE CELL' SUBTYPE INITIALLY TREATED AS CELLULITIS

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Purpose: We report the first case of small cell neuroendocrine carcinoma involving both orbits and causing complete blindness.

Methods: Case report and review of literature

Results: A 35-year-old Caucasian female presented to hospital with right periorbital pain and fullness for 3 weeks. She was initially prescribed oral antibiotics for presumed cellulitis but developed progressive swelling and blurred vision in the right eye.

Magnetic resonance imaging demonstrated abnormal tissue in the superomedial and superior right orbit with destruction of the anterior lamina papyracea, extension into the anterior ethmoid air cells, right frontal bone and right frontal lobe. Histopathology from the orbital tissue and right frontal subdural empyema demonstrated small cell neuroendocrine carcinoma of 'small blue cell' subtype. The patient received a multidisciplinary treatment protocol combining surgery, chemotherapy (both systemic and intra-arterial) and radiotherapy.

Conclusion: 10 months after surgery, there was further progression of the disease to involve the left eye, with tumour extending from the cavernous sinus to the left orbital apex and involving the optic nerve, leading to complete loss of vision. The patient passed away 22 months after initial diagnosis. Neuroendocrine carcinoma is an extremely rare disease with a very poor prognosis. Further research is required for more effective treatment options.

73. MATRIX METALLOPROTEINASES AND THEIR INHIBITORS IN THE PATHOPHYSIOLOGY OF SQUAMOUS CELL CARCINOMA OF THE CONJUNCTIVA

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Purpose: Squamous cell carcinoma of the conjunctiva (SCCC) belongs to a spectrum of diseases collectively known as ocular surface squamous neoplasia (OSSN). Epidemiological evidence suggests environmental ultraviolet radiation (UVR) exposure as the principal triggering agent. Despite this indirect, but convincing evidence, the pathogenesis of SCCC remains poorly understood, likely due to the absence of robust molecular, cellular and animal models. We postulated that matrix metalloproteinases (MMPs) and tissue inhibitor of metalloproteinases (TIMPs) are upregulated in SCCC and this could account for the invasive activity associated with this disease.

Methods: Archival tissue specimens from 10 patients with SCCC were acquired to assess the expression of seven MMPs, three TIMPs and two growth factors and their receptor by immunohistochemistry using specific antibodies.

Results: All MMPs and TIMP-2 were over-expressed in the tumour component compared to adjacent normal conjunctiva and cornea. Active MMP-7 was detected in diseased tissue, suggesting at least some members of this family of enzymes are functionally involved. Moreover, active epidermal growth factor receptor and its ligands were detected within the tumour compartment.

Conclusion: This data suggests that UVR-induced downstream cellular signaling events, including activation of cell-surface receptors and the induction of downstream effector molecules, such as MMPs and growth factors, are involved in the pathogenesis of SCCC. Mapping and inhibiting these pathways may aid in delineating the pathogenesis of SCCC and provide clues for optimising current, or developing novel treatment strategies.

74. THE EFFECT OF SUPPLEMENTAL TRANSPUPILLARY THERMAL THERAPY ON VISUAL OUTCOMES IN CHOROIDAL MELANOMA

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Purpose: To compare the visual outcomes in patients treated with primary Iodine-125 brachytherapy with and without supplementary transpupillary thermal therapy (TTT).

Methods: Retrospective case controlled trial consisting of 54 patients in the TTT group (group A) and 57 patients in the non-TTT group (group B). Indications for supplementary TTT included radiotherapy related complications and signs of early tumour recurrence. Mean follow up was 65.3 and 60.1 months for group A and B respectively.

Results: Poorer levels of visual acuity were found in group A compared to group B with the difference reaching statistical significance at 1 and 4 years during follow up. With respect to the visual acuity immediately following primary brachytherapy, the amount of visual acuity loss was equal between group A and B.

Conclusion: When TTT was used for the treatment of radiotherapy related complications or signs of early tumour recurrence, there was a trend towards poorer visual outcomes but this difference only reached significance at 1 and 4 years during follow up. Accounting for early radiotherapy related vision loss, the addition of TTT did not result in significantly worse visual acuity.

OCULOPLASTIC/ORBIT

75. POST-OPERATIVE HAEMORRHAGE IN ENDOSCOPIC DACRYOCYSTORHINOSTOMY

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Purpose: To assess the perioperative use of agents with antiplatelet or anticoagulant activity, and the rate of post-operative haemorrhage, in patients undergoing powered endoscopic dacryocystorhinostomy (DCR).

Method: A prospective, non-randomised, non-comparative, single-surgeon, interventional case series.

Results: 174 consecutive cases were included. The mean age of patients at the time of surgery was 62.9 years with a range of 19–94 years, 65% were female. 28 cases (16%) were taking a medication with anti-

platelet or anticoagulant activity and these agents were completely stopped pre-operatively in 14 cases. Three cases (1.7%) were complicated by post-operative haemorrhage but only one case (0.6%) required physician intervention.

Conclusion: The risk of significant haemorrhage following endoscopic DCR is low. The decision to stop antithrombotic agents prior to surgery should be made in consultation with the patient's internal physician.

76. PLEOMORPHIC ADENOMA OF THE LACRIMAL GLAND: LONG TERM FOLLOW UP IN A JAPANESE COHORT

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Purpose: To review the clinical characteristics and outcome of lacrimal gland pleomorphic adenomas.

Method: A retrospective, interventional case series reviewing all patients with histologically confirmed pleomorphic adenomas of the lacrimal gland that presented to the Kyoto Prefectural University of Medicine, Japan, between January 2004 and March 2012.

Results: 41 consecutive cases were included (mean age 53.0 years, 61% female). The cohort included 35 de novo cases, and six cases (14.6%) that presented with recurrent pleomorphic adenoma following excision a mean of 148 months prior. Diplopia was the most common presenting symptom in both groups (78%), pain was uncommon (4.9%). Proptosis (78%), inferonasal dystopia (75.6%), and ptosis (27.8%) were the most common signs. All de novo cases presented with a single lesion, located in the orbital lobe in 82.8% and the palpebral lobe in 17.2%. All recurrent cases presented with two lesions (50.0%), or more (50.0%). In the de novo group, there was intraoperative capsular breach in 10 cases (28.6%), however all of the surgical margins were free from tumour histologically. There were no recurrences in this group during a mean follow-up time of 53.3 months. In the recurrent group, five cases (83.3%) were managed with incomplete excision and had surgical margins involved by tumour. One case (16.6%) developed repeat recurrence during a mean follow-up time of 42.0 months. Malignant transformation was not observed in any of the 41 cases.

Conclusion: The proportion of pleomorphic adenomas arising in the palpebral lobe is not insignificant. Recurrent tumour is more commonly multifocal.

77. ORBITAL BLOWOUT FRACTURES IN CHINESE AND JAPANESE INDIVIDUALS: A MULTICENTRE STUDY

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Purpose: Previous studies have suggested that the characteristics of orbital blowout fractures vary according to race. We thus sought to characterise the location of orbital blowout fractures in Chinese and Japanese individuals.

Methods: Consecutive Chinese and Japanese patients presenting to the Eye Hospital of Wenzhou Medical College, the Kyoto Prefectural University of Medicine, Hamamatsu University School of Medicine, and Aichi Medical University Hospital with orbital blowout fractures were included in this study. Computed tomography characterised the location and severity of fractures involving the medial wall, orbital floor, and/or the maxillo-ethmoidal strut. Predictors of fracture location were determined using multi-variable analyses.

Results: From 688 Asian patients, a total of 696 orbital blowout fractures were identified. More than one fracture location was involved in 150 cases (22%). Isolated medial wall fractures were the most common type of fracture within Chinese patients, presenting in 128 cases (54%), followed by fractures of the maxilla-ethmoidal strut, occurring in 37 (16%) cases. Within the Japanese cohort, isolated orbital floor fractures were the most commonly encountered fracture type, presenting in 223 cases (48%), followed by isolated medial wall fractures, occurring in 130 (28%) of cases. Race remained as an independent predictor of fracture location in multi-variate analyses ($p < 0.05$ for all).

Conclusion: There is significant inter-racial variation of orbital blowout fracture location. Orbital blowout fractures within the Chinese population most commonly involve the medial wall, while Japanese patients were more likely to fracture their orbital floor.

78. BACTERIAL BIOFILMS ASSOCIATED WITH OCULAR PROSTHESES

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Purpose: Biofilm growth on ocular implants has been shown to demonstrate a broad spectrum of clinical manifestations, from no clinical signs of infection, to endophthalmitis and microbial keratitis. Despite the identification of biofilm on the surface of multiple different types of ocular implants, there has yet to be a study investigating ocular prostheses. We therefore sought to demonstrate the presence of biofilm on the surface of ocular prostheses.

Methods: Long-term ocular prostheses and other orbital implants that were no longer required were obtained from patients attending the Royal Adelaide Hospital. Each specimen was fixed and coated with platinum prior to imaging. Scanning electron microscopy was used to image the surface of specimens.

Results: Three prostheses, one implant and one conformer were obtained and imaged. The duration of implantation ranged from 3 months, to over 10 years. All patients were asymptomatic prior to removal and ophthalmic examination was unremarkable. Extensive biofilm growth was demonstrated on all specimen surfaces except for the conformer. Numerous small colonies of rod-shaped bacilli were identified adherent to the surface of two prostheses. Ongoing investigations into the culture, identification and eradication of these biofilms are in progress.

Conclusion: Bacterial biofilms develop on ocular prostheses and can be detected via scanning electron microscopy. This may explain the presence of socket inflammation in some patients with long-term ocular prostheses, and further studies are required to investigate methods to eradicate these biofilm growths.

79. OSTIUM SHRINKAGE FOLLOWING ENDOSCOPIC DACRYOCYSTORHINOSTOMY

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Purpose: Ostium closure following endoscopic dacryocystorhinostomy (DCR) is the commonest cause for anatomical failure. We aim to determine the changes in size of the dacryocystorhinostomy ostium over time

and investigate the correlation of ostium size and surgical outcome.

Methods: This is a single surgeon, prospective, non-randomised, non-comparative, interventional case series on consecutive series of patients who underwent powered endonasal DCR. All patients had radiologically confirmed nasolacrimal duct or canalicular obstruction. The patients were operated on by one surgeon (DS) and follow-up was at 4 weeks and 12 months. Ostium sizes were measured at the end of surgery and at 4 weeks and 12 months after surgery.

Results: 161 patients fulfilled the inclusion criteria. 3 patients were lost to follow-up. The ostium measured 8.6 (95% confidence interval [CI] = 5.0–12.2) by 13.4 (95% CI = 10.3–16.5) at the time of surgery and 5.7 (95% CI = 2.3–9.0) by 9.5 (95% CI = 6.0–13.0) at 4 weeks, and 4.8 (95% CI = 1.9–7.7) by 8.2 (95% CI = 4.5–11.9) at 12 months. There were statistically significant ostial shrinkage from surgery to 4 weeks (mean shrinkage of 50%) and from 4 weeks to 12 months (mean shrinkage of further 15%).

Conclusion: Following endoscopic DCR, the final ostium size on average is 35% of the original at 12 months. The majority of the ostium shrinkage occurs within 4 weeks post-operatively with a lesser degree of shrinkage between 1–12 months post-operatively.

80. ASSISTED LOCAL ANAESTHESIA FOR POWERED ENDOSCOPIC DACRYOCYSTORHINOSTOMY

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Purpose: The role of assisted local anaesthetic (aLA) for both endoscopic and external dacryocystorhinostomy is a well-tolerated and established approach. However the tolerability of aLA is unclear with powered burrs used in powered endoscopic DCR (PEDCR). We aim to evaluate the acceptability of aLA for PEDCR.

Methods: This is a prospective, interventional, non-randomized, non-comparative, single surgeon study. Consecutive series of patients that underwent PEDCR performed under aLA were included in the study. Tolerability was assessed by intra-operative pain score on 100 point visual analogue scale (VAS) and if patients were willing to have aLA-PEDCR again.

Results: A total of 44 PEDCR was performed on 42 patients. 56% of patients reported 0 on 100 point VAS, 65.9% (29/44) reported <10, 88.6% (39/44) reported <20 and no patients had score of >30/100. 97.7%

(43/44) of patients are happy to have PEDCR performed again under aLA. The one patient unwilling to have a repeat aLA-PEDCR was not because of pain but intolerance to the sound of drilling.

Conclusion: PEDCR with assisted local anaesthetic is well tolerated and accepted by patients.

81. THE MUSTARDÉ CHEEK ROTATION FLAP FOR RECONSTRUCTION OF LARGE LOWER EYELID DEFECTS

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Purpose: To describe the technique & outcomes of lower lid reconstruction with the Mustardé cheek rotation flap.

Method: A retrospective review of 24 cases of lower lid reconstruction using a Mustardé cheek rotation flap performed at a single centre. Outcome measures were ocular function, appearance, patient satisfaction, complications and additional surgery.

Results: There were 24 cases of lower lid reconstruction using a Mustardé cheek rotation flap over a 10 year period. Patient age ranged from 42 to 92 years. The majority of tumour excisions (79%) were performed using Mohs micrographic surgery. The remainder were staged excisions using permanent section histological control. The commonest preoperative diagnosis was basal cell carcinoma (n = 17). The size of the lower lid defects ranged from 2 × 1 cm to 6 × 4 cm. Eighteen Mustardé flap reconstructions were combined with other flaps (Hughes, glabellar, rhomboid, & periosteal flaps), horizontal lid tightening, and autologous grafts (hard palate & full thickness skin). The mean follow-up was 6 months. Early complications included bleeding (n = 2) and infection (n = 2). Late complications included ectropion and/or lower lid retraction (n = 6) and epiphora (n = 2). Overall patient satisfaction ranged from satisfactory to very satisfied.

Conclusion: The Mustardé flap is useful in the reconstruction of large lower lid defects after tumour resection. It can be combined with other periocular flaps, skin grafts & other autologous grafts. Ectropion and lower lid retraction are the commonest side effects especially when defects involve loss of the whole lower lid and/or there is significant loss of subcutaneous cheek tissue.

82. AN UNEXPECTED COMPLICATION OF CRAWFORD TUBE INSERTION

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Purpose: To describe an unusual and challenging case of complicated bicanalicular (Crawford) stent insertion.

Method: Noncomparative case study and literature review.

Results: A 27-year-old man sustained multiple lacerations to the left (L) periocular and facial areas with glass bottle. He was admitted under the care of the Plastic Surgical team and underwent repair of the facial, eyelid and canalicular lacerations in the operating room. During the surgery, a Crawford bicanalicular stent was guided into the L inferior canaliculus, but was not retrieved nasally. Neuro-imaging revealed that the metallic probe of the Crawford tube was within the orbit extending via the superior orbital fissure into the middle cranial fossa. The patient was then transferred under the care of the neurosurgical team and underwent combined orbital/neurosurgical surgery to retrieve the metallic probe. This complication has never previously been reported. Other complications of bicanalicular stent insertion will also be discussed.

Conclusion: Crawford stent insertion can be challenging in the setting of traumatic lid lacerations. An understanding of the potential complications and increased surgical experience with insertion reduces the likelihood of adverse outcomes.

83. ONE-STEP RECONSTRUCTION OF LARGE LOWER EYELID DEFECTS: TECHNIQUE AND OUTCOMES

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Purpose: To describe a one-step technique of lower eyelid reconstruction using a free tarsoconjunctival graft combined with an upper eyelid myocutaneous transposition flap.

Method: Retrospective case series analysis of twelve lower eyelid reconstructions between 2006–2010 at two tertiary referral hospitals. All cases of lower eyelid reconstruction used a free tarsoconjunctival graft combined with an upper eyelid myocutaneous flap and

were assessed for functional and cosmetic result. Surgical techniques were described for both medial and lateral approaches, with use of intra- and post-operative photographs.

Results: Successful reconstruction was achieved in all twelve patients, with no graft failure, flap necrosis or major complications of entropion/ectropion or trichiasis. Follow-up revealed minor complications in approximately half of patients, notably notching (8.3% upper lid; 25% lower lid) and granuloma formation (25% lower lid only), though these did not affect vision or functional closure. One case required minimal revisional surgery.

Conclusion: The authors contend that the technique of free tarsoconjunctival graft together with upper eyelid myocutaneous transposition flap best mimics the natural form and function of the lower eyelid and is a useful technique for reconstruction of moderate to large lower eyelid defects. It has an advantage over the traditional Hughes technique of being a one-stage procedure, therefore limiting time the eye is occluded from weeks to days, and overall provides excellent long-term functional and cosmetic results.

84. TRAUMATIC PTOSIS DUE TO PRESUMED NEURAPRAXIA

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Purpose: To describe a small case series of patients with traumatic ptosis due to presumed isolated neurapraxia of the nerve of innervation to the levator palpebrae superioris.

Method: Retrospective case note review.

Results: We describe 3 patients who sustained significant tractional or blunt injury to the eyelid resulting in complete ptosis and markedly reduced or no levator function. Extraocular movements were full and the remainder of the ocular examination was normal. All patients showed recovery of levator function and resolution of the ptosis over a 2–6 month period. The presumed mechanism of ptosis in all cases was neurapraxia of the nerve of innervation to the levator muscle.

Conclusions: Neurapraxia of the nerve of innervation to the levator muscle is a rare cause of traumatic ptosis. The mechanism of injury in these cases is thought to be secondary to traction on the eyelid and levator muscle with secondary traction on the nerve to the levator muscle. Resolution of the ptosis may take up to 6 months.

85. PREVALENCE OF GRAVE'S OPHTHALMOPATHY AND OTHER OCULAR MANIFESTATIONS IN AUTOIMMUNE THYROID DISORDER PATIENTS IN MALAYSIA

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Purpose: This study was initiated to investigate the prevalence of Graves' Ophthalmopathy and other ocular manifestations in autoimmune thyroid disorder patients in Malaysia.

Method: Ninety seven (97) patients who have been diagnosed with autoimmune thyroid disorders from a main referral Endocrine Centre were recruited. History and complete ocular examination was done. Pearson's correlation statistical tests were used to evaluate correlations between the findings.

Results: Occurrence rate of Graves' ophthalmopathy in this study was 42.3%. We also found that higher rate of symptoms of tearing (38.1%), photophobia (36.1%), retrobulbar discomfort and foreign body sensation (17.5%) and diplopia (22.7%). On examination, 32.0% of the patients had lid lag, 18.6% had upper lid retraction and 27.8% had lower lid retraction. Only 3% of the patients had exophthalmos. Pearson's correlation statistical test showed that females had higher risk of diplopia (odds ratio 8.197, 95% CI 1.032–66.667). Patients diagnosed with Graves' ophthalmopathy had a significant correlation with symptoms of photophobia ($p = 0.003$), epiphora ($p = 0.034$) and retrobulbar discomfort ($p = 0.014$). Eyelid retraction was also found to have correlations with lid lag ($p = 8.8 \times 10^{-5}$) and retrobulbar discomfort ($p = 0.004$).

Conclusion: Rate of Grave's ophthalmopathy in the study was consistent with previous studied done in the western countries. However the spectrum of signs and symptoms were different whereby upper lid retraction was less seen and other symptoms such as photophobia, epiphora and retrobulbar discomfort is twice more than previous studies. Further investigations for lid lag and retrobulbar discomfort as a predictor to the development of Grave's Ophthalmopathy should be initiated.

86. ACUTE POST OPERATIVE RHINOSINUSITIS FOLLOWING ENDONASAL DACRYOCYSTORHINOSTOMY

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Purpose: Prospective, non-randomised, single surgeon interventional case series reporting incidence of acute rhinosinusitis following routine endoscopic dacryocystorhinostomy (DCR) and the prevalence of chronic rhinosinusitis (CRS) in the same cohort of patients.

Methods: 196 consecutive patients undergoing 203 endonasal DCRs with clinical and radiological evidence of nasolacrimal duct obstruction (NLDO) or stenosis or canalicular pathology underwent lacrimal and sinonasal clinical assessment including lacrimal imaging and nasal endoscopy. Patients were questioned about a history of sinonasal disease, CRS and sinonasal surgery. Cases of acute post operative sinusitis, their management and surgical outcomes at 12 months were recorded.

Results: The rate of CRS in this cohort of 196 patients was 10.2% ($n = 20$). Three patients (1.5% of all or 15% of those with CRS) developed acute rhinosinusitis within the first 6 post operative days, none had experienced per-operative complications and none had symptoms of CRS at the time of surgery. All three patients had a history of CRS and one had undergone previous sinus surgery. All 3 responded to oral antibiotic therapy.

Conclusion: Following endoscopic DCR, the rate of acute rhinosinusitis was 1.5% or 15% among patients with a history of CRS ($p < 0.0001$). Acute rhinosinusitis developed in the first post operative week. All 3 patients had a history of CRS but were asymptomatic at the time of surgery, and responded to oral antibiotics. CRS may be a risk factor for the development of post-operative acute rhinosinusitis.

87. TURN-OVER ORBITAL SEPTAL FLAP COMBINED WITH LEVATOR RECESSION FOR UPPER EYELID RETRACTION

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Purpose: This retrospective multicenter case series reports the outcomes of levator lengthening using a turn-over orbital septal flap to treat upper lid retraction due to a variety of causes.

Methods: Fifteen eyelids of 13 patients (average age 63 years(24–83)) with upper lid retraction associated with TED, facial nerve palsy or previous eyelid surgery causing symptomatic corneal exposure, with or without keratitis. Surgery involved a transcutaneous levator lengthening technique using the reflected orbital septum as a spacer to achieve eyelid lowering. Two patients with TED were excluded from the study where the orbital septum was found to be too thin for use as a flap.

Results: At a mean follow-up period of 9.4 (3–19) months all patients were satisfied with the surgical outcome and resolution of symptoms of exposure keratopathy. The PA was reduced by an average of 2.7 mm ($p < 0.0001$) and the upper MRD by 2.7 mm ($p < 0.001$). The position of the SC remained unaltered at final follow up in 80%, and the remaining 20% were within 1 mm of the height set intra-operatively. The difference in PA between the operated and contralateral eyelid was reduced from 1.75 mm to 0.3 mm ($p < 0.0001$). The proportion of eyes with symmetry of upper eyelid contour was unchanged by surgery. Complications included one case of overcorrection necessitating further surgery and one case of late recurrent lateral flare.

Conclusions: The turn-over orbital septal flap technique may be a viable option as an autogenous spacer for upper eyelid retraction surgery in selected cases.

88. ARE WE THERE YET? THE JOURNEY OF FDG PET IN THE STAGING OF OCULAR ADNEXAL LYMPHOPROLIFERATIVE DISEASE

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Purpose: To further evaluate fluorine 18 deoxyglucose positron emission tomography (FDG PET) in the initial staging of ocular adnexal lymphoproliferative disease (OALD).

Method: Retrospective non-randomised case series with review of clinical and imaging records including computed tomography (CT), FDG PET and/or magnetic resonance imaging (MRI).

Results: Thirty four patients with biopsy proven OALD and FDG PET at initial staging were retrospectively reviewed. Of 34 patients, 17 (50%) had primary disease, 17 (50%) had secondary and of these, 11 patients (32%) had OALD as their presenting complaint. FDG PET found orbital disease in 27 of 34 patients (79%) versus 33 of 34 patients with orbital CT (97%). Systemic disease was demonstrated by FDG PET in 15 of 16 (94%) patients and 11 of 16 (69%) patients with CT.

Conclusion: This study further confirms FDG PET as an essential part of initial staging. Our findings suggest FDG PET detection for local OALD is less sensitive than CT with data echoing current literature estimates. Our study suggests FDG PET detects systemic disease more reliably than CT and results in significant changes in management. Given the prevalence of secondary simultaneous presentations of OALD, FDG PET in this regard is especially important.

89. THYROID EYE DISEASE SECONDARY TO CAMPATH (ALEMTUZUMAB) USED IN TREATMENT OF MULTIPLE SCLEROSIS

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Purpose: Alemtuzumab is an anti-CD52 monoclonal antibody (humanized) which has demonstrated some benefit in multiple sclerosis treatment. Its safety profile

lists infusion-associated reactions, infections and secondary autoimmunity as being the main adverse events for consideration. Thyroid disease is the most common secondary auto-immune event (up to thirty percent of treated patients) and is a major safety concern, however it is readily treatable. We report on a case of thyroid eye disease being induced by Alemtuzumab and its response to management.

Methods: A single case report detailing the presenting history, ocular examination, sequential CT and serum thyroid hormone levels (TSH, T3, T4, antibodies) as well as further management is discussed.

Results: In this case the patient's thyroid eye disease stabilized post treatment with carbimazole and oral steroids. Eyelid surgery was withheld until the options for hyperthyroid treatment had been exhausted. At the time of publication the patient was awaiting thyroidectomy and had not progressed to eyelid surgery. Visual acuity was stable with minimal signs of inflammation and no optic neuropathy. Cosmetically, there was still significant proptosis bilaterally and lid retraction.

Conclusions: Thyroid orbitopathy as a result of Alemtuzumab induced secondary auto-immunity presents another new challenge to consider during management and monitoring of response. With good patient and physician education, the condition can be reduced if detected early by vigilant monitoring, thoughtful consideration given to evaluate thyroid orbitopathy, and then treated accordingly. Eyelid surgery should be approached cautiously on these patients and only after every chance of spontaneous resolution has been allowed.

OTHER

90. SQUASH(ED): MULTIPLE CRANIOMAXILLOFACIAL AND VERTEBRAL FRACTURES FROM SQUASH COURT COLLISION

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Purpose: To describe a case of extensive craniomaxillofacial and vertebral fractures secondary to collision during a squash game, without use of protective eye-wear.

Methods: Case report.

Results: We present a case of significant ocular, head and spinal injury resulting from a game of squash, after the patient collided with the corner of the court.

He sustained fractures of the orbits, frontal bone, maxillae, zygomas, the first and second cervical vertebrae and the spinous process of the seventh cervical vertebra. This is the first case of squash-related injury with such extensive craniofacial and vertebral involvement.

Conclusion: The patient underwent craniotomy, obliteration of the frontal sinus and open reduction internal fixation (ORIF) of the frontal bone. ORIF of the maxilla bilaterally, ORIF of the infra-orbital rim and supra-orbital rim bilaterally and inter-maxillary fixation was also performed. The patient was admitted to the neurosurgical intensive care unit post-operatively and underwent cervical fusion one week after surgical correction of the head injuries. This case highlights the risk of significant trauma in squash, a popular racquet sport where the use of protective eye gear is strongly recommended.

91. SLIT LAMP PHOTOGRAPHY WITH MOBILE PHONE CAMERA

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Purpose: The capabilities of cameras in smart phones have improved exponentially over the last decade. Camera phones are increasingly used for slit lamp photography to produce high quality images and videos. We present our experience with using mobile phone camera for slit lamp photography.

Methods: We will first describe the various methods of attaching a phone to the slit lamp. We will also present various techniques that can improve image quality. Finally we will present our experience of using this system in outback Australia and other potential application in Ophthalmology.

Results: There are various DIY and commercially available phone to slit lamp adaptors. These systems can be rapidly set up and the qualities of the images are good. Real time imaging capabilities on phone display are invaluable in patient and student education. Cellular and internet connection also facilitate real time telemedicine.

Conclusion: Slit lamp photography with camera phones can provide high quality images that are conducive for telemedicine and education.

92. AN UNEXPECTED COMPLICATION OF BEHAVIOURAL VISION THERAPY

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Purpose: To describe an unfortunate complication of behavioural vision therapy.

Methods: Single case report concerning the occurrence of a penetrating eye injury in the context of performing behavioural vision therapy.

Results: A 43 year old female suffered a large ragged, shelved corneal laceration as a result of a vision therapy apparatus prescribed for the treatment of reading dysfunction and hyperopia.

Conclusion: Vision therapy equipment should be made as safe as possible to avoid the possibility of adverse events.

93. THE IPHONE AS A PHOTOGRAPHIC TOOL IN OPHTHALMOLOGY

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Purpose: To demonstrate the use of the iphone as a photographic tool in an ophthalmic practice including tips on its usage.

Method: A poster presentation will cover the following: the adapter used for slit lamp iphone photography, applications used to capture the photos and transfer them to an electronic medical record, photos taken, useful tips for better photographs.

Results: The results are the photographs demonstrated.

Conclusion: The increasing popularity of the iphone is making it worthwhile for further investigation of its use in ophthalmology. Whilst its function is never going to match dedicated equipment for this task, it may have some role in low cost photography to enable images to be sent over the internet.

94. REMODELLING IN ADVANCED RETINAL DEGENERATION IN rd1 MICE

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Purpose: Over recent years there has been a surge of interest in vision restoring therapies that replace the

function of dead or dying photoreceptors in patients with photoreceptor diseases. But for any of these treatments to succeed, a basic requirement is that inner retinal neurons, especially the ganglion cells remain functionally intact. In mid to late stages of the disease when most photoreceptors have been lost, significant remodelling of inner retinal neurons occurs. We aimed to characterise the cause and progression of events during late phase degeneration in a mouse model of retinitis pigmentosa, the rd1 mouse. We specifically were interested in the mechanisms leading to ganglion cell remodelling and death.

Method: We used two double mutant transgenic mice: 1.rdl-FTL mice, containing the rd1 mutation and an axon-targeted β -galactosidase (β -gal) reporter system under the regulation of the c-fos gene. In these mice neurons that express c-fos also express β -gal in the entire cell. 2.rdl-Thy1-YFP mice which in addition to rd1 express YFP under the control of Thy1 promoter in a subpopulation of ganglion cells. Retinae from mice between postnatal days P30 to P365 were immunocytochemically analysed as wholemounts or cryostat sections.

Results: Following complete loss of cones/cone pedicles, Müller cells change their function and switch to proliferative gliosis. In patches of proliferative gliosis, ganglion cells upregulate c-fos expression and remodel.

Conclusion: These results suggest that change in Müller cell function and switch to proliferative gliosis might lead to ganglion cell dysfunction and gliosis.

95. ESTABLISHING A REGIONAL OPHTHALMOLOGY CLINICAL TRIAL NETWORK

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Purpose: This paper describes the possible structure and benefits of an investigator led ophthalmologic clinical trial network.

Method: A number of existing clinical trial networks are examined and critiqued, including the Australasian Stroke Trials Network (ASTN) and the European Vision Institute – Clinical Trials – Sites of Excellence (EVI.CT.SE)

Results: Clinical trial networks can be structured to provide important research tools and services to promote and support clinical research activities across academic and commercially sponsored clinical trials. Such a network could provide common and harmonised organisational and technical Standard Operating Procedures, quality control and staff training according

to ICH GCP-Guidelines, and play an important advocacy role.

Conclusion: Several local and international clinical trial networks have demonstrated the utility in providing various centralised services to support clinical research, particularly investigator-initiated clinical trials.

96. A SIMPLE TECHNIQUE FOR CONTROLLING THE NON VIEWING EYE DURING DDIRECT OPHTHALMOSCOPY

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Purpose: To describe a simple technique that allows for direct ophthalmoscopy without having to close or suppress the non viewing eye.

Method: Whilst having the thumb still anchored on the brow, the dorsal aspect of the wrist is brought towards the examiner's non viewing eye. By doing this, the viewing eye is blocked efficiently by the anatomical snuffbox. As the examiner gets closer to the subject the wrist occluding the non viewing eye relaxes to allow the viewer to approach the subject. The rest of the direct ophthalmoscopy method is unchanged.

Results: We find this technique very useful for teaching direct ophthalmoscopy as dominant eye closure or suppression problems are common amongst students. Those who may not be practising ophthalmoscopy frequently, such as general practitioners or physicians, may also find the technique helpful to use with either eye.

Conclusion: This simple technique allows easy occlusion of the dominant eye, making direct ophthalmoscopy easier hence improving its efficiency.

97. INTEROCULAR INTERACTION IN ANISOMETROPIC AMBLYOPIA

WITHDRAWN

98. VALIDATION OF A NOVEL FAST AUTOMATED TECHNIQUE FOR EOG ARDEN RATIO CALCULATION

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Purpose: To develop and validate an automated fast method of calculating the Arden Ratio from EOG recordings.

Methods: Raw EOG traces were exported from the RVEEH visual electrophysiology system. 3rd order harmonic approximations to square waves were fitted as models via non-linear least square (NLS) algorithm and again with a robust linear model (RLM). Calculations were performed in the open source statistical language R using a routine developed by the authors. Arden ratios were calculated by fitting 3rd degree polynomials to the model amplitudes in the dark and light phases.

Results: Arden ratios from the two methods were compared against manually calculated ratios in 54 eyes of 27 patients with a variety of ophthalmic diagnoses. The NLS and RLM results correlated closely with the manual method (NLS adjusted R squared = 0.89, $p < 0.01$; RLM = 0.90, $p < 0.01$). The automated method classified the EOG results with a sensitivity of 89% and specificity of 92% NLS; sensitivity 96% and specificity of 92% RLM, compared to the manual ("gold standard") method whilst reducing technician processing time from 20 to 2 minutes. Advantages of this algorithm: no saccades need be detected and the information of the fully determined time course of the gaze target is optimally utilized making it robust to irregular gaze changes.

Conclusions: The results validate this novel automated method against the manual method, while considerably saving time. The authors intend to make this available as part of an open source R package.

99. CHOROIDAL THICKNESS ASSOCIATED WITH SPHERICAL EQUIVALENT IN HEALTHY YOUNG ADULTS: THE RAINE EYE HEALTH STUDY

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Purpose: To measure subfoveal choroidal thickness (SFCT) in healthy young adults and to analyse correlation with other measures of ocular biometry, anthropometric measures and birth parameters.

Methods: Prospective study of 502 eyes of 502 participants from the Raine Eye Health Study, aged 19 to 22 years old. Participants underwent enhanced-depth imaging optical coherence tomography (EDI-OCT) imaging with the Spectralis HRA-OCT, with 2 vertical and 2 horizontal high-resolution scans through the fovea. An average measurement was taken from the 4 individual EDI-OCT scans. Measurements of axial length, central corneal thickness, intraocular pressure,

lens thickness, refractive error, height and weight were also taken. Birthweight and gestational age were prospectively collected from the participant's antenatal records. Data from the right eye were used in multivariate linear regression analysis to identify statistically significant associations.

Results: Subfoveal choroidal thickness was normally distributed, with a mean and standard deviation of $364 \pm 83 \mu\text{m}$ (Range: 162–660 μm). Spherical equivalent ($\beta = 16.3$, $p < 0.001$) and axial length ($\beta = -12.6$, $p = 0.02$) were significantly associated with subfoveal choroidal thickness.

Conclusions: This is the largest study investigating the relationship between SFCT and ocular biometry. It confirms that spherical equivalent and axial length are independent predictors of SFCT. This finding opens a new avenue of research into the aetiology of myopia.

100. OPHTHALMIC EVALUATION IN PATIENTS SUFFERING FROM TEMPOROMANDIBULAR JOINT DISORDER* (TMJD) A PILOT INTERDISCIPLINARY STUDY

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Purpose: To evaluate ocular involvement in facial pain and temporomandibular joint disorders (TMJD) in order to propose a new rationale for eradication or reduction of ocular or periocular pain.

Method: A group of 30 patients, aged between 30 and 73 years, suffering from **ocular or periocular pain** were included in this study. They initially presented with various degrees of neck pain, dorsalgia, dysesthesia, occasional vertigo and tinnitus. Being a multidisciplinary study, complete ophthalmological and orthoptic examinations were performed, including gnathologic and psychiatric evaluation and therapy. Any patients suffering from concomitant glaucoma, dry eye disorders, uveitis, sinusitis, cranial trauma, uncontrolled systemic hypertension and manifest scheletic disease of the cervical and dorsal tract were excluded.

Results: All patients showed various degrees of improvement in subjective symptoms and facial or cranio cervical pain after receiving gnathologic therapy. Ophthalmologic analysis showed anomalous response in **Contrast Sensibility test** (sinusoid examination) at low frequency low sensibility, with normal or excess degree in higher frequency. Gnathologic examination found anomalous changes in mandibular jaw occlusion and consequent compression/overload of the

temporomandibular joint, displacement of the articular condyle and meniscus, with pathological electromiographic temporomandibular aspects.

Conclusion: The study confirms that abnormal nervous stimulation may be caused by masticatory muscles and that any disturbance of the stimulus transmission could interfere with head and eye posture, proprioceptive perception and quality of vision, especially in the case of prolonged facial pain. Alterations in Sensibility Contrast examination confirms this hypothesis. Further studies, including statistical evaluation, will be required to confirm.

PAEDIATRIC OPHTHALMOLOGY

101. CEREBRAL VISUAL IMPAIRMENT: A RETROSPECTIVE REVIEW OF CHILDREN ATTENDING THE PAEDIATRIC LOW VISION CLINIC, QUEENSLAND, AUSTRALIA

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Purpose: To review a cohort of children with a primary diagnosis of cerebral visual impairment (CVI) attending the Paediatric Low Vision Clinic in Queensland.

Methods: A retrospective chart review of 168 children with a diagnosis of Cerebral Visual Impairment seen at the Paediatric Low Vision Clinic in Queensland between 1995 and 2011 was performed to determine the nature & frequency of co-morbidities, causes, associated clinical features & visual outcomes.

Results: The largest subgroup (29%) of children referred to the PLVC during the period of our study was comprised of children diagnosed as suffering from CVI. 111 (66%) of our cohort of 168 children were males. The most common co-morbidities were epilepsy (67%) & cerebral palsy (48%). A wide variety of causes was identified with hypoxic-ischaemic brain injury (39%), CNS infections (14%) and congenital CNS malformations (16%) occurring most frequently. While children with CVI may have a normal eye examination, optic atrophy (30%), strabismus (55%), nystagmus (22%) & field defects (19%) were common associated clinical findings in our group. 145 (86%) of children examined had a visual acuity of 6/60 or worse.

Conclusion: CVI is the commonest cause of decreased visual acuity in the paediatric age group in developed

countries. In the majority of children in our study, CVI was associated with systemic diseases particularly cerebral palsy & epilepsy. While reduced vision can occur in a setting of a normal ocular examination, optic atrophy, strabismus & sensory nystagmus are frequent clinical findings.

102. BILATERAL OPTIC NERVEHEAD COLOBOMA, MITTENDORF SPOT, ANOMALOUS CLOQUET'S CANAL AND SPONTANEOUSLY REAPPLIED RETINAL DETACHMENT

WITHDRAWN

103. IMPROVED VISUAL ACUITY IN PATIENTS WITH CONGENITAL NYSTAGMUS FOLLOWING ANDERSON-KESTENBAUM PROCEDURES

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Purpose: To assess the effectiveness of Anderson-Kestenbaum procedure on patients with congenital nystagmus, in improving both the size of the compensatory head posture and the level of visual acuity.

Method: This retrospective study conducted within a large paediatric clinical practice identified nine patients who were operated on during the period of 1996–2007, using the Anderson-Kestenbaum procedure. These patients, aged 4–17 were operated on by the same surgeon. Visual acuity was measured using age dependent linear or single optotypes and results were compared pre and post operatively. Observation of head posture by ophthalmologist, orthoptist and parent was recorded pre and post operatively.

Results: Visual acuities improved by at least 3 letters in 7 of the 9 patients post operatively. Visual acuity in one patient remained the same whilst another decreased by 5 letters. It should be noted that all patients had a visual acuity of 3/9 or better and therefore no patients had poor visual acuity pre or post operatively. All patients had some amount of residual head posture but all were observed to have significant improvement cosmetically. Head postures were not reversed in any instance.

Conclusions: Anderson-Kestenbaum procedure reduced the compensatory head posture and improved visual acuity in 78% of patients, with an average improvement in visual acuity of 6 letters. Patients showed minimal residual head posture. This study was

limited by its retrospective nature and small subject numbers.

104. WAARDENBURG SYNDROME: A RARE CASE WITH BILATERAL CONGENITAL CATARACT: AN UNUSUAL ENTITY

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Purpose: Waardenburg syndrome, a rare genetic disorder caused by mutations in the *PAX3* gene located on 2q35. It is characterized by deafness, defects in neural crest cells and pigmentation anomalies. Ocular manifestations commonly seen are dystopia canthorum, blue iris and retinal pigmentary changes.

We report a 2 year old boy who presented with characteristic features of Waardenburg syndrome with congenital cataract in both eyes. Although different subtypes and phenotypic variations are known, association with congenital cataract is unusual.

Method: 2 year old boy was brought by parents with white reflex in right eye. On general examination brilliant blue color of iris was striking. Child had centrally placed white forelock with patches of depigmentation over body.

On ocular evaluation, child had broad nasal bridge, shortened palpebral fissure (19 mm) with increased intercanthal distance (45 mm). Pupils briskly reacting to light. Lamellar cataract was present in both eyes more dense in right eye than left eye. Dilated fundus evaluation showed normal fundus in both eyes without any obvious pigmentary changes.

Child was evaluated for cataract surgery and underwent uneventful cataract extraction. Posterior capsulorhexis performed with implantation of hydrophobic foldable IOL in both eyes.

Results: Child had uneventful post operative period with rapid recovery.

Conclusion: Congenital cataract is not a common association noted with this disorder. Bilateral lamellar cataract involving visual axis was the presenting feature in our case and required surgical intervention. Early diagnosis with definitive intervention helped in visual rehabilitation of the patient.

105. CUTANEOUS AND OCULAR ASSESSMENT WITH THE USE OF FUNDUS FLUORESCEIN ANGIOGRAPHY IN INCONTINENTIA PIGMENTI

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Purpose: To describe the case of a mother and child, both of whom have incontinentia pigmenti with cutaneous and ocular manifestations.

Method: A three week old female Caucasian infant presented with an erythematous papulovesicular rash. Her 23 year old mother had a previous medical history of incontinentia pigmenti.

Results: The vesicular rash of the infant was evident along Blaschko lines. Ocular examination demonstrated peripheral retinal avascularisation with an abnormal retinal vascular pattern in both eyes. Compared to clinical examination, examination under anaesthesia with fundus fluorescein angiography revealed significantly greater areas of peripheral retinal non-perfusion. The infant was subsequently treated with peripheral confluent retinal photocoagulation. The mother had scarring alopecia and dermal hyperpigmented patches in a Blaschkoid distribution. Her vision in the right eye was reduced to hand movements due to amblyopia associated with significant myopia and a peripheral tractional retinal detachment. Both mother and infant had genetic testing which demonstrated deletion of the NEMO gene, confirming the diagnosis of incontinentia pigmenti.

Conclusion: Incontinentia pigmenti can be associated with significant visual loss. Early ocular screening involving the use of fundus fluorescein angiography when there is clinical suspicion of incontinentia pigmenti is vital in improving visual outcomes and ultimately, reducing patient morbidity.

REFRACTIVE SURGERY

106. AXIAL MYOPIC PROGRESSION FOLLOWING IMPLANTABLE COLLAMER LENS PLACEMENT

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Purpose: To report the clinical case of a 45-year old female who was found to have axial myopic progression following Implantable Collamer Lens placement.

Methods: A review of the clinical history and investigations of a single patient was undertaken to form a case report.

Results: A 45-year old female presented to her ophthalmologist complaining of worsening vision on the background of an Implantable Collamer Lens (ICL) placement ten years earlier. Prior to her ICL surgery she was highly myopic with a refractive error of -11.25D OD and -10.00D OS and axial lengths of 28.28 mm and 27.9 mm respectively. Post-operatively there were no surgical complications and she had a recorded visual acuity of 6/6 ou. When she was reviewed 10 years following her surgery she complained of deteriorating vision with an uncorrected visual acuity of 6/60 OD and 6/24 OS and it was noted that her axial lengths had increased to 30.25 mm and 29.22 mm respectively. She subsequently underwent bilateral Photorefractive Keratectomy producing a post-operative uncorrected visual acuity of 6/9 OD and 6/6 OS.

Conclusions: Firstly, age and recent myopic stability are insufficient criteria to conclude that a highly myopic patient is non-progressive. Secondly, the placement of an Implantable Collamer Lens and the induction of a stable, constant and focused foveal image are not sufficient to prevent further axial myopic progression in all individuals. Finally, patients should be advised that there is a small but real risk that myopia will continue to progress despite Implantable Collamer Lens placement.

107. SIX-MONTH POSTOPERATIVE VISUAL PERFORMANCE OF AMETROPIC PREBYOPIC PATIENTS IMPLANTED WITH A CORNEAL INLAY

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Purpose: To measure the visual performance of patients implanted with KAMRA corneal inlay in combination with simultaneous LASIK for the treatment of presbyopia.

Methods: Prospective study in 36 presbyopes treated with LASIK and implantation of a small aperture inlay. Inlays were implanted in the non-dominant eye and target refractive outcome was -0.75D. Spherical equivalent refraction (MRSE), monocular uncorrected distance (UCDVA) and uncorrected near (UCNVA) visual acuities of the inlay eye were measured at 1W (n = 34), 1M (n = 35), 3M (n = 31) & 6M (n = 9). DVA was measured in Snellen and NVA in J-notation. Mid-point refraction was used to assess refraction in inlay eyes. All results are shown in decimal acuity and are represented in mean $\pm$ stdev.

Results: Mean age of the patients was 51.1 ± 12.2 yr. MRSE was $0.2D \pm 1.4D$ preoperatively, $-0.6D \pm 0.8D$ at 1W, $-1.3D \pm 0.7D$ at 1M, $-1.3D \pm 0.6D$ at 3M and $-1.5D \pm 0.6D$ at 6M. The mean UCDVA was 0.7 ± 0.4

preoperatively, 0.6 ± 0.2 at 1W, 0.5 ± 0.2 at 1M, 0.5 ± 0.3 at 3M and 0.6 ± 0.3 at 6M. The mean UCNVA was 0.4 ± 0.3 preoperatively, 0.7 ± 0.2 at 1W, 0.9 ± 0.2 at 1M, 1.0 ± 0.2 at 3M and 0.9 ± 0.2 at 6M.

Conclusion: Inlay patients achieved significant improvement in near vision (J6 preoperatively to J1, J2 or J3 postoperatively). Mean UCDVA varied between 20/34 and 20/39 which is consistent with a postoperative MRSE that is more myopic than the intended $-0.75D$ target.

108. THE EFFECT OF SMOKING ON EXCIMER LASER REFRACTIVE OUTCOME

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Purpose: The wound healing response is known to be affected by several environmental factors, including cigarette smoking. This study aims to determine the effect of smoking on refractive outcomes in patients undergoing excimer laser refractive corneal ablation.

Methods: In this case control study, 168 participants were included, consisting of 90 having undergone LASIK and 78 PRK. Of these participants 128 were myopic and 40 hyperopic. 84 participants positive for smoking status were paired with an age, spherical equivalent and cylinder power matched control. The refractive outcomes at 3 months follow up were compared. Excimer laser ablation was carried out with a LADARVision 4000 excimer laser in all cases.

Results: Smoking status did not significantly affect the refractive outcomes, including in sub-analysis for refractive procedure, (LASIK or PRK) myopia or hyperopia status. Including all refractive errors, mean achieved sphere equivalent change ($2.96D$, SD 1.93 for non-smokers, $3.03D$, SD 2.02 , for smokers, $p = 0.834$) and mean surgically induced astigmatism ($0.77D$, SD 0.60 for non-smokers, $0.91D$, SD 0.76 for smokers, $p = 0.226$) were not significantly different.

Conclusion: This is the first study to report the effect of smoking on refractive outcomes in those undergoing LASIK. No association was found between smoking status and refractive outcome in participants who had undergone LASIK and PRK.

109. MEASURED VISUAL OUTCOMES OF PATIENTS UNDERGOING LASIK TREATMENT FOR LOW TO MODERATE-GRADE HYPERMETROPIA VERSUS LOW TO HIGH-GRADE MYOPIA

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Purpose: The effectiveness of myopic LASIK (M-LASIK) is well established, however hypermetropic LASIK (H-LASIK) is less so. H-LASIK presents challenges; pre-operative measurement is complicated by the accommodative reflex, patient presentation is often linked to the onset of presbyopia and it is more difficult to steepen a cornea than to flatten it with ablation. This has led to the sentiment that H-LASIK is less rewarding. The purpose of this retrospective analysis is to evaluate the differences in outcomes.

Method: Data was collected from a retrospective review of patients with low (0 to $+2.75$) or moderate ($>+2.75$ to $+5.75$) grade hypermetropia, or low (0 to -2.75), moderate (<-2.75 to -5.75) or high (<-5.75) grade myopia. Each group underwent LASIK with outcomes measured after 3 months. Data from Hobart Eye Surgeons covered 2009 to 2012. 49 hypermetropic and 57 myopic eyes were included.

Results: Post-operative UCVA was $\geq 6/6$ in 61% of hypermetropic and 88% of myopic eyes and $\geq 6/9$ in 90% and 100% respectively. When plotting achieved versus attempted refraction, the myopic group showed a smoother line of best fit, within which the moderate subgroup was best. Evaluating safety, all groups had one eye lose 1 line of BCVA.

Conclusion: H-LASIK presents challenges. Despite sample size, it appears that low-grade H-LASIK corrections are comparable to M-LASIK. For moderate-grade, it is difficult to accomplish results comparable to myopic. H-LASIK treatment requires development, specifically for $> +2.75$ to $+5.75$ refractions.

RETINA

110. RANIBIZUMAB FOR NEOVASCULAR AGE-RELATED MACULAR DEGENERATION IN A REAL WORLD CLINICAL SETTING

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Purpose: To evaluate safety and efficacy of treatment with intravitreal ranibizumab for neovascular age-related macular degeneration in a clinical setting.

Methods: A nonrandomised, single-surgeon, retrospective, interventional cases series of 522 patients (626 eyes) with neovascular age-related macular degeneration treated with intravitreal ranibizumab between September 2006 and May 2011.

Results: In total 5,983 injections were administered at mean interval of 6.9 weeks and mean follow-up of 15.7 months. Overall, BCVA improved a mean of 2.1 letters from baseline to endpoint of study period. Mean BCVA improvement at 1-year follow-up was 11 letters. Nearly 50% of patients maintained driving standard and 1/3 were living alone independently. Smaller lesion size (<1 disc area) was associated with significant improvement ($p = 0.0039$).

Conclusion: Cautious inject and extend (capped extend) methods with less frequent comprehensive examinations in selected patients can result in good visual acuity outcomes in a clinical setting but does not match monthly clinical trial results. Under treatment (evidenced by disease recurrence) remains an important obstacle to obtaining best outcomes as we seek to individualise therapy for patients and reduce treatment burden. Small lesion size and better baseline vision are proxy markers for time to treatment and increased education and resources are needed to improve time to treatment initiation.

111. A FELINE MODEL FOR CHRONIC ACTIVE STIMULATION WITH A SUPRACHOROIDAL ELECTRODE ARRAY

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Purpose: To develop a surgical approach for implanting a suprachoroidal electrode array for chronically stimulating the retina in a feline model and assessing function and safety.

Methods: Normally sighted adult cats were implanted with a suprachoroidal electrode array via an approach developed from initial acute surgeries. The devices consist of a suprachoroidal electrode array, with a percutaneous lead, two dural recording electrodes, a return electrode and fixation arm. Following recovery the cats were fitted with wearable multichannel stimulators.

Ocular health was assessed by regular fundus photography, OCT and ERG during the study. Behavioural and cortical responses were measured regularly. At the end of the study multichannel "Utah" arrays were

used to confirm thresholds and verify the origin of the responses was primary visual cortex. Histopathological assessments of the eyes were performed.

Results: The implant surgeries took 4–5 hours however they were successful with minimal complications. The electrodes and lead assemblies remained viable throughout the period of stimulation. Behavioural responses were able to be elicited reliably and cortical responses were elicited in most animals. The array position within the eye was stable and the devices were tolerated well. Evaluation of the eye with fundus photography, OCT and ERG did not reveal any evidence of retinal damage due to stimulation, nor did histopathology.

Conclusion: An animal model for assessing safety and efficacy of chronic electrical stimulation of the retina by a suprachoroidal array has been developed. The one-piece array and lead wire described is robust and reliable over a 3 month period.

112. BLIND FELINE MODEL FOR RETINAL PROSTHESIS

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Purpose: Retinal prosthesis development relies on the testing of prosthesis safety and efficacy in animal models of retinal degeneration. Large-eyed models of retinal degeneration are often expensive to house and slow to develop. The objective of this research is to develop a rapid, cheap and effective unilateral blind model suitable for an implanted retinal prosthesis.

Method: Acute loss of photoreceptors was induced in 10 cats via intravitreal injection of 1M adenosine triphosphate (ATP) solution. Six cats were terminated 30 hours post injection and the retinas removed, sectioned and labeled for cell death using TUNEL. In four cats, retinal structure and function was monitored for 6 weeks post injection using electroretinograms (ERG), optical coherence tomography (OCT), fundus photography and immunohistochemistry.

Results: Cell death selective to the outer nuclear layer (ONL) was visible in the 30 hour cohort (1092 ± 388 TUNEL positive cell nuclei per mm² retina). Two weeks post injection the outer nuclear layer had significantly thinned and there was a significant reduction in ERG a-wave amplitude. The injected retina displayed remodelling-like events including bipolar cell changes and formation of a glial scar.

Conclusion: These data demonstrate that it is possible to produce photoreceptor death in cat retina via

intravitreal injection of ATP. Once the reliability of the model is improved, acute electrophysiological experiments will be used to determine the effect of degenerate retina on prosthesis stimulation efficacy. This research will contribute to the development of an effective bionic eye for implantation in blind patients.

113. FEASIBILITY OF REMOVING SUPRACHOROIDAL ELECTRODE ARRAYS IN AN ANIMAL MODEL

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Purpose: The current technology of retinal visual prostheses, or “bionic eyes”, has limited number of electrodes and hence restricted resolution capabilities. One of the main challenges in the development of these devices is to manufacture a device which can be safely explanted from the eye if new technology becomes available, in order to “upgrade” the device. The aim of this study was to examine the feasibility of removing suprachoroidal electrode arrays in a chronic animal model, by assessing any changes in the retinal structure post explantation.

Methods: Suprachoroidal electrode arrays were implanted in six adult animals, and then explanted one month later. Retinal examination, consisting of fundus photography and optical coherence tomography (OCT), was completed pre-surgery, post-implant, post-explant and two months after explantation. We compared fundus photographs at each time point for a qualitative analysis of change. Quantitatively, we compared retinal thickness on the OCT at various locations surrounding the electrode array.

Results: The array explantation procedure was performed successfully in all subjects. OCT showed localised tapetal damage near the implant's tip in all subjects two weeks post-implantation. There was no significant change in the thickness of the choroid, photoreceptors or inner retina adjacent to the array compared to the contralateral unimplanted control eye.

Conclusion: Explantation was well tolerated with minimal damage to the suprachoroidal space, retina and choroid. These findings suggest that suprachoroidal retinal prostheses may be easier to remove and replace than epiretinal devices.

114. INFLUENCE OF 23-GAUGE VITRECTOMY AND MEMBRANE PEELING ON RETINAL OXYGEN SATURATION

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Purpose: Proliferation of fibrocellular membranes on the internal surface of the retina, at the macula, in previously healthy eyes is known as idiopathic epiretinal membrane (ERM). ERMs cause structural retinal changes including macular oedema. Combined vitrectomy and membrane peeling improves anatomical and functional outcomes in these patients; however, little is known about changes in retinal haemodynamics after vitrectomy. In the present study we examined the effect of vitrectomy on retinal haemodynamics and morphology in patients with idiopathic ERM.

Method: Idiopathic ERM patients with macula oedema were included. The main outcome measure was the change in retinal vascular oxygenation. The IMEDOS Retinal Vessel analyser and a custom built retinal laser Doppler velocimeter measured retinal vessel oxygenation and blood flow respectively. Cirrus OCT and ETDRS visual acuity were performed at each visit. Patients were followed up for at least 6 months.

Results: Retinal venous oxygen saturation reduced significantly postoperatively. There was no significant change in retinal arterial oxygen saturation. There was a significant positive correlation between venous oxygen saturation and retinal thickness and a negative correlation between venous oxygen saturation and visual acuity.

Conclusion: The decrease in venous oxygen saturation postoperatively suggests greater retinal oxygen perfusion after vitrectomy. This improvement in retinal perfusion was associated with a decrease in central macular thickness, and an improvement in visual acuity, suggesting that retinal perfusion is an important factor in macula oedema caused by idiopathic epiretinal membrane.

115. PATIENT EXPERIENCES OF ONGOING INTRAVITREAL RANIBIZUMAB INJECTIONS: RATES OF NON-SERIOUS OCULAR ADVERSE EVENTS OVER 4 YEARS

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Purpose: To investigate rates of non-serious ocular adverse events (NSOAE) associated with intravitreal injections and factors affecting this rate.

Method: Retrospective chart review of 88 injection naïve patients (1987 total injections) receiving ranibizumab (Lucentis) injections performed by a single operator commencing after July 2007 with four years follow up. Number of injections per year and number of non-serious ocular adverse events, including significant pain, epiphora, foreign body sensation and conjunctivitis were recorded. To account for improved operator technique, 4-year ratios were compared with rates over two years in 72 patients (1028 total injections) who commenced treatment between June 2009 and May 2010, after the operator had gained further experience.

Results: Most patients reported one or less non-serious ocular adverse events over a four year period, with a subset of patients (16%) reporting two or more side effects. Average number of injections per year decreased over four years, as did average number of adverse events. There was a lower rate of adverse events per injection in later years (0.05 average adverse events per injection year one, 0.01 year four, p -value 0.001). No significant increase in the rate of adverse events was seen between the initial four year group and the later two year group, and baseline characteristics between the groups were similar.

Conclusion: The rate of non-serious ocular adverse events correlates with number of years of treatment. The rate of adverse events per injection decreases with time since first injection, implying that patient experiences of injections may change over time

116. THE NEVER-ENDING TRANSIT OF VENUS: A CASE OF SOLAR RETINOPATHY

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Purpose: Solar retinopathy is an uncommon clinical entity in the developed world. Despite being widely

understood within the ophthalmological community, cases continue to surface following publicised solar astronomical events such as the recent Transit of Venus. A case of solar retinopathy in an 11 year-old school-girl is presented. While this case fits well in an historical context of sun-gazing practices, no cases should occur in any Australian school-children who are surrounded by educated adults. Measures to prevent future occurrences in this age group are discussed.

Method: An 11 year-old girl presented to the emergency department with metamorphopsia and a central scotoma in her right eye following one-eye gazing into the sun for an accumulative period of one hour. A full ocular examination was performed along with spectral-domain OCT and Goldmann visual fields.

Results: The examination and investigation findings were consistent with solar retinopathy. This case mirrors sun-induced blindness and eye damage in communities that practiced regular sun-gazing and in notable individuals in history.

Conclusion: Solar retinopathy is a rare but important clinical phenomenon with the potential of causing permanent damage to vision. The educational importance of solar astronomical events cannot be over-stated, however, much more can be done to protect children's eyes. Increased public awareness by the ophthalmological community together with increased accessibility of suitable protective viewing methods prior to astronomical events should prevent this condition.

117. IS ASPIRIN INTAKE ASSOCIATED WITH EARLY AGE-RELATED MACULAR DEGENERATION?

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Purpose: To examine the relationship between aspirin intake and early age-related macular degeneration (AMD) among an Asian population.

Methods: A population-based, cross-sectional study of 3,207 ethnic Indians aged 40 years or older residing in Singapore. AMD signs were graded from retinal images following the modified Wisconsin grading system. Information on aspirin intake was obtained from a standardized questionnaire.

Results: The prevalence of early AMD was 5.6%. Aspirin use was reported by 11.4% of participants. Early AMD signs were present among 10.9% of aspirin users and 4.9% of non-aspirin users ($p < 0.001$). After adjusting for potential confounders including age,

smoking and previous cataract surgery, aspirin use was associated with early AMD (odds ratio [OR] 1.50; 95% confidence interval [CI]: 1.01, 2.22). The association was stronger and significant in participants with cardiovascular disease (adjusted OR 2.64, 95% CI: 1.31, 5.36) but not seen in those without a history of cardiovascular disease (OR 0.73; 95% CI: 0.36, 1.51) (interaction term, $p = 0.011$).

Conclusions: Aspirin intake was associated with early AMD in this sample of Asian Indians with a history of cardiovascular disease. Further evaluation of the longitudinal relationship between aspirin use and progression of early to late AMD is warranted.

118. THE AVASTIN ADVANTAGE IN THE TOP END – A RETROSPECTIVE STUDY ON THE USE OF AVASTIN IN THE TREATMENT OF MACULAR OEDEMA IN RETINAL VEIN OCCLUSION

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Purpose: The use of Avastin as a treatment for macular oedema secondary to central retinal vein occlusion (CRVO) or branch retinal vein occlusion (BRVO) have been encouraging. This study looks at the results of the use of Avastin for BRVO and CRVO in a tertiary eye unit.

Methods: A retrospective review of 21 eyes was done with a mean follow-up of 12 months, seven eyes with BRVO and 14 eyes with CRVO. We compared changes in visual acuity (VA) and central retinal thickness (CRT) and looked at the average number of injections used for the treatment of macular oedema in this case series.

Results: Improvement in VA was seen after a single dose of Avastin with a median LogMAR of 0.2 (two lines of letters) over a mean follow-up of 31 days. Comparison with the final VA recorded showed a median improvement of 0.5 (five lines of letters) over a mean follow-up of 398 days. 19 eyes (90%) showed improvement with treatment on final readings. Median reduction in recorded CRT was 150 μ m post first dose (mean value 32 days) and 273 μ m at final follow-up (mean value 379 days). Throughout the follow-up period 84% showed a reduction in CRT values.

The average Avastin injections required was three at six months (21 eyes) and five at twelve months (14 eyes).

Conclusions: Within the constraints of a retrospective analysis, these results support recent studies that Avastin as a treatment for macular oedema in CRVO and BRVO is an effective agent to improve VA and reduce macular oedema.

119. 4 YEAR REAL-WORLD OUTCOMES OF RANIBIZUMAB TREATMENT FOR NEOVASCULAR AGE RELATED MACULAR DEGENERATION

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Purpose: Intravitreal Ranibizumab treatment remains the current standard of care for exudative age related macular degeneration (AMD) based on randomised clinical trials. Limited long term data exists addressing treatment in a real-world setting, where patients are often managed by strategies to reduce treatment burden.

Methods: Retrospective case series of patients commenced on Ranibizumab for exudative ARMD between July 2007 and March 2008, all with 4 years follow-up at a single Australian retinal clinic. Snellen Visual acuity (VA) was converted to VA scale, complications, number of injections and the development of neovascularisation in the fellow eye were recorded.

Results: Baseline mean VA scale for the 216 patients was 67 letters. Peak VA was achieved at month 3 (71 letters, $p < 0.0001$) and maintained until month 18 (70 letters, $p = 0.006$.) VA declined slightly to 66 letters at month 36 ($p = 0.30$) and 65 letters at month 48 ($p = 0.06$). Rate of ocular adverse events was 4.3% (minor) and 0.34% (serious) of total injections. 34% of patients developed sub-foveal neovascularisation in the fellow eye within 4 years, 56% of which were asymptomatic, detected with screening optical coherence tomography (OCT), and diagnosed by fluorescein angiography.

Conclusion: Ranibizumab was effective at improving and maintaining visual acuity over the first 2 years after which VA declined slightly until year 4. The risk of developing neovascularisation in fellow eyes is significant, warranting rigorous follow up.

120. MANAGEMENT OF FINGOLIMOD ASSOCIATED MACULAR EDEMA

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Purpose: Cystoid macular edema is an uncommon complication in patients with multiple sclerosis (MS) on fingolimod with symptoms resolving upon cessation of fingolimod. This is the first report describing

management of fingolimod associated macular edema (FAME) in a patient with MS without cessation of fingolimod.

Method: Case report.

Results: A 67 year old woman with MS presented with decreased vision in her right eye after commencing on fingolimod six months ago. Her best corrected visual acuity (BCVA) was 6/7.5 and 6/6 respectively in her right and left eye. Fundoscopy and optical coherence tomography (OCT) showed cystic changes affecting her right macula. Her fluorescein angiogram showed late phase leakage of dye in the subfoveal region. A provisional diagnosis of FAME was made. Since the patient wished to remain on fingolimod, she was commenced on topical 0.5% ketorolac QID and 0.1% dexamethasone QID for her right eye. After one month, BCVA remained at 6/7.5 in her right eye but worsened to 6/9 in her left eye, corresponding to OCT findings of resolving macular edema in her right eye with progression of macular edema in her untreated left eye. At this stage, topical ketorolac and dexamethasone were prescribed for both eyes with gradual improvement of BCVA to 6/7.5 bilaterally at three weeks and 6/6 bilaterally at three months.

Conclusion: Our observations suggest that inflammation may contribute to pathogenesis of FAME. Topical anti-inflammatory drugs may be useful in management of patients with FAME when withdrawal of Fingolimod is not desired.

121. CLINICAL UTILITY OF MULTIFOCAL PUPILLOGRAPHIC OBJECTIVE PERIMETRY IN TYPE 1 DIABETES

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Purpose: To evaluate the sensitivity and specificity of five variants of multifocal pupillographic objective perimetry (mfPOP) for discriminating controls from patients with type 1 diabetes (T1D) presenting with nil to non-proliferative diabetic retinopathy (NPDR).

Methods: 25 T1D patients (37.9 ± 2.44 years, 12 females, mean duration 22.5 ± 10.1 years) and 23 normal (36.0 ± 2.58 years, 12 females) subjects were recruited. Pupillary contraction amplitudes and times-to-peak responses to red/green, yellow and blue dichoptic stimuli with 44 test regions/eye extending,

to either ± 15 or ± 30. Disease severity was based on either ETDRS scores from fundus photos; or scores based on factors from correlations between 18 patient variables, which were mainly based on blood analyses, but included peripheral neuropathy, and advanced glycation end-products.

Results: Factor scores based mainly on renal function, HbA1c and BMI were the best indicator of diabetic damage as measured by mfPOP. In eyes with no or mild NPDR, red/green mfPOP stimuli extending to ±30 achieved the highest ROC area under the curve (AUC) of 0.95 ± 0.04 (mean ± SE) for delay deviations and 0.91 ± 0.05 when amplitude deviations were considered. In moderate NPDR eyes diagnostic accuracy improved to 1.00 ± 0.00 for blue and yellow protocols.

Conclusion: The results suggest that red/green stimuli presented within the central 60° of fixation achieved good diagnostic power to determine diabetes status in eyes with minimal to no retinopathy. MfPOP may be a useful clinical tool in evaluating and monitoring eyes with early NPDR.

122. VISUAL FUNCTION IN EARLY AMD

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Purpose: Retinal function was studied across a spectrum of early Age-related Macular Degeneration (AMD).

Methods: In this cross-sectional study, uniocular visual function was assessed in 293 AMD participants having different levels of AMD, classified into 12 groups based upon the International Classification and Grading System; visual acuity [VA] better than 6/18) and 64 age-similar individuals with normal fundi (VA, 6/6 and better). Visual function tests included achromatic 14 Hz flicker (F14 Hz), isoluminant Blue color (BCT) and two adaptation measures (Photo-Stress Cone Recovery [PSCR] and Dark Adaptation Rod Recovery [DARR]). Functional deficiency was evaluated in each of the AMD groups and the groups were ranked accordingly.

Results: Flicker and Color thresholds (F14 Hz and BCT) were associated with AMD grade, declining gradually as the severity of fundus change increased (linear

trend, $p < 0.001$). In contrast, adaptation tests (PSCR and DARR) worsened rapidly across the early AMD groups (linear trend, $p < 0.001$) and stabilizing in more advanced forms of early AMD (linear trend, $p > 0.05$).

Conclusion: Adaptation measurements are sensitive to detect functional changes at the very early stages of the disease and could therefore provide a useful tool for AMD screening before clinical signs are visible. Blue Color and 14 Hz Flicker thresholds are effective in identifying a spectrum of AMD changes and may prove useful for tracking progression of the disease along the entire spectrum.

123. CHOROIDAL THICKNESS IN MYOPIC PATIENTS WITH CENTRAL SEROUS CHORIORETINOPATHY

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Purpose: Central serous chorioretinopathy (CSC) is uncommon in myopes. While increased choroidal thickness is a hallmark of CSC in emmetropes, it has not been described in myopes, who are known to have thinner choroids. The purpose of this study is to investigate choroidal thickness in these patients.

Methods: Imaging analysis of patients diagnosed with CSC and moderate to high myopia. The diagnosis of CSC was confirmed by clinical examination, fluorescein and indocyanine green angiography, fundus autofluorescence imaging and enhanced depth imaging optical coherence tomography EDI-OCT.

Results: Seven eyes of 7 patients (5 male; 2 female) with a mean age $\pm$ SD of 41.9 ± 10.8 years (range, 29 to 53 years) were included. Mean refractive error $\pm$ SD was -5.8 ± 2.6 diopters (range, -3 to -9 diopters). Mean Snellen visual acuity was 6/9 (range, 6/6 to 6/18). All patients had subretinal fluid and/or pigment epithelial detachment on EDI-OCT. Mean subfoveal choroidal thickness $\pm$ SD measured 317.8 ± 102.9 μ m (range, 215 to > 500 μ m). Subfoveal choroidal thickness in all but 1 patient in our study would have been considered "normal" in emmetropes (mean 287 μ m, SD 76 μ m). Age, refractive error and visual acuity did not correlate with choroidal thickness in our study group.

Conclusion: Myopic eyes with CSC have an increased choroidal thickness relative to normal eyes with similar degrees of myopia. When evaluating myopic patients with suspected CSC, awareness of normal EDI-OCT choroidal thickness values in eyes with similar degrees of myopia is important for the correct interpretation of results.

124. DIRECT INHIBITION PLAYS AN IMPORTANT ROLE IN SURROUND INHIBITION AND DIRECTION SELECTIVITY OF RETINAL GANGLION CELLS

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Purpose: Neuronal responses depend on the balance of excitation and inhibition. Using whole-mount mouse retinae, we set out to determine whether different levels of inhibition modulate the spatial-tuning and direction selectivity (DS) properties of ganglion cells.

Method: Using dynamic-clamp techniques, light-evoked excitatory and inhibitory synaptic conductances recorded in response to different stimulus sizes were injected into ganglion cells. The ratios of excitatory and inhibitory conductances injected were varied between 1:0 and 1:2. To elucidate the origin of DS, we injected digitally synthesised directionally-tuned conductances (modelled on physiological responses of ON-OFF DS ganglion cells). We also tested whether or not DS can be achieved by modifying the amplitude and timing of directional-tuned excitatory and inhibitory conductances.

Results: Injection of light-evoked excitatory and inhibitory conductances recorded for different stimulus sizes at equal strength (ratio 1:1), resulted in area-response functions typical of physiological responses displaying surround inhibition. For the other ratios, surround inhibition increased as the level of the injected inhibitory conductance increased ($n = 14$ cells). Neuronal responses to excitatory and inhibitory conductances tuned to preferred direction produced strong responses, which were increased in the absence of inhibition. Injection of null direction conductances produced no/little response but provoked a much larger response after inhibition was removed ($n = 32$). Injection of directionally-tuned conductances showed that DS is reduced when inhibition is delayed or removed ($n = 21$).

Conclusion: The sharpness of spatial-tuning curves of ganglion cells was increased by direct inhibition. Directional-tuning of inhibitory conductance is critical for direction selectivity.

125. SUTURELESS LARGE INCISION WOUND IN VITREORETINAL SURGERY

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Purpose: To describe the sutureless surgical techniques in vitrectomy where a large wound is required.

Method: Long scleral tunnel wound employed in sutureless ECCE was adapted for vitreoretinal cases, for which a large wound was required. After a limited superior peritomy, a 6 mm scleral tunnel extending into the clear cornea was constructed. A conventional small-gauge sutureless vitrectomy was performed. The scleral tunnel wound was extended into the anterior chamber to allow large-aperture access for removal of retained lens, intraocular lens exchange or secondary intraocular lens implantation. Both the scleral tunnel and sclerotomy ports were left unsutured.

Results: Long scleral tunnel wounds are self-sealing despite their large size, and provide a greater stability and safety during vitrectomy procedures that require a large incision. This approach shortens surgical time and causes less suture-related problems such as astigmatism, postoperative inflammation and discomfort.

Conclusion: Long scleral tunnel wounds offer distinct advantages to conventional sutured scleral or corneal wounds in vitrectomy procedures that require large incisions.

126. CLINICAL OUTCOMES AND COMPLICATIONS FROM OPEN GLOBE INJURY REPAIR

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Purpose: To determine the baseline features, aspects of primary management and clinical outcomes of open globe injuries.

Method: A retrospective chart review of open eye injuries. Cases were categorized according to the Birmingham classification of open eye injury. Baseline features, primary management, and key outcome measures were recorded.

Results: A total of 267 eyes (263 patients) from 1 January 2009 to 31 December 2011 were included, of which 82.8% were men with a mean age of 44.8 years (4–97). There were 183 (68.5%) penetrating, 83 (31.1%) globe ruptures and two (0.7%) perforating eye injuries. Forty-three (16.1%) eyes had intraocular foreign body. Average follow up was 6.9 months. Vitrectomy was required in 94 (35.2%) cases. Final visual acuity (VA) was $\geq 6/12$ in 123 (46.1%) eyes. Factors contributing to final VA $< 6/60$ were: male (OR 3.75, [95%CI:1.18–11.9]), poor presenting VA (OR 16.0, [95%CI:4.81–53.1]), globe rupture (OR 19.4, [95%CI:1.40–267]), retinal detachment (OR

3.90, [95%CI:1.16–13.10]) and ≥ 50 years of age (OR 2.45, [95%CI:1.05–5.74]).

Post-repair wound leak occurred in 44 eyes (16.5%). Factors contributing to wound leak were stellate shaped wound (OR 3.28, [95%CI:1.39–7.73]) and delayed closure of ≥ 48 hr (OR 2.70, 95%CI: 1.01–7.27).

Ten eyes developed endophthalmitis (3.7%). Factors associated with endophthalmitis were delayed presentation ≥ 48 hr (OR 8.50, [95%CI:1.69–42.7]) and breach of lens capsule (OR 9.61, [95%CI:1.47–63.0]). Enucleation was performed in 14 (5.2%) eyes of which 4 had endophthalmitis.

Conclusion: Following open globe injuries, a number of features are associated with the need for re-suturing, endophthalmitis and final VA outcomes. Understanding these factors may lead to measures to reduce the morbidity from such injuries.

127. AMYL NITRITE INHALATION CAUSING TEMPORARY BILATERAL LOSS OF VISION

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Purpose: We report the first case of amyl nitrite-induced vision loss in Australasia.

Methods: Case report and review of literature.

Results: A 38-year-old male presented to hospital with sudden-onset bilateral 'blurred vision' involving all visual fields. Four days prior to presentation, he had inhaled amyl nitrite 'poppers'. Visual acuity was 6/24 on the right and 6/18 on the left. The remainder of the ophthalmic examination was normal. Optical coherence tomography (OCT), computed tomography (CT) and magnetic resonance imaging (MRI) of the brain also revealed no abnormalities. However, OCT changes in amyl nitrite toxicity are thought to be associated with longstanding exposure and may not be evident with minute foveal damage. The patient completely recovered his vision over the following 24 hours. Despite vision loss occurring four days after exposure to amyl nitrite, we note similar temporal relationships in other reports.

Conclusion: Recreational amyl nitrite abuse is increasingly common around the world, including approximately 60% of Australia's male homosexual demographic. Our patient's recent commencement and infrequent use of amyl nitrite (fortnightly for 3 months) indicates that even occasional use can cause acute loss of vision. To our knowledge, this is only the 7th case of amyl nitrite-induced vision loss in the world and the first reported in Australasia.

128. BLOOD PRESSURE AND VISUAL ACUITY CHANGES IN AUSTRALIAN PATIENTS RECEIVING INTRAVITREAL RANIBIZUMAB INJECTIONS FOR ARMD

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Background: Intravitreal Ranibizumab (ivR) injections have been shown to preserve and improve the visual acuity of patients with neovascular age related macular degeneration (ARMD). Systemic antiVEGF therapy induces hypertensive vascular changes and cause arterial thromboembolic events (ATEs). The CATT trial confirmed the efficacy of Ranibizumab and gave further insight into its systemic and ocular safety profile. We present prospectively collected data on blood pressure in patients receiving ivR.

Method: 13-month data of 71 patients (75 eyes) with ARMD treated with ivR injections, on as-needed regimen that had blood pressure measured at every clinic visit. Outcome measures also included the difference in the visual acuity (VA) and CMT at the time of diagnosis compared to that of most recent review. The trends in VA, CMT and BP, along with adverse events, over the 13-month period have also been statistically analyzed.

Results: Mean and median systolic/diastolic blood pressure remained stable during the study period. Rate of serious systemic adverse events was very low. No significant difference in VA, assessed in logMAR equivalents, (t-test, p value = 0.83) and in CMT (t-test, p value = 0.34), had been observed from the time of diagnosis to the latest recording. Mean and median VA remained stable, with average increase of 15 μ m in CMT, over the 13 months.

Conclusion: IvR injections, on as required basis, stabilize the visual acuity and CMT of patients with ARMD. No significant increase in BP has been observed.

129. PLASMA LEVELS OF AMYLOID-BETA IN PATIENTS WITH AGE RELATED MACULAR DEGENERATION

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Purpose: To investigate plasma levels of Amyloid-Beta (A β) in patients with various stages of AMD compared to that of age-matched controls.

Method: Plasma samples were obtained from controls with no AMD, patients with early (drusen only) AMD, geographic atrophy (GA), or neovascular AMD (CNV). Levels of A β Isotypes were quantified from collected plasma samples using immune-precipitation LC/MS/MS. Analysis of covariance (ANCOVA) was conducted to compare each amyloid isotype across the AMD groups and in comparison to the control group while adjusting for age and sex. Statistical significance was established at p-value < 0.05. Receiver operating characteristics plots were generated for the strongest predictor variables.

Results: The levels of A β isotypes were found to increase with increasing severity of AMD stage (p values ranged from 0.0521 to 0.0016) (ANCOVA). When adjusted for multiple comparisons' analysis, Plasma A β 1–42 levels were the most significantly associated with late AMD stages. Consistently with the ANCOVA results for A β isotypes, the ROC curve showed a moderate prediction of AMD vs. controls using the A β 1–42 isotype.

Conclusions: Plasma A β 1–42 may have utility as a systemic biomarker for advanced AMD.

130. RETINAL FUNCTION OF A FAMILY WITH RETICULAR PSEUDODRUSEN

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Purpose: Reticular pseudodrusen (RPD) is considered a significant risk factor for the development of late age-related macular degeneration (AMD). We investigated retinal function of a family with RPD as it has been shown that functional markers are important for the prediction of late AMD.

Methods: A 62 year-old female with RPD and her 2 daughters (aged 41 and 44 years) were participated in this functional study. In addition to the routine clinical eye examination including best corrected visual acuity (BCVA) assessment and dilated fundus examination, infrared reflectance (IR) imaging, fundus autofluorescence imaging and optical coherence tomography were also performed to confirm the presence of RPD. The retinal function was assessed using multifocal electroretinography (mfERG).

Results: BCVA for the RPD index case was 6/30 OD and 6/6 OS. RPD was not detectable on any imaging modalities in either of the daughters. Their BCVA was 6/5 in each eye. Retinal function of the index case was severely abnormal with markedly reduced in mfERG response amplitude and significantly increased in the implicit time. Despite both daughters having a normal

fundus appearance their mfERG implicit time was significantly increased in both eyes. However, the mfERG response amplitude was normal.

Conclusion: Offspring of a RPD index case could have a significant delay in retinal response despite the retina appearing normal. The significance of the increase in mfERG implicit time is unclear and warrants further study.

131. QUIET PLEASE! EFFECT OF DISTRACTION ON SIMULATED POSTERIOR SEGMENT SURGICAL PERFORMANCE

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Purpose: To determine the effect of distraction on posterior segment surgical performance using a virtual reality simulator in expert and novice surgeons.

Setting: Tennent Institute of Ophthalmology, Gartnavel General Hospital Glasgow, Scotland, UK.

Design: This is a randomised cross-over trial involving three levels of operator under two randomly assigned operating conditions (distracted and not distracted).

Methods: Subjects were given 6 minutes to read an unpublished research paper. Randomised into 2 groups and asked to complete a standardised vitreoretinal simulated surgical task $\times 2$ (+ 1 practice run) on a virtual reality simulator. Group 1 – 'undistracted 1st attempt' were allowed a maximum of 3 minutes to complete the surgical task undistracted. Group 2 – 'distracted first attempt' were asked a series of 6 randomised questions on the research paper whilst completing the surgical task in a maximum of 3 minutes. Each subject then performed the alternate scenario in their 2nd attempt. All participants were then asked 6 randomised questions on the research paper whilst not operating.

Results: *Overall score:* There was no evidence of a difference in the overall scores between the two groups – Wilcoxon $p = 0.113$, estimated median difference (undistracted – distracted) was 5 with a 95% CI (–0.5, 14.0).

Time: The time taken by the distracted group was significantly greater than the undistracted group – paired t-test $p = 0.028$, 95% CI (–26.03, –1.67) – for undistracted – distracted.

Odometer: There was no evidence of a difference in the odometer values between the two groups – paired t-test $p = 0.127$, 95% CI (–44.6, 6.0) for undistracted – distracted.

Instrument slipped: There was some evidence to suggest this was higher in the distracted group – 95% confidence interval for undistracted – distracted (–7.35,

0.65) – but this difference did not reach statistical significance at the 5% level – paired t-test $p = 0.096$.

Cognitive task score: The distraction of surgical simulation had no statistically significant effect on cognitive task score.

Conclusion: Distraction significantly increases the time taken to perform a simulated VR task, irrespective of the level of experience of the surgeon. This is likely attributable to slower surgery rather than inefficient surgery as there is no difference in the odometer score. We suggest that best practice excludes any communication with the surgeon not immediately related to the case in hand.

132. CHARACTERISTICS OF PATIENTS WHO DROPOUT FROM RANIBIZUMAB THERAPY

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Purpose: Anti-vascular endothelial growth factor (VEG-F) therapy is a first line treatment for age-related macular degeneration but frequent visits and injections can be a burden for patients. The purpose of this study is the estimate the remission rate and tolerability of Ranibizumab therapy in a clinical setting.

Method: We investigated 251 eyes of 251 consecutive patients with choroidal neovascular membranes who underwent anti-VEGF therapy with Ranibizumab in a single practice from 01 July 2008 to 30 June 2010. Clinical characteristics of patients who dropped out of treatment ($n = 86$) were recorded and compared with patients who continued with follow up ($n = 165$).

Results: The dropout rate from follow-up was 34.2%. The dropout rate increased with increasing age, to a maximum of 42.8% in the 91–100 years age group. Dropout rate was slightly higher in females than males. Dropout rate was higher in those living within 100 km of the clinic, compared to those who had to travel greater than 100 km. Having private health insurance or Veterans Affairs assistance did not appear to affect dropout rate. Overall, patients who were more likely to dropout were older patients, those who lived close to the clinic, those without a family history of ARMD and patients with other health problems.

Conclusions: With Ranibizumab therapy, remission was achieved in 22.7% of cases. The dropout rate from this clinic was 34.2%. These findings have implications in the planning of interventions to improve the follow up of patients with blinding eye disease.

133. THE REFLUX BLEB ASSOCIATED WITH INTRA-VITREAL ANTI-VEGF INJECTIONS

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Purpose: To record and quantify the occurrence of a sub conjunctival/sub Tenon's reflux bleb associated with intra vitreal anti-vascular endothelial growth factor (VEGF) injections.

Method: Prospective, consecutive, case series of patients administered with anti-VEGF therapy for all approved indications for VEGF inhibition.

Results: Sixty one eyes of 61 patients were injected with anti-VEGF (ranibizumab and bevacizumab) agents (data collection is ongoing). The mean age of the patients was 71 years and there were 25 males (42%) and 36 females (57%). The mean number of injections per patient was 6.25. Bevacizumab was used in 39 patients (64%) and ranibizumab in 22 patients (36%). The indications for treatment were neo vascular age related macular degeneration in 28 patient (54.9%), retinal vein occlusions in 17 patients (33.3%) and exudative diabetic maculopathy in 6 patients (11.8%).

A subconjunctival/sub Tenon's reflux bleb was observed in 31 patients (51%) and absent in 30 patients (49%).

Conclusion: A subconjunctival/sub Tenon's reflux bleb was observed in 51% of intra vitreal injections with anti-VEGF agent.

The composition and significance of the reflux bleb on the efficacy of treatment for the indications is not known. This study was undertaken as a preliminary investigation to quantify the occurrence of the reflux bleb to investigate the effects of efficacy and composition.

134. PROGNOSTIC FACTORS AND OUTCOME OF INTRAVITREAL RANIBIZUMAB TREATMENT IN NEOVASCULAR AGE-RELATED MACULAR DEGENERATION

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Purpose: To assess the efficacy of intravitreal ranibizumab treatment for neovascular age-related macular degeneration in a clinical setting, and to identify prognostic factors associated with a successful response to treatment.

Method: Retrospective study of 159 cases of choroidal neovascularization treated with 0.5 mg intravitreal ranibizumab injections monthly for three consecutive

doses, followed by a variable dose schedule with final follow-up after 12 months. Repeat injections were administered in the event of persistent or recurrent intraretinal or subretinal fluid, or the occurrence of a new macular hemorrhage on clinical examination or ocular coherence tomography.

Results: The mean number of injections administered over 12 months was 6.3 ± 3.1 . The mean change in visual acuity after 12 months was a gain of 4.0 letters ± 15 letters, which was associated with a mean decrease in central retinal thickness of $88 \mu\text{m} \pm 148 \mu\text{m}$. A total of 85.5% of patients lost fewer than 3 lines of vision, and 20.8% gained more than 3 lines of vision. Visual improvement after 3 months and visual improvement after 6 months were both associated with higher change in final visual acuity ($p = 0.001$ and $p < 0.0001$ respectively). Patients who were not responsive to treatment after 3 months were highly unlikely to become responsive to continued treatment by 6 months (OR = 1.29, 95% CI = 0.479 to 3.45, $p = 0.8036$).

Conclusion: Early visual improvement after 3 months and 6 months can be predictive of sustained visual gains in patients treated with intravitreal ranibizumab.

135. PREPAPILLARY VASCULAR LOOPS – AN UNUSUAL CASE OF RETINAL ARTERY OCCLUSION IN A YOUNG PATIENT AND A POPULATION-BASED SURVEY

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Purpose: Prepapillary vascular loops are uncommon congenital anomalies of the retinal vasculature which project from the optic disc into the vitreous. We report the youngest case in the literature of branch retinal artery occlusions secondary to prepapillary vascular loops. We also aim to determine the prevalence of prepapillary vascular loops in a tertiary referral population.

Methods:

a) A retrospective case review of a young patient with unilateral branch retinal artery occlusions secondary to prepapillary vascular loops.

b) Colour fundus and red-free photos of 1255 consecutive patients from a tertiary referral clinic were reviewed by 2 independent graders for the presence of prepapillary vascular loops.

Results:

- a) A 12 year old girl presented with an acute superior scotoma and headache. Fluorescein angiography revealed bilateral prepapillary vascular loops of arterial origin with thrombus and arterial occlusion at the loop apex in the right eye only. Systemic investigations were negative and she was managed conservatively. After 3 years of follow-up there is persistent scotoma and no evidence of vascular occlusion in the prepapillary loop of the fellow eye.
- b) 2 of 1255 consecutive patients (0.16%, approx. 1 in 628) were found to have prepapillary vascular loops. These patients were asymptomatic and fundus photo analysis determined that the loops in both patients were of venous origin.

Conclusion:

Prepapillary vascular loops are uncommon yet serious causes of retinal artery occlusion which may occur in young patients. The prevalence in the general population may be more than previously estimated.

136. PROCEDURE-BASED ASSESSMENTS (PBA) FOR SURGICAL RETINA RESIDENTS

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Purpose: To introduce a procedure-based assessment (PBA) tool for common vitreo-retinal procedures.

Method: Three different PBA tools for including complex vitrectomy procedures for epiretinal membrane, macular hole surgery and retinal detachment repair with vitrectomy.

This PBA tool for vitreo-retinal surgery includes assessments for pre-operative planning and preparation, global and task-specific intra-operative techniques and post-operative management.

The rating system for each specific task would be as follows: N = not observed or not appropriate; D = development required; S = satisfactory standard (no prompting or intervention required)

Results: The performance of the surgical retina resident can be assessed by an experienced vitreo-retinal surgeon who supervises the trainee using the proposed PBA tools for these vitreo-retinal procedures. Thus, the trainer is able to provide structured, objective feedback to the trainees at different periods of their training.

Conclusion: This procedure-based assessment (PBA) tool for vitreo-retinal surgery is a novel and useful tool for trainers to assess the competency of vitreo-retinal residents during their training. It can be utilised to identify and focus attention on specific surgical tasks that requires development. In addition, trainees are able to continually reflect and improve on their surgi-

cal skills using this tool. This in turn will facilitate interaction and feedback between both trainers and trainees.

137. THE EFFECT OF CATARACT ON OPTICAL COHERENCE TOMOGRAPHY SEGMENTATION ERROR

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Purpose: Spectral-domain optical coherence tomography devices (SD-OCT) produce less measurement artifacts due to higher resolution and faster scanning speeds compared to time-domain. Nevertheless, the images produced by SD-OCT are vulnerable to the interference caused by lens opacities. In this study, the effect of cataract type and severity on image quality and measurement errors was examined.

Method: OCT images of 43 eyes were made preoperatively and 2 days after routine phacoemulsification using Cirrus and Stratus OCT (Carl Zeiss Meditec, Dublin, CA, USA), Topcon OCT (3D OCT-1000, Topcon Corp., Tokyo, Japan), Spectralis HRA+OCT (Heidelberg Engineering Inc., Heidelberg, Germany). For each device, the magnitude and frequency of automatic segmentation errors was analyzed. Additionally, the segmentation errors were correlated to cataract type and severity (according to the LOCS-III classification system).

Results: Error rates were dependent not only on OCT device, but also on cataract type and severity. Measurements were least affected by nuclear cataracts and most affected by posterior subcapsular cataracts. The comparison of segmentation errors between different OCT devices showed the highest rate for Topcon and lowest for Spectralis and Cirrus.

Conclusion: Image quality and measurements for each OCT device were affected to variable extents by cataract. Unfortunately, no reliable segmentation algorithm yet exists for reliable retinal thickness measurements in the presence of cataract. Time-consuming manual post-segmentation analysis would be required to eliminate measurement error. Therefore, with current algorithms, automatically generated OCT measurements are of questionable accuracy in the context of cataract.

138. SPECTRAL-DOMAIN OPTICAL COHERENCE TOMOGRAPHY ANALYSIS OF POSTOPERATIVE CYSTOID MACULAR OEDEMA FOLLOWING PHACOEMULSIFICATION WITH VITREOUS LOSS BETWEEN DIABETIC AND NON-DIABETIC PATIENTS

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Purpose: To assess the incidence of postoperative cystoid macular oedema (CMO) following phacoemulsification with vitreous loss between diabetic patients without pre-existing diabetic retinopathy and non-diabetic patients using spectral-domain optical coherence tomography (SD-OCT).

Methods: This is a prospective cohort study. Patients who underwent complicated phacoemulsification with vitreous loss in University of Malaya Medical Centre between July 2009 and June 2010 were recruited. Subjects with history of uveitis, present use of topical or systemic steroids or non-steroidal anti-inflammatory drugs, pre-existing macular or retinal pathologies were excluded. The average macular thickness was recorded preoperatively, 1 week, 6 weeks and 16 weeks postoperatively. The presence of CMO was detected using SD-OCT (Cirrus; Carl Zeiss Meditech Inc, Jena, Germany). Statistical analyses were performed using SPSS 15.0.

Results: 20 patients (51.3%) with diabetes mellitus (DM) and 19 patients without DM were recruited. The incidence of CMO was 55% in the DM group and 21% in the non-DM group ($p = 0.032$). The mean age, gender and laterality of eyes were not significant between DM and non-DM groups: 66.5 vs 68.0 years ($p = 0.681$), 75% vs 84.2% females ($p = 0.482$) and 55% vs 42.1% left eyes ($p = 0.427$) respectively. The mean macular thickness $\pm$ standard deviation in the DM group and non-DM group at baseline, 1 week, 6 weeks and 16 weeks postoperatively were $275.73 \pm 25.2 \mu\text{m}$, $286.07 \pm 29.6 \mu\text{m}$, $305.67 \pm 28.86 \mu\text{m}$, $313.27 \pm 31.49 \mu\text{m}$ and $262.29 \pm 18.5 \mu\text{m}$, $266.35 \pm 20.7 \mu\text{m}$, $279.24 \pm 23.70 \mu\text{m}$, $284.35 \pm 27.0 \mu\text{m}$ respectively.

Conclusion: Following phacoemulsification with vitreous loss, diabetic patients have higher incidence of developing postoperative CMO compared to non-diabetic patients.

139. THE EFFICACY OF INTRAVITREAL RANIBIZUMAB COMPARED WITH STANDARD GRID LASER FOR THE TREATMENT OF MACULAR OEDEMA OVER A 12 MONTH PERIOD FOLLOWING BRANCH RETINAL VEIN OCCLUSION (BRVO)

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Purpose: To assess the efficacy of intravitreal injections of 0.5 mg ranibizumab to treat foveal-involving macular oedema secondary to BRVO compared with conventional grid laser.

Method: A prospective, phase III randomised, multi-centre trial. Total of 36 patients were randomised 1:1 to a treatment group (15 patients) that received monthly ranibizumab injections for first 6 months and thereafter monthly as needed, or sham group (21 patients) that received monthly sham injections. Grid laser was administered at 13 and 25 weeks in both groups if criteria for laser treatment (BVOS) were met.

Results: Mean BCVA change from baseline (EDTRS letters) was statistically different between ranibizumab and sham groups at 4 months (8.7 versus -3.4), 7 months (8.9 versus -3.9) and 12 months (12.4 versus -1.6) ($p < 0.05$). The percentage of patients who gained ≥ 15 letters in BCVA at 12 months was 53.3% in the ranibizumab group and 19% in the sham group ($p < 0.005$). At 12 months, significantly more ranibizumab-treated patients (60%) achieved BCVA of 6/12 compared with sham-treated patients (28.6%), whilst CFT had decreased by a mean of $361.7 \mu\text{m}$ and $175.6 \mu\text{m}$ in the ranibizumab and sham group respectively ($p < 0.05$). At 13 weeks, more patients in the sham group (47.4%) received grid laser compared with ranibizumab group (6.7%) ($p < 0.0001$), and similarly at 25 weeks 31.3% of sham-treated versus 8.3% ranibizumab-treated patients received laser ($p < 0.005$).

Conclusions: Compared to standard of care, intravitreal ranibizumab provided significant and sustained benefits in visual function and anatomical improvement in macular oedema secondary to BRVO over a 12 month period.

140. TRANSIENT EXACERBATION OF RENAL INSUFFICIENCY ASSOCIATED WITH INTRAVITREAL THERAPY WITH VASCULAR ENDOTHELIAL GROWTH FACTOR INHIBITORS

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Purpose: To report on two cases of transient exacerbation of renal function associated with intravitreal injections of vascular endothelial growth factor (VEGF) inhibitors.

Methods: Since the renal glomerulus is dependent on VEGF for normal functioning, we have routinely measured renal function in patients receiving this treatment. We report two cases, one receiving treatment for neovascular Age-related Macular Degeneration (AMD) and one for diabetic macular edema who, suffered exacerbation of their renal insufficiency after intravitreal injections of VEGF inhibitors. Both the patients were then followed with regular monitoring of their renal function.

Results: The patient with neovascular AMD had mild renal insufficiency at baseline due to hypertensive nephrosclerosis, however his renal function deteriorated from serum creatinine of 156 $\mu\text{mol/L}$ at baseline (normal range: 45–90 $\mu\text{mol/L}$) to 182 $\mu\text{mol/L}$ one month after the eighth intraocular injection of aflibercept. His serum creatinine values improved to near baseline levels seven months after cessation of the treatment. The patient with diabetic macular edema also had mild renal insufficiency at baseline due to diabetic nephropathy. She suffered progressive deterioration of renal function from creatinine values of 140 $\mu\text{mol/L}$ at baseline to a peak value of 225 $\mu\text{mol/L}$ two and a half months after her sixth injection of ranibizumab. Her renal function had improved to near baseline levels four months after cessation of the treatment.

Conclusion: Treatment with intravitreal VEGF inhibitors maybe associated with exacerbation of renal insufficiency. Renal function returned to pre-treatment levels after cessation of intravitreal therapy in the two cases described.

141. AN INTER-DEVICE, INTRA-SESSION COMPARISON OF 2 AUTOMATED MICROPERIMETERS: THE NIDEK MP-1S AND THE CENTERVUE MAIA

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Purpose: To compare the mean retinal sensitivity in the macular region as measured by 2 automated microperimeters – the Nidek MP-1S and the CenterVue MAIA.

Methods: In this prospective study, 34 patients with a range of macular pathology were enrolled. Using the same testing strategy, each patient performed microperimetry on both devices in a randomised order. The difference in mean retinal sensitivity between the devices was analysed by Bland-Altman plots and 95% limits of agreement.

Results: Mean (SD) age was 55 (18) years. Mean (SD) macular sensitivity derived from the MAIA device was 9.9 (1.9) dB greater than that recorded on the MP-1S instrument ($p = 0.000$, Wilcoxon rank test). The 95% limits of agreement (dB) were 6.2 and 13.6. The range of mean sensitivity (dB) in this cohort was 2.7 to 30.6 and 0.0 to 20.0 dB for MAIA and MP-1S respectively.

Conclusions: Based on differences in luminance calibration of the microperimeters, the expected difference between the MP-1S and MAIA sensitivity should be +4 dB. Our observed difference of approximately +10 dB was larger than expected. The MAIA is able to measure a wider range of retinal sensitivity compared to the MP-1S.

142. A COMPARISON OF FEASIBILITY: THE NIDEK MP-1S AND THE CENTERVUE MAIA MICROPERIMETERS

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Purpose: To compare the feasibility and time required for performing microperimetry between two microperimeters.

Methods: In this prospective study, 36 participants with a range of macular pathology were enrolled. Using identical test strategies, one eye of each participant underwent microperimetry on both devices in a randomised order. One of two testing grids were chosen – Grid A (68 loci/ 20° diameter/ Cartesian layout) and Grid B (37 loci/ 6° diameter/ radial layout). Parameters assessed: need for dilation, ability to complete testing, and duration.

Results: 20 subjects were randomised to perform the MP-1S first, 18/20 were able to commence undilated and 2 required mydriasis. However, 1/18 undilated subjects failed to complete the test due to excessive eye movement. Both subjects requiring mydriasis for the MP-1S completed the test. In contrast all 16 subjects randomised to perform MAIA first were able to commence without mydriasis. However, 1/16 subjects failed to complete testing.

For Grid A, mean (SD) duration was 8"47 (0"14) and 8"58 (2"33) for the MAIA and MP-1S respectively ($p = 0.499$, Wilcoxon Rank Test). For Grid B, mean (SD) duration was 5"07 (0"23) and 5"44 (2"09) for the MAIA and MP-1S respectively ($p = 0.062$, Wilcoxon Rank Test).

Conclusions: Microperimetry is a feasible test to perform as part of routine assessment and management of macular conditions. Most subjects are able to perform the test without the need for dilation. On average, the test takes less than 10 minutes for each eye when utilising either of the standard test grids chosen in this study.

STRABISMUS

143. DUANE'S RETRACTION SYNDROME ASSOCIATED WITH MARCUS GUNN JAW WINKING

WITHDRAWN

UVEITIS

144. THE ROLE OF CHLAMYDIA IN THE PATHOGENESIS OF HLA-B27 ACUTE ANTERIOR UVEITIS AND SERONEGATIVE ARTHRITIS

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Purpose: To examine the role of Chlamydia infection in HLA-B27-related disease

Methods: Anterior uveitis (AU) and Seronegative arthritis (SNA) are associated with both HLA-B27 and preceding microbial infections, particularly *C. trachomatis*. It has been hypothesised that molecular mimicry between bacterial and self-antigens presented by HLA-B27 may provide a mechanism for disease. In order to further examine this association peripheral blood mononuclear cells (PBMC) were obtained from 29 patients with a history AU and or SNA. Serology for Chlamydia was assessed by Elisa. In order to measure the HLA restricted cytotoxic T cell response to Chlamydia PBMCs were stimulated with *C. trachomatis* antigens and extracellular components in an ELISPOT assay.

Results: 50% of patients had positive serology for Chlamydia compared to 14% of controls ($N = 7$) ($P < 0.05$, Fisher's exact test). However, there was no difference between patients (24% positive) and controls (29% positive) in the ELISPOT assay ($p = 0.60$, Fisher exact test).

Conclusion: A significant proportion of patients with AAU and SNA have evidence of preceding *C. trachomatis* infection and are HLA-B27 positive. However we were unable to detect a specific HLA B27 restricted cytotoxic T cell-mediated response to *C. Trachomatis*.

145. MULTIFOCAL CHOROIDITIS AND PANUVEITIS IN A PATIENT WITH DISSEMINATED MYCOBACTERIUM AVIUM COMPLEX

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Purpose: We present a case of multifocal choroiditis and panuveitis secondary to disseminated mycobacterium avium complex infection in a patient with HIV/AIDS.

Method: Case report and review of relevant literature.

Results: A 48 yo female with HIV/AIDS was admitted for the work up of abdominal pain, fevers and weight loss. She was non-compliant with anti-retroviral therapy with a CD4 count of $30 \times 10^6/L$ on admission. She reported bilateral reduction in vision with corrected acuities of OD 6/18 and OS 6/36. Bilateral granulomatous panuveitis with multifocal choroiditis was present. A vitreous biopsy of the left eye confirmed MAC, also identified on mycobacterium blood cultures. She was started on oral Azithromycin and Ethambutol and after 6 months her corrected acuities

were OD 6/9 bilaterally with an improvement in uveitis and choroiditis.

Conclusion: Mycobacterium avium complex infection should be considered as a cause of visual impairment in patients with HIV/AIDS. Early detection and prompt management are essential for the prevention of visual and other systemic complications.

146. VEMURAFENIB (B-RAF INHIBITOR) ASSOCIATED UVEITIS IN PATIENTS WITH METASTATIC CUTANEOUS MELANOMA

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Purpose: To report the incidence of uveitis with vemurafenib (BRAF-inhibitor) treatment for metastatic cutaneous melanoma.

Method: Retrospective case review of patients with metastatic cutaneous melanoma enrolled in a clinical trial of vemurafenib, referred to Ocular Immunology Clinic.

Results: Across four centres, 23 (4.5%) patients of 516 treated with vemurafenib presented with uveitis.

Seven presented to our clinic. The mean age was 41.9 years (range:26–59). Five were males. The mean time to uveitis from commencement of treatment was 191.4 days (range:5–594). Most presentations of uveitis were insidiously symptomatic with redness, mild photophobia and visual blurring presenting with signs of mild-moderate bilateral anterior uveitis, fine KPs, and multiple iris nodules. Three patients had 1–2 clock hours of posterior-synechiae. Topical corticosteroid resolved most cases but persistence/chronicity (mean 7.7 months, range:3–17) was a feature. Two cases had signs of intermediate uveitis (one with cystoid macular oedema) requiring oral/peri-ocular corticosteroid. Intraocular pressure rise was not a feature. All eyes resolved with better than 6/6 visual acuity and were treated in the presence of ongoing vemurafenib therapy due to good response to cutaneous disease.

Multiple theories of uveitis include a direct vemurafenib inflammatory response on subclinical metastatic cells within the uveal tract, indirect inflammatory response with antigens shared by melanocytes in the melanoma and choroid, and direct/inadvertent/paradoxical inhibition/activation of downstream MEK-ERK pathways.

Conclusion: Patients have a good response to vemurafenib for cutaneous melanoma and clinicians should be aware that uveitis can be a side-effect. Fortunately, most cases are controlled without discontinuation of this drug.